

FC7xxx ICU User Manual

Rev.0.6

Revision History

Revision	Date	Changes
0.1	2023/07/14	Initial release for MCAL V0.1.0
0.3	2023/10/20	Updated for MCAL V0.3.0
0.6	2023/03/29	Updated for MCAL V0.6.0 - Added support for FC7240

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Chapter 1 ICU Introduction

1.1 Requirements

The design of this module follows the specifications of the ICU driver specified in AUTOSAR Classic Platform Release 4.3.1. For detailed requirements, refer to the *AUTOSAR_SWS_ICUDriver*.

1.2 Hardware Summary

The ICU driver provides logical channels which are configured by hardware including Flexible Timer Unit (FTU), Port Control(PORT) and Always-on Timer (AONTIMER).

FTU supports signal edge detection, signal measurement, timestamp and edge counter.

PORT supports signal edge detection and edge counter.

AONTIMER supports signal edge detection and edge counter.

FTU provides up to 96 channels (for 7240 Only Ftu_0-Ftu_7) .

PORT provides up to 124 channels (for 7240 Only Port_A-Port_E) .

AONTIMER provides 1 channel.

Cmp provides 3 channels (for 7240 Only Cmp_0-Cmp_1) .

Chapter 2 Software Design

2.1 Rejected Requirements

Rejected Requirement 1 **SWS_Icu_00006**

Description	The function Icu_Init shall initialize all relevant registers of the configured hardware with the values of the structure referenced by the parameter ConfigPtr.
Rejection Reason	Some relevant registers are configured in: - Icu_SetActivationCondition - Icu_StartTimeStamp - Icu_EnableEdgeCount - Icu_EnableEdgeDetection - Icu_StartSignalMeasurement

Rejected Requirement 2 **SWS_Icu_00246**

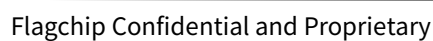
Description	Icu_Irq.c shall include Icu.h for the function which shall be called in the interrupt function and Icu_Irq.h for the declaration of interrupt functions.
Rejection Reason	Icu_Irq.c Icu_Irq.h is not created, interrupt function is located in Icu_Ftu.c Icu_Port.c and Icu_Aontimer.

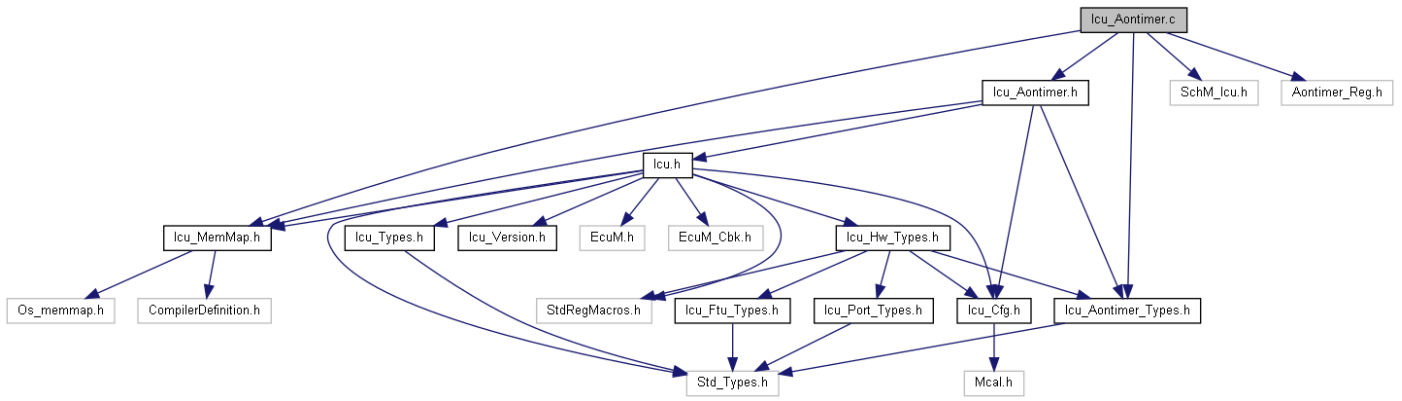
Rejected Requirement 3 **SWS_Icu_00250**

Description	Icu.c shall include Icu_Cbk.h for pre-compile time configuration
Rejection Reason	Icu_Cbk.h is not created

Rejected Requirement 4 **SWS_Icu_00252**

Description	Icu_Irq.c shall include Icu_MemMap.h.
Rejection Reason	Icu_Irq.c is not created





2.3 Macros

2.3.1 Macros in Icu.h

- **#define ICU_E_PARAM_POINTER ((uint8)0x0A)**
API IS called with an invalid pointer.
- **#define ICU_E_PARAM_CHANNEL ((uint8)0x0B)**
API service used with an invalid channel identifier or channel was not configured for the functionality of the calling API.
- **#define ICU_E_PARAM_ACTIVATION ((uint8)0x0C)**
API service used with an invalid or not feasible activation.
- **#define ICU_E_INIT_FAILED ((uint8)0x0D)**
Init function failed.
- **#define ICU_E_PARAM_BUFFER_SIZE ((uint8)0x0E)**
API service used with an invalid buffer size.
- **#define ICU_E_PARAM_MODE ((uint8)0x0F)**
API service Icu_SetMode used with an invalid mode.
- **#define ICU_E_UNINIT ((uint8)0x14)**
API service used without module initialization.
- **#define ICU_E_NOT_STARTED ((uint8)0x15)**
API service Icu_StopTimestamp called on a channel which was not started or already stopped.
- **#define ICU_E_BUSY_OPERATION ((uint8)0x16)**
API service Icu_SetMode is called while a running operation.
- **#define ICU_E_ALREADY_INITIALIZED ((uint8)0x17)**
API Icu_Init service is called and when the ICU driver and the Hardware are already initialized.

- **#define ICU_E_PARAM_NOTIFY_INTERVAL** ((uint8)0x18)
API Icu_StartTimeStamp is called and the parameter NotifyInterval is invalid (e.g. NotifyInterval < 1).
- **#define ICU_E_PARAM_VINFO** ((uint8)0x19)
The pointer to notification function is null.
- **#define ICU_E_PARAM_PARTITION** ((uint8)0x20)
API Check invalid partition.
- **#define ICU_E_PARAM_DMACHANNEL_BUSY** ((uint8)0x22)
Icu dma running time error.
- **#define ICU_INIT_ID** ((uint8)0x00)
API service ID for Icu_Init function.
- **#define ICU_DEINIT_ID** ((uint8)0x01)
API service ID for Icu_DeInit function.
- **#define ICU_SETMODE_ID** ((uint8)0x02)
API service ID for Icu_SetMode function.
- **#define ICU_DISABLEWAKEUP_ID** ((uint8)0x03)
API service ID for Icu_DisableWakeup function.
- **#define ICU_ENABLEWAKEUP_ID** ((uint8)0x04)
API service ID for Icu_EnableWakeup function.
- **#define ICU_SETACTIVATIONCONDITION_ID** ((uint8)0x05)
API service ID for Icu_SetActivationCondition function.
- **#define ICU_DISABLENOTIFICATION_ID** ((uint8)0x06)
API service ID for Icu_DisableNotification function.
- **#define ICU_ENABLENOTIFICATION_ID** ((uint8)0x07)
API service ID for Icu_EnableNotification function.
- **#define ICU_GETINPUTSTATE_ID** ((uint8)0x08)
API service ID for Icu_GetInputState function.
- **#define ICU_STARTTIMESTAMP_ID** ((uint8)0x09)
API service ID for Icu_StartTimestamp function.

- **#define ICU_STOPTIMESTAMP_ID** ((uint8)0x0A)
API service ID for Icu_StopTimestamp function.
- **#define ICU_GETTIMESTAMPINDEX_ID** ((uint8)0x0B)
API service ID for Icu_GetTimestampIndex function.
- **#define ICU_ENABLEEDGECOUNT_ID** ((uint8)0x0D)
API service ID for Icu_EnableEdgeCount function.
- **#define ICU_DISABLEEDGECOUNT_ID** ((uint8)0x0E)
API service ID for Icu_DisableEdgeCount function.
- **#define ICU_GETTIMEELAPSED_ID** ((uint8)0x10)
API service ID for Icu_GetTimeElapsed function.
- **#define ICU_GETEDGENUMBERS_ID** ((uint8)0x0F)
API service ID for Icu_GetEdgeNumbers function.
- **#define ICU_RESETEGECOUNT_ID** ((uint8)0x0C)
API service ID for Icu_ResetEdgeCount function.
- **#define ICU_GETDUTYCYCLEVALUES_ID** ((uint8)0x11)
API service ID for Icu_GetDutyCycleValues function.
- **#define ICU_GETVERSIONINFO_ID** ((uint8)0x12)
API service ID for Icu_GetVersionInfo function.
- **#define ICU_STARTSIGNALMEASUREMENT_ID** ((uint8)0x13)
API service ID for Icu_StartSignalMeasurement function.
- **#define ICU_STOPSIGNALMEASUREMENT_ID** ((uint8)0x14)
API service ID for Icu_StopSignalMeasurement function.
- **#define ICU_CHECKWAKEUP_ID** ((uint8)0x15)
API service ID for Icu_CheckWakeup function.
- **#define ICU_ENABLEEDGEDETECTION_ID** ((uint8)0x16)
API service ID for Icu_EnableEdgeDetection function.
- **#define ICU_DISABLEEDGEDETECTION_ID** ((uint8)0x17)
API service ID for Icu_EnableEdgeDetection function.
-
- **#define ICU_GET_INPUT_LEVEL_ID** ((uint8)0x80)
API service ID for Icu_GetInputLevel function.

- **#define ICU_INSTANCE_ID** ((uint8)0U)
Instance ID of this ICU driver.
- **#define ICU_PARAM_TIME_STAMP_BUFFER_TYPE_MASK** ((uint32)(BIT0))
Definition of bit mask to isolate the time stamp buffer type bitfield.
- **#define ICU_PARAM_MEASUREMENT_PROPERTY_MASK** ((uint32)(BIT1 | BIT2 | BIT3))
Definition of bit mask to isolate the measurement type bitfield.
- **#define ICU_PARAM_MEASUREMENT_PROPERTY_SHIFT** (1U)
Definition of bit shift to isolate the measurement type bitfield.
- **#define ICU_PARAM_MEASUREMENT_GET_PROPERTY(v)** (1U)
Definition of get measurement property

2.3.2 Macros in Icu_Version.h

- **#define ICU_AR_RELEASE_MAJOR_VERSION** 4
- **#define ICU_AR_RELEASE_MINOR_VERSION** 6
- **#define ICU_AR_RELEASE_REVISION_VERSION** 0
- **#define ICU_SW_MAJOR_VERSION** 0
- **#define ICU_SW_MINOR_VERSION** 6
- **#define ICU_SW_PATCH_VERSION** 0
- **#define ICU_VENDOR_ID** 174
- **#define ICU_MODULE_ID** 122

2.3.3 Macros in Icu_Cfg.h

- **#define ICU_CFG_AR_RELEASE_MAJOR_VERSION** 4
- **#define ICU_CFG_AR_RELEASE_MINOR_VERSION** 6
- **#define ICU_CFG_AR_RELEASE_REVISION_VERSION** 0
- **#define ICU_CFG_SW_MAJOR_VERSION** 0

- **#define ICU_CFG_SW_MINOR_VERSION 6**

- **#define ICU_CFG_SW_PATCH_VERSION 0**

- **#define ICU_MAX_CHANNEL ((Icu_ChannelType)XU)**
Maximum number of ICU channels configured.

- **#define ICU_PRECOMPILE_SUPPORT**
Pre-compile configuration constants.

- **#define ICU_DEV_ERROR_DETECT (STD_ON)**
Switches the Development Error Detection and Notification on or off.
STD_ON: Enabled.
STD_OFF: Disabled.

- **#define ICU_REPORT_WAKEUP_SOURCE (STD_ON)**
Switch for enabling Wakeup source reporting.
STD_ON: Enabled.
STD_OFF: Disabled.

- **#define ICU_OVERFLOW_TIME_STATISTICS (STD_ON)**
Switch for enabling Wakeup source reporting.
STD_ON: The number of overflows will be added to the statistics in ICU_MODE_SIGNAL_MEASUREMENT mode..
STD_OFF: The number of overflows will not be added to the statistics.

- **#define ICU_GET_VERSION_INFO_API (STD_ON)**
Adds or removes the service Icu_GetVersionInfo() from the code.
STD_ON: Icu_GetVersionInfo() can be used.
STD_OFF: Icu_GetVersionInfo() can not be used.

- **#define ICU_DE_INIT_API (STD_ON)**
Adds or removes the service Icu_DeInit() from the code.
STD_ON: Icu_DeInit() can be used.
STD_OFF: Icu_DeInit() can not be used.

- **#define ICU_SET_MODE_API (STD_ON)**
Adds or removes the service Icu_SetMode() from the code.
STD_ON: Icu_SetMode() can be used.
STD_OFF: Icu_SetMode() can not be used.

- **#define ICU_DISABLE_WAKEUP_API (STD_ON)**
Adds or removes the service Icu_DisableWakeup() from the code.
STD_ON: Icu_DisableWakeup () can be used.

STD_OFF: Icu_DisableWakeup () can not be used.

- **#define ICU_ENABLE_WAKEUP_API (STD_ON)**
Adds or removes the service Icu_EnableWakeup() from the code.
STD_ON: Icu_EnableWakeup () can be used.
STD_OFF: Icu_EnableWakeup () can not be used
- **#define ICU_TIMESTAMP_API (STD_ON)**
Adds or removes all services related to the timestamping functionality as listed below from the code:
Icu_StartTimestamp(), Icu_StopTimestamp(), Icu_GetTimestampIndex().
STD_ON: The services listed above can be used.
STD_OFF: The services listed above can not be used.
- **#define ICU_EDGE_COUNT_API (STD_ON)**
Adds or removes all services related to the edge counting functionality as listed below, from the code:
Icu_ResetEdgeCount(), Icu_EnableEdgeCount(), Icu_DisableEdgeCount(), Icu_GetEdgeNumbers().
STD_ON: The services listed above can be used.
STD_OFF: The services listed above can not be used.
- **#define ICU_GET_TIME_ELAPSED_API (STD_ON)**
Adds or removes the service Icu_GetTimeElapsed() from the code.
STD_ON: Icu_GetTimeElapsed() can be used.
STD_OFF: Icu_GetTimeElapsed() can not be used.
- **#define ICU_GET_DUTY_CYCLE_VALUES_API (STD_ON)**
Adds or removes the service Icu_GetDutyCycleValues() from the code.
STD_ON: Icu_GetDutyCycleValues () can be used.
STD_OFF: Icu_GetDutyCycleValues () can not be used.
- **#define ICU_GET_INPUT_STATE_API (STD_ON)**
Adds or removes the service Icu_GetInputState() from the code.
STD_ON: Icu_GetInputState () can be used.
STD_OFF: Icu_GetInputState () can not be used.
- **#define ICU_SIGNAL_MEASUREMENT_API (STD_ON)**
Adds or removes the services Icu_StartSignalMeasurement() and Icu_StopSignalMeasurement() from the code.
STD_ON: The services listed above can be used.
STD_OFF: The services listed above cannot be used.
- **#define ICU_WAKEUP_FUNCTIONALITY_API (STD_ON)**
Adds or removes the service Icu_CheckWakeup() from the code.
STD_ON: Icu_CheckWakeup() can be used.
STD_OFF: Icu_CheckWakeup() can not be used.

- **#define ICU_EDGE_DETECT_API (STD_ON)**
Adds or removes the services Icu_EnableEdgeDetection() and Icu_DisableEdgeDetection() from the code.
STD_ON: The services listed above can be used.
STD_OFF: The services listed above cannot be used.
- **#define ICU_GET_INPUT_LEVEL_API (STD_ON)**
Adds or removes the service Icu_GetInputLevel() from the code.
STD_ON: Icu_GetInputLevel() can be used.
STD_OFF: Icu_GetInputLevel() can not be used
- **#define ICU_OVERFLOW_NOTIFICATION_API (STD_ON)**
Add / removes Overflow Notification functionality.
STD_ON: Add Overflow Notification functionality
STD_OFF: Removes Overflow Notification functionality.
- **#define ICU_OVERFLOW_TIME_STATISTICS (STD_ON)**
Switch for enabling overflow time statistics
STD_ON: The number of overflows will be added to the statistics in ICU_MODE_SIGNAL_MEASUREMENT mode.
STD_OFF: The number of overflows will be added to the statistics.
- **#define ICU_MULTICORE_SUPPORT (STD_OFF)**
Switch for enabling multicore feature.
STD_ON: Enable Icu multicore feature.
STD_OFF: Do not enable Icu multicore feature.
- **ICU_FTU_GTB_SUPPORT**
Switch for FtU Gtb feature.
STD_ON: Enable Icu FtU Gtb feature.
STD_OFF: Do not enable Icu FtU Gtb feature.
- **#define ICU_PARTITIONS_NB (XU)**
The partition number.
- **ICU_FTU_GTBGROUPMAXNB**
The FtU Group number.
- **IcuConf_IcuFtuGtbGroup_X (XU)**
Symbolic Names for configured Icu global time base group.
- **#define ICU_FTU_ISR_USED (STD_ON)**
ICU FTU driver interrupt used.

- #define **ICU_PORT_ISR_USED** (STD_ON)
ICU PORT driver interrupt used.
- #define **ICU_AONTMER_ISR_USED** (STD_ON)
ICU AONTIMER driver interrupt used.
- #define **ICU_CMP_ISR_USED**(STD_ON)
ICU CMP driver interrupt used.

2.4 Enums

2.4.1 Enumerations in Icu_Types.h

2.4.1.1 Icu_ModeType

Enumeration	Icu_ModeType	
Description	Allow enabling / disabling of all interrupts which are not required for the ECU wakeup.	
Values	Value	Description
	ICU_MODE_NORMAL = 0	Normal operation, all used interrupts are enabled according to the notification requests.
	ICU_MODE_SLEEP = 1	Reduced power operation. In sleep mode only those notifications are available which are configured as wakeup capable.

2.4.1.2 Icu_InputStateType

Enumeration	Icu_InputStateType	
Description	Input state of an ICU channel.	
Values	Value	Description
	ICU_ACTIVE = 0	An activation edge has been detected.
	ICU_IDLE = 1	No activation edge has been detected since the last call of Icu_GetInputState() or Icu_Init().

2.4.1.3 Icu_InputLevelType

Enumeration	Icu_InputLevelType	
Description	Input level of an ICU channel.	
Values	Value	Description
	ICU_INPUT_LOW = 0	Power state function call sequence undefined.
	ICU_INPUT_HIGH = 1	Power state function has call prepared power s.tate first.

2.4.1.4 Icu_ActivationType

Enumeration	Icu_ActivationType	
Description	Definition of the type of activation of an ICU channel.	
Values	Value	Description

	ICU_RISING_EDGE = 0	An appropriate action shall be executed when a rising edge occurs on the ICU input signal.
	ICU_FALLING_EDGE = 1	An appropriate action shall be executed when a falling edge occurs on the ICU input signal.
	ICU_BOTH_EDGES = 2	An appropriate action shall be executed when either a rising or falling edge occurs on the ICU input signal.

2.4.1.5 Icu_MeasurementModeType

Enumeration	Icu_MeasurementModeType	
Description	Definition of the measurement mode type.	
Values	Value	Description
	ICU_MODE_SIGNAL_EDGE_DETECT = 0	Mode for detecting edges.
	ICU_MODE_SIGNAL_MEASUREMENT = 1	Mode for measuring different times between various configurable edges.
	ICU_MODE_TIMESTAMP = 2	Mode for capturing timer values on configurable edges.
	ICU_MODE_EDGE_COUNTER = 3	Mode for counting edges on configurable edges.

2.4.1.6 Icu_SignalMeasurementPropertyType

Enumeration	Icu_SignalMeasurementPropertyType	
Description	Definition of the measurement property type.	
Values	Value	Description
	ICU_LOW_TIME = 0	The channel is configured for reading the elapsed Signal Low Time.
	ICU_HIGH_TIME = 1	The channel is configured for reading the elapsed Signal High Time.
	ICU_PERIOD_TIME = 2	The channel is configured for reading the elapsed Signal Period Time.
	ICU_DUTY_CYCLE = 3	The channel is configured to read values that are needed for calculating the duty cycle (coherent Active and Period Time).

2.4.1.7 Icu_TimestampBufferType

Enumeration	Icu_TimestampBufferType	
Description	Definition of the timestamp measurement property type.	
Values	Value	Description
	ICU_LINEAR_BUFFER = 0	The buffer will just be filled once.
	ICU_CIRCULAR_BUFFER = 1	After reaching the end of the buffer, the driver restarts at the beginning of the buffer.

2.4.1.8 Icu__TimestampDmaChannelState

Enumeration	Icu__TimestampDmaChannelState	
Description	Definition of the Dma channel state.	

Values	Value	Description
	ICU_DMA_CHANNEL_STATE_IDLE = 0	The dma channel state is idle.
	ICU_DMA_CHANNEL_STATE_ACTIVE = 1	The dma channel state is active

2.5 Typedefs

2.5.1 Typedefs in Icu_Types.h

- typedef **uint32 Icu_ChannelType**
Numeric identifier of an ICU channel.
- typedef **uint32 Icu_ValueType**
Width of the buffer for timestamp ticks and measured elapsed time ticks.
- typedef **uint32 Icu_IndexType**
Type, to abstract the return value of the service Icu_GetTimestampIndex().
- typedef **uint16 Icu_EdgeNumberType**
Type, to abstract the return value of the service Icu_GetEdgeNumbers().
- typedef **void (*Icu_NotifyType)(void)**
ICU notify function type.

2.6 Structures

2.6.1 Icu_AontimerHwConfigType

Structure	Icu_AontimerHwConfigType
Description	This type contains initialization data of ICU driver.
Diagram	<pre> classDiagram class Icu_AontimerModuleConfigType class Icu_AontimerHwConfigType Icu_AontimerHwConfigType .. > Icu_AontimerModuleConfigType : pAontimerModuleConfig </pre>
Data Fields	• const uint8 u8ModuleCount • const Icu_AontimerModuleConfigType (*pAontimerModuleConfig)[]

2.6.2 Icu_AontimerModuleConfigType

Structure	Icu_AontimerModuleConfigType
Description	Aontimer Module Config Type
Diagram	N/A
Data Fields	• const Icu_AontimerInstanceType eAontimerInstance • const Icu_AontimerClkSrcType eClkSrc

- const Icu_AontimerPrescalerEnableType ePrescalerEnable
- const Icu_AontimerGlitchFilterType eGlitchFilter
- const Icu_AontimerPulseClkSrcType ePinSelect

2.6.3 Icu_ConfigType

Structure	Icu_ConfigType
Description	This type contains initialization data of ICU driver.
Diagram	
Data Fields	• const Icu_ChannelType tChannelCount The number of configured channels. • const Icu_ChannelConfigType (*pChannelsConfig)[] Pointer to the ICU channel configuration array. • const Icu_HwConfigType tHwConfig Combined HW specific configuration structure. • const Icu_ChannelType (*pIcuHwMap)[ICU_HW_CHANNEL_COUNT] Index table to translate HW channels to logical used to process interrupts for notifications. • const uint8 *pIcuChannelPartitionMappingPtr Multicore channel partition map table. • const uint32 *pIcuFtuGroupPtr Icu Ftu Gtb Group. Index table to Dma channel.

2.6.4 Icu_ChannelConfigType

Structure	Icu_ChannelConfigType
Description	ICU channel configuration type.
Diagram	N/A
Data Fields	• Icu_MeasurementModeType eMode Measurement mode type of channel. • Icu_HwType eHwType Hardware module type of channel. • uint32 u32ChannelOfHw Sub channel of module of channel. • Icu_ActivationType eActivationEdge Activation type of channel. • Icu_NotifyType pChannelNotification Notification of channel. • uint32 u32Param Parameters of channel (timestamp buffer type or signal measurement property). • boolean bChannelDmaEn

	Enable the dma mode of the channel. - uint8 u8DmachannelRef Ref to the dma channel index. - uint32 u32WakeupSource EcuM wakeup source ID. - Icu_NotifyType pOverflowNotification Notification of channel overflow.
--	---

2.6.5 Icu_DutyCycleType

Structure	Icu_DutyCycleType
Description	ICU channel configuration type.
Diagram	N/A
Data Fields	- Icu_ValueType ActiveTime - Icu_ValueType PeriodTime

2.6.6 Icu_HwConfigType

Structure	Icu_HwConfigType
Description	Hardware configuration type.
Diagram	
Data Fields	- Icu_Ftu_HwConfigType - const Icu_FtuModuleConfigType(* pFtuModuleConfig)[]

2.6.7 Icu_Ftu_HwConfigType

Structure	Icu_Ftu_HwConfigType
Description	Ftu Hw Config Type.
Diagram	
Data Fields	- const Icu_Ftu_HwConfigType *pFtuHwConfig - Point to FTU modules configuration. - const Icu_Port_HwConfigType *pPortHwConfig - Point to PORT modules configuration. - const Icu_AontimerHwConfigType *pAontimerHwConfig - Point to AONTIMER modules configuration.

2.6.8 Icu_FtuModuleConfigType

Structure	Icu_FtuModuleConfigType
Description	FTU Module Config Type
Diagram	N/A
Data Fields	• FTU Module Config Type • const Icu_FtuClkSrcType eClkSrc • const Icu_FtuPrescalerType ePrescaler • const Icu_FtuFilterPrescalerType efilterPrescaler • const uint8 aInputFilter [ICU_FTU_INPUT_FILTER_COUNT] • const uint16 u16Period

2.6.9 Icu_Port_HwConfigType

Structure	Icu_Port_HwConfigType
Description	Port Hw Config Type.
Diagram	<pre> graph TD A[Icu_PortModuleConfigType] -.-> pPortModuleConfig B[Icu_Port_HwConfigType] </pre>
Data Fields	• const uint8 u8ModuleCount • const Icu_PortModuleConfigType(* pPortModuleConfig)[]

2.6.10 Icu_PortModuleConfigType

Structure	Icu_PortModuleConfigType
Description	Port Module Config Type
Diagram	N/A
Data Fields	• const Icu_PortInstanceType ePortInstance

2.6.11 Icu_CmpHwConfigType

Structure	Icu_CmpHwConfigType
Description	Cmp Module Config Type
Diagram	<pre> graph LR A[Icu_CmpHwConfigType] -- pCmpModuleConfig --> B[Icu_CmpmoduleConfigType] B -.-> sCmpWinConfig C[Icu_CmpWinConfigType] B -.-> sCmplnPconfig, sCmplnNconfig D[Icu_CmplnChannelConfigType] B -.-> sCmpDacConfig, sCmpOutPinConfig E[Icu_CmpDacConfigType] B -.-> sCmpOutPinConfig F[Icu_CmpOutPinConfigType] </pre>
Data Fields	• const uint8 u8ModuleCount • const Icu_CmpmoduleConfigType (*pCmpModuleConfig)[];

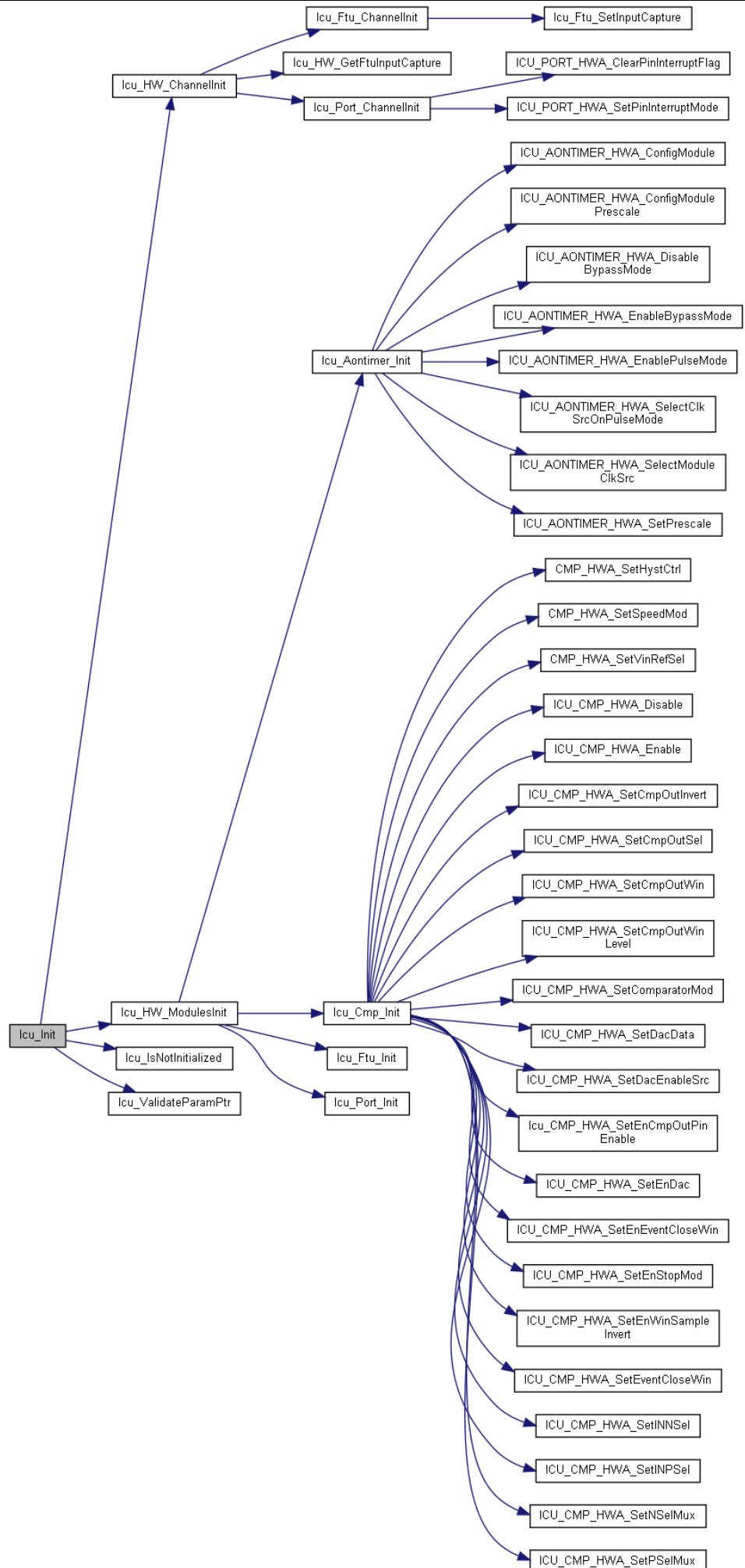
2.7 API Functions

2.7.1 Functions in Icu.h

2.7.1.1 void Icu_Init (const Icu_ConfigType * ConfigPtr)

Function	void Icu_Init (const Icu_ConfigType * ConfigPtr)
Description	Initialize the ICU driver.

Diagram



Parameters	Parameter	Parameters
	ConfigPtr	Pointer to a selected configuration structure.
Returns	N/A	

2.7.1.2 void Icu_DelInit (void)

Function	void Icu_DelInit (void)	
Description	De-initialize the ICU module.	
Diagram	<pre> graph LR Icu_DelInit --> Icu_IsInitialized Icu_DelInit --> Icu_HW_ModulesDelInit Icu_IsInitialized --> Icu_DelInit_Return[N/A] Icu_HW_ModulesDelInit --> Icu_HW_ChannelDelInit Icu_HW_ModulesDelInit --> Icu_Ftu_DelInit Icu_HW_ModulesDelInit --> Icu_Port_DelInit Icu_HW_ChannelDelInit --> Icu_Ftu_StopChannel Icu_Ftu_StopChannel --> ICU_PORT_HWA_ClearPinInterruptFlag Icu_Ftu_StopChannel --> ICU_PORT_HWA_SetPinInterruptMode Icu_Ftu_StopChannel --> Icu_Aontimer_DelInit Icu_Aontimer_DelInit --> ICU_AONTIMER_HWA_ConfigModule ICU_AONTIMER_HWA_ConfigModule --> ICU_AONTIMER_HWA_ConfigModule_Prescale ICU_AONTIMER_HWA_ConfigModule_Prescale --> ICU_AONTIMER_HWA_SetModule_CompareValue ICU_AONTIMER_HWA_SetModule_CompareValue --> Icu_DelInit_Return Icu_Cmp_DelInit --> ICU_CMP_HWA_ClearCCR1 ICU_CMP_HWA_ClearCCR1 --> ICU_CMP_HWA_ClearCCR2 ICU_CMP_HWA_ClearCCR2 --> ICU_CMP_HWA_ClearCmpout ICU_CMP_HWA_ClearCmpout --> ICU_CMP_HWA_ClearDCR ICU_CMP_HWA_ClearDCR --> ICU_CMP_HWA_ClearIntFallingFlag ICU_CMP_HWA_ClearIntFallingFlag --> ICU_CMP_HWA_ClearIntRisingFlag ICU_CMP_HWA_ClearIntRisingFlag --> ICU_CMP_HWA_Disable ICU_CMP_HWA_Disable --> ICU_CMP_HWA_SetComparatorMod ICU_CMP_HWA_SetComparatorMod --> ICU_CMP_HWA_SetDacEnableSrc ICU_CMP_HWA_SetDacEnableSrc --> ICU_CMP_HWA_SetEnStopMod ICU_CMP_HWA_SetEnStopMod --> ICU_CMP_HWA_SetIntEn_Falling ICU_CMP_HWA_SetIntEn_Falling --> ICU_CMP_HWA_SetIntEn_Rising ICU_CMP_HWA_SetIntEn_Rising --> Icu_DelInit_Return </pre>	
Parameters	N/A	
Returns	N/A	

2.7.1.3 void Icu_SetMode (Icu_ModeType Mode)

Function	void Icu_SetMode (Icu_ModeType Mode)	
Description	Set the ICU mode.	

Diagram	<pre>graph LR Icu_SetMode --> Icu_GetBitChState Icu_SetMode --> Icu_HW_SetSleepMode Icu_SetMode --> Icu_HW_SetNormalMode Icu_SetMode --> Icu_IsInitialized Icu_GetBitChState --> Icu_Cmp_SetSleepMode Icu_GetBitChState --> Icu_Cmp_SetNormalMode Icu_HW_SetSleepMode --> Icu_Port_SetSleepMode Icu_HW_SetSleepMode --> Icu_Ftu_SetSleepMode Icu_HW_SetSleepMode --> Icu_Cmp_SetSleepMode Icu_HW_SetSleepMode --> Icu_Aontimer_SetSleepMode Icu_HW_SetNormalMode --> Icu_Port_SetNormalMode Icu_HW_SetNormalMode --> Icu_Ftu_SetNormalMode Icu_HW_SetNormalMode --> Icu_Cmp_SetNormalMode Icu_HW_SetNormalMode --> Icu_Aontimer_SetNormalMode Icu_HW_SetNormalMode --> Icu_HW_GetPortIntConfig Icu_HW_SetNormalMode --> Icu_Ftu_SetInputCapture Icu_HW_SetNormalMode --> Icu_HW_GetAontimerPolarity Icu_HW_SetNormalMode --> Icu_HW_GetCmpPolarity Icu_HW_SetNormalMode --> Icu_HW_GetFtuInputCapture Icu_Port_SetSleepMode --> ICU_PORT_HWA_SetPinInterruptMode Icu_Ftu_SetSleepMode --> ICU_PORT_HWA_SetPinInterruptMode Icu_Cmp_SetSleepMode --> ICU_CMP_HWA_SetIntEn_Falling Icu_Cmp_SetSleepMode --> ICU_CMP_HWA_SetIntEn_Rising Icu_Cmp_SetSleepMode --> ICU_CMP_HWA_ClearIntFallingFlag Icu_Cmp_SetSleepMode --> ICU_CMP_HWA_ClearIntRisingFlag Icu_Aontimer_SetSleepMode --> ICU_AONTIMER_HWA_DisableModuleInterrupt Icu_Aontimer_SetNormalMode --> ICU_AONTIMER_HWA_ClearInterruptFlag Icu_Aontimer_SetNormalMode --> ICU_AONTIMER_HWA_EnableModuleInterrupt Icu_Aontimer_SetNormalMode --> ICU_AONTIMER_HWA_ConfigModulePolarity</pre>					
Parameters	<table><tr><th>Parameter</th><th>Description</th></tr><tr><td>Mode</td><td>ICU_MODE_NORMAL: Normal operation, all used interrupts are enabled according to the notification requests. ICU_MODE_SLEEP: Reduced power mode. In sleep mode, only those notifications are available which are configured as wakeup capable.</td></tr></table>	Parameter	Description	Mode	ICU_MODE_NORMAL: Normal operation, all used interrupts are enabled according to the notification requests. ICU_MODE_SLEEP: Reduced power mode. In sleep mode, only those notifications are available which are configured as wakeup capable.	
Parameter	Description					
Mode	ICU_MODE_NORMAL: Normal operation, all used interrupts are enabled according to the notification requests. ICU_MODE_SLEEP: Reduced power mode. In sleep mode, only those notifications are available which are configured as wakeup capable.					
Returns	N/A					

2.7.1.4 void Icu_DisableWakeup (Icu_ChannelType Channel)

Function	void Icu_DisableWakeup (Icu_ChannelType Channel)					
Description	Disable the wakeup capability of a single ICU channel.					
Diagram	<pre>graph LR; A[Icu_DisableWakeup] --> B[Icu_ClearBitChState]; A --> C[Icu_ValidateWakeup];</pre>					
Parameters	<table><tr><th>Parameter</th><th>Description</th></tr><tr><td>Channel</td><td>Numeric identifier of the ICU channel.</td></tr></table>	Parameter	Description	Channel	Numeric identifier of the ICU channel.	
Parameter	Description					
Channel	Numeric identifier of the ICU channel.					
Returns	N/A					

2.7.1.5 void Icu_EnableWakeup (Icu_ChannelType Channel)

Function	void Icu_EnableWakeup (Icu_ChannelType Channel)	
Description	(re-)enable the wakeup capability of the given ICU channel.	
Diagram		

Parameters	Parameter	Description
	Channel	Numeric identifier of the ICU channel.
Returns	N/A	

2.7.1.6 void Icu_CheckWakeup (EcuM_WakeupSourceType WakeupSource)

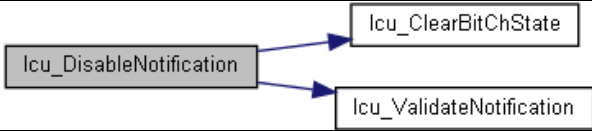
Function	void Icu_CheckWakeup (EcuM_WakeupSourceType WakeupSource)	
Description	Checks if a wakeup capable ICU channel is the source for a wakeup event and calls the ECU state manager service EcuM_SetWakeupEvent in case of a valid ICU channel wakeup event.	
Diagram	<pre> graph LR Icu_CheckWakeup --> Icu_ClearBitChState Icu_CheckWakeup --> Icu_GetBitChState Icu_CheckWakeup --> Icu_IsInitialized </pre>	
Parameters	Parameter	Description
	WakeupSource	Information on wakeup source to be checked.
Returns	N/A	

2.7.1.7 void Icu_SetActivationCondition (Icu_ChannelType Channel, Icu_ActivationType Activation)

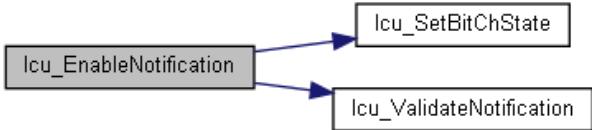
Function	void Icu_SetActivationCondition (Icu_ChannelType Channel, Icu_ActivationType Activation)	
Description	Set the activation-edge for the given channel.	
Diagram	<pre> graph LR Icu_SetActivationCondition --> Icu_ClearBitChState Icu_SetActivationCondition --> Icu_HW_SetActivationCondition Icu_SetActivationCondition --> Icu_IsInitialized Icu_HW_SetActivationCondition --> Icu_Aontimer_SetActivationCondition Icu_HW_SetActivationCondition --> Icu_Cmp_SetActivationCondition Icu_HW_SetActivationCondition --> Icu_Ftu_SetActivationCondition Icu_HW_SetActivationCondition --> Icu_HW_GetAontimerPolarity Icu_HW_SetActivationCondition --> Icu_HW_GetCmpPolarity Icu_HW_SetActivationCondition --> Icu_HW_GetFtuInputCapture Icu_HW_SetActivationCondition --> Icu_HW_GetPortIntConfig Icu_HW_SetActivationCondition --> Icu_Port_SetActivationCondition Icu_Aontimer_SetActivationCondition --> ICU_AONTIMER_HWA_ConfigModulePolarity Icu_Cmp_SetActivationCondition --> ICU_CMP_HWA_SetIntEn_Falling Icu_Cmp_SetActivationCondition --> ICU_CMP_HWA_SetIntEn_Rising Icu_Ftu_SetActivationCondition --> Icu_Ftu_SetInputCapture Icu_Port_SetActivationCondition --> ICU_PORT_HWA_GetPinInterruptMode Icu_Port_SetActivationCondition --> ICU_PORT_HWA_SetPinInterruptMode </pre>	
Parameters	Parameter	Description
	Channel	Numeric identifier of the ICU channel.
	Activation	Type of activation (if supported by hardware) • ICU_RISING_EDGE • ICU_FALLING_EDGE • ICU_BOTH_EDGES
Returns	N/A	

2.7.1.8 void Icu_DisableNotification (Icu_ChannelType Channel)

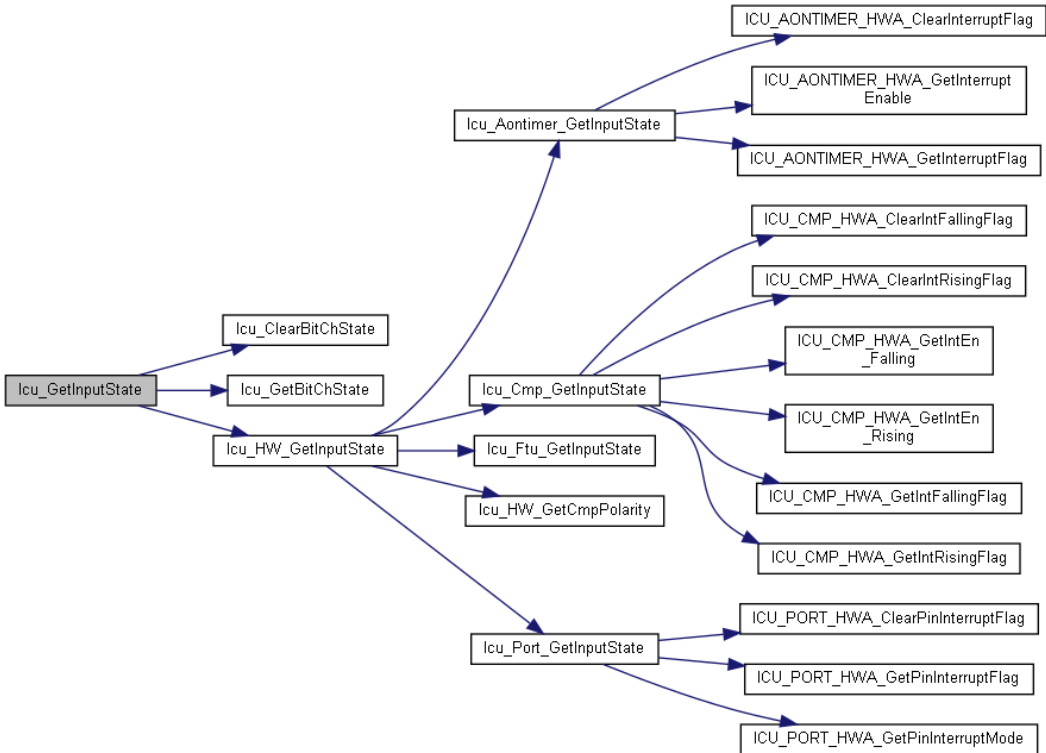
Function	void Icu_DisableNotification (Icu_ChannelType Channel)
----------	--

Description	Disable the notification of a channel.	
Diagram		
Parameters	Parameter	Description
	Channel	ICU channel.
Returns	N/A	

2.7.1.9 void Icu_EnableNotification (Icu_ChannelType Channel)

Function	void Icu_EnableNotification (Icu_ChannelType Channel)	
Description	Enable the notification on the given channel	
Diagram		
Parameters	Parameter	Description
	Channel	ICU channel.
Returns	N/A	

2.7.1.10 Icu_InputStateType Icu_GetInputState (Icu_ChannelType Channel)

Function	Icu_InputStateType Icu_GetInputState (Icu_ChannelType Channel)	
Description	Returns the status of the ICU input.	
Diagram		
Parameters	Parameter	Description
	Channel	ICU channel.
Returns	ICU_ACTIVE: An activation edge has been detected ICU_IDLE: No activation edge has been detected since the last call of Icu_GetInputState() or Icu_Init().	

2.7.1.11 void Icu_StartTimestamp (Icu_ChannelType Channel, Icu_ValueType * BufferPtr, uint16 BufferSize, uint16 NotifyInterval)

Function	void Icu_StartTimestamp (Icu_ChannelType Channel, Icu_ValueType * BufferPtr, uint16 BufferSize, uint16 NotifyInterval)	
Description	Start the capturing of timer values on the edges.	
Diagram	<pre> graph LR Icu_StartTimestamp --> Icu_HW_GetSrcAddr Icu_StartTimestamp --> Icu_HW_StartTimestamp Icu_StartTimestamp --> Icu_SetBitChState Icu_StartTimestamp --> Icu_ValidateParamPtr Icu_StartTimestamp --> Icu_ValidateTimestamp Icu_HW_GetSrcAddr --> Icu_Ftu_GetSrcAddr Icu_HW_StartTimestamp --> Icu_Ftu_EnableChnlDma Icu_HW_StartTimestamp --> Icu_Ftu_EnableOverflow Icu_HW_StartTimestamp --> Icu_Ftu_StartChannel Icu_HW_StartTimestamp --> Icu_HW_GetFtuInputCapture Icu_Ftu_StartChannel --> Icu_Ftu_SetInputCapture </pre>	
Parameters	Parameter	Description
	Channel	ICU channel.
	BufferPtr	Pointer to the buffer-array where the timestamp values shall be placed.
	BufferSize	Size of the external buffer (number of entries).
	NotifyInterval	Notification interval (number of events).
Returns	N/A	

2.7.1.12 void Icu_StopTimestamp (Icu_ChannelType Channel)

Function	void Icu_StopTimestamp (Icu_ChannelType Channel)	
Description	Stop the timestamp measurement of the given channel.	
Diagram	<pre> graph LR Icu_StopTimestamp --> Icu_ClearBitChState Icu_StopTimestamp --> Icu_GetBitChState Icu_StopTimestamp --> Icu_HW_StopTimeStam Icu_StopTimestamp --> Icu_ValidateTimestamp Icu_HW_StopTimeStam --> Icu_Ftu_DisableChnlDma Icu_HW_StopTimeStam --> Icu_Ftu_DisableOverflow Icu_HW_StopTimeStam --> Icu_Ftu_StopChannel </pre>	
Parameters	Parameter	Description
	Channel	ICU channel
Returns	N/A	

2.7.1.13 Icu_IndexType Icu_GetTimestampIndex (Icu_ChannelType Channel)

Function	Icu_IndexType Icu_GetTimestampIndex (Icu_ChannelType Channel)	
Description	Read the timestamp index of the given channel.	
Diagram	<pre> graph LR Icu_GetTimestampIndex --> Icu_ValidateTimestamp </pre>	
Parameters	Parameter	Description
	Channel	ICU channel.
Returns	Timestamp index of the given channel.	

2.7.1.14 void Icu_ResetEdgeCount (Icu_ChannelType Channel)

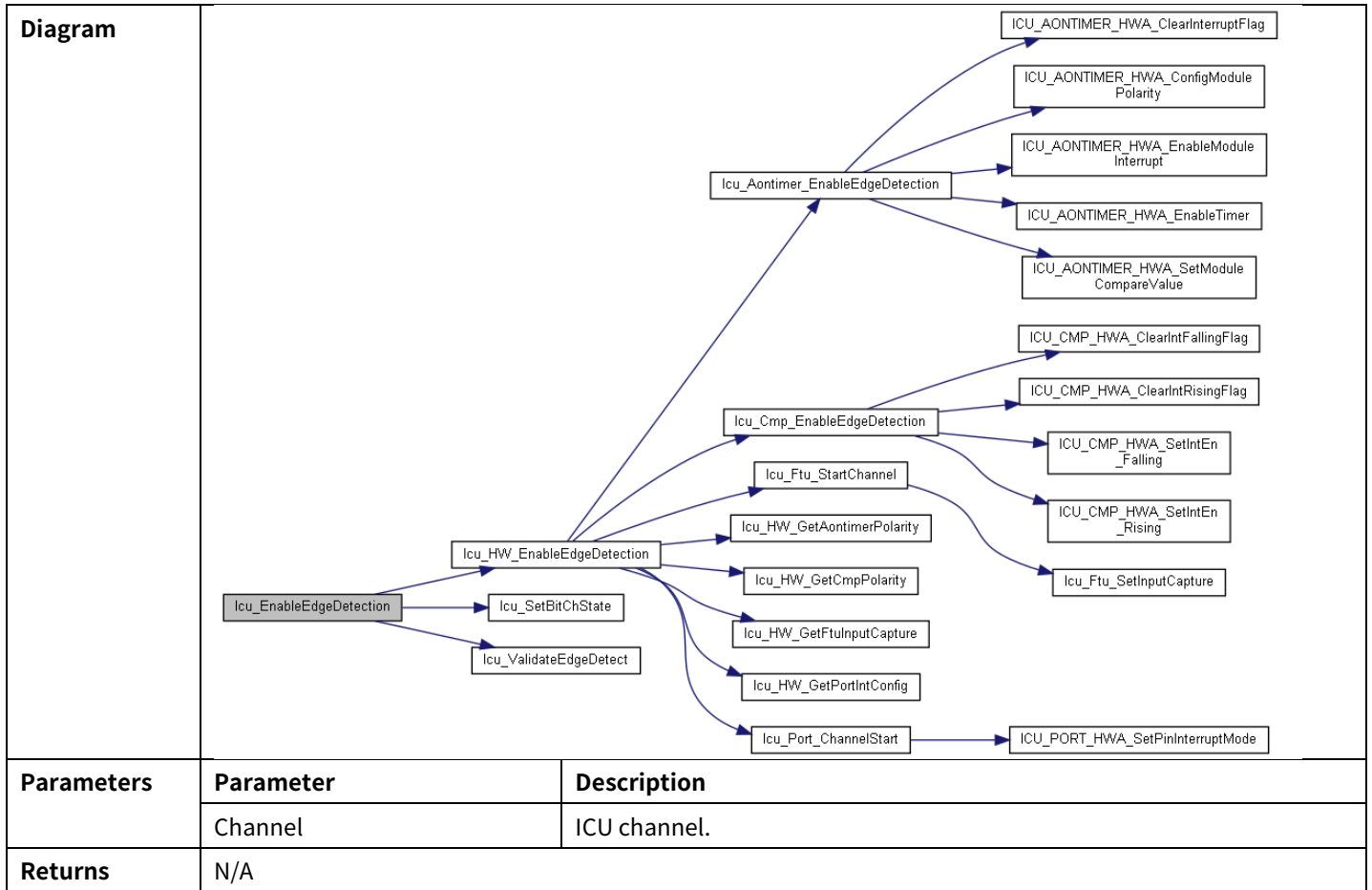
Function	void Icu_ResetEdgeCount (Icu_ChannelType Channel)	
Description	Reset the value of the counted edges to zero.	
Diagram	<pre> graph LR Icu_ResetEdgeCount --> Icu_HW_ResetEdgeNumbers Icu_ResetEdgeCount --> Icu_ValidateEdgeCount Icu_HW_ResetEdgeNumbers --> Icu_Aontimer_ResetEdgeNumbers Icu_Aontimer_ResetEdgeNumbers --> ICU_AONTIMER_HWA_DisableTimer Icu_Aontimer_ResetEdgeNumbers --> ICU_AONTIMER_HWA_EnableTimer </pre>	
Parameters	Parameter	Description
	Channel	ICU channel.
Returns	N/A	

2.7.1.15 void Icu_EnableEdgeCount (Icu_ChannelType Channel)

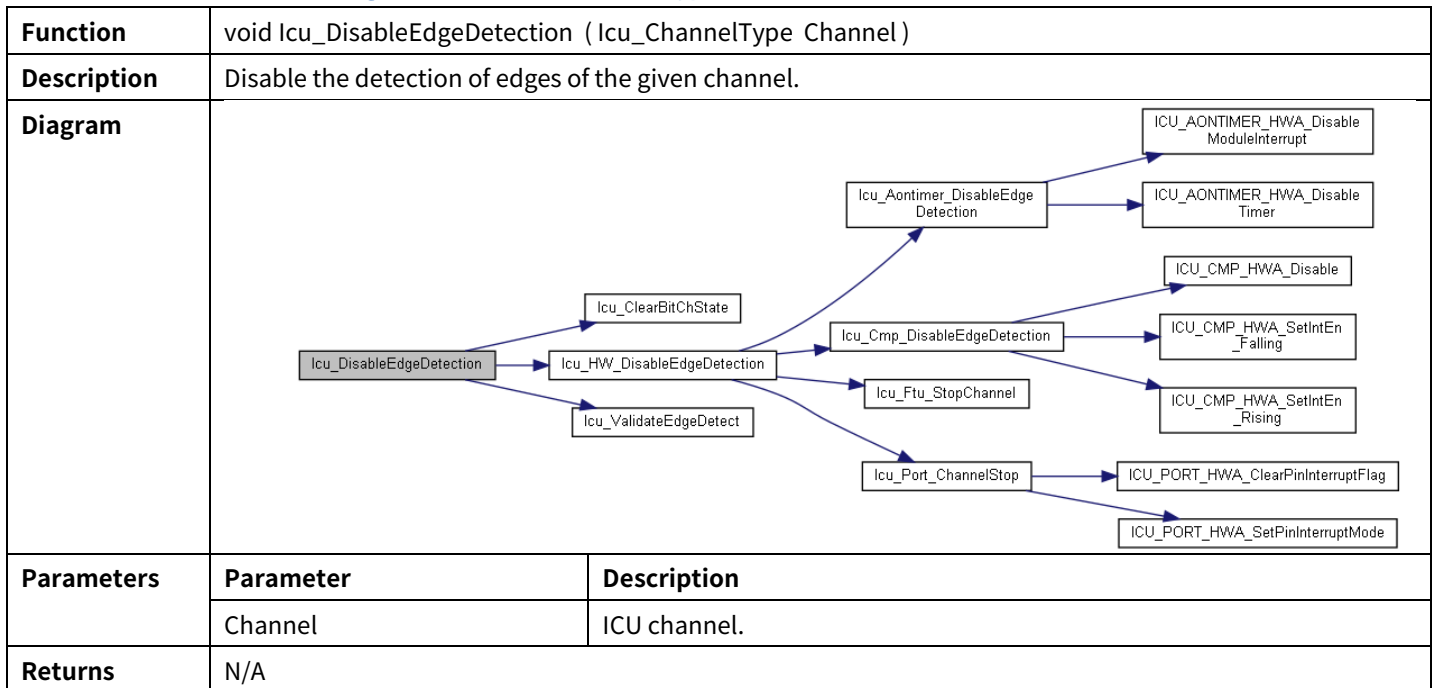
Function	void Icu_EnableEdgeCount (Icu_ChannelType Channel)	
Description	Enable the counting of edges of the given channel.	
Diagram	<pre> graph LR Icu_EnableEdgeCount --> Icu_SetBitChState Icu_EnableEdgeCount --> Icu_ValidateEdgeCount Icu_SetBitChState --> Icu_HW_EnableEdgeCount Icu_HW_EnableEdgeCount --> Icu_Aontimer_EnableEdgeCount Icu_HW_EnableEdgeCount --> Icu_Cmp_EnableEdgeCount Icu_HW_EnableEdgeCount --> Icu_Ftu_StartChannel Icu_HW_EnableEdgeCount --> Icu_HW_GetAontimerPolarity Icu_HW_EnableEdgeCount --> Icu_HW_GetCmpPolarity Icu_HW_EnableEdgeCount --> Icu_HW_GetFtuInputCapture Icu_HW_EnableEdgeCount --> Icu_HW_GetPortIntConfig Icu_HW_EnableEdgeCount --> Icu_Port_ChannelStart Icu_Aontimer_EnableEdgeCount --> ICU_AONTIMER_HWA_ClearInterruptFlag Icu_Aontimer_EnableEdgeCount --> ICU_AONTIMER_HWA_ConfigModulePolarity Icu_Aontimer_EnableEdgeCount --> ICU_AONTIMER_HWA_EnableTimer Icu_Aontimer_EnableEdgeCount --> ICU_AONTIMER_HWA_SetModuleCompareValue Icu_Cmp_EnableEdgeCount --> ICU_CMP_HWA_ClearIntFallingFlag Icu_Cmp_EnableEdgeCount --> ICU_CMP_HWA_ClearIntRisingFlag Icu_Cmp_EnableEdgeCount --> ICU_CMP_HWA_SetIntEnFalling Icu_Cmp_EnableEdgeCount --> ICU_CMP_HWA_SetIntEnRising Icu_Ftu_StartChannel --> Icu_Ftu_SetInputCapture Icu_Port_ChannelStart --> ICU_PORT_HWA_SetPinInterruptMode </pre>	
Parameters	Parameter	Description
	Channel	ICU channel.
Returns	N/A	

2.7.1.16 void Icu_EnableEdgeDetection (Icu_ChannelType Channel)

Function	void Icu_EnableEdgeDetection (Icu_ChannelType Channel)	
Description	Enable / re-enable the detection of edges of the given channel.	



2.7.1.17 void Icu_DisableEdgeDetection (Icu_ChannelType Channel)



2.7.1.18 void Icu_DisableEdgeCount (Icu_ChannelType Channel)

Function	void Icu_DisableEdgeCount (Icu_ChannelType Channel)	
Description	Disable the counting of edges of the given channel.	

Diagram		
Parameters	Parameter	Description
	Channel	ICU channel.
Returns	N/A	

2.7.1.19 Icu_EdgeNumberType Icu_GetEdgeNumbers (Icu_ChannelType Channel)

Function	Icu_EdgeNumberType Icu_GetEdgeNumbers (Icu_ChannelType Channel)	
Description	Read the number of counted edges.	
Diagram		
Parameters	Parameter	Description
	Channel	ICU channel.
Returns	N/A	

2.7.1.20 void Icu_StartSignalMeasurement (Icu_ChannelType Channel)

Function	void Icu_StartSignalMeasurement (Icu_ChannelType Channel)	
Description	Start the measurement of signals.	
Diagram		
Parameters	Parameter	Description

	Channel	ICU channel.
Returns	N/A	

2.7.1.21 void Icu_StopSignalMeasurement (Icu_ChannelType Channel)

Function	void Icu_StopSignalMeasurement (Icu_ChannelType Channel)	
Description	Stop the measurement of signals of the given channel.	
Diagram	<pre> graph LR Icu_StopSignalMeasurement --> Icu_ClearBitChState Icu_StopSignalMeasurement --> Icu_HW_StopSignalMeasurement Icu_StopSignalMeasurement --> Icu_ValidateMeasurement Icu_HW_StopSignalMeasurement --> Icu_Ftu_DisableOverflow Icu_HW_StopSignalMeasurement --> Icu_Ftu_StopChannel Icu_HW_StopSignalMeasurement --> FTU_HWA_ClearChannelEdge_Config[FTU_HWA_ClearChannelEdge Config] Icu_Ftu_DisableOverflow --> FtUCommon_DisableOverflow FtUCommon_DisableOverflow --> FTU_HWA_DisableOverflowInterrupt Icu_Ftu_StopChannel --> FTU_HWA_DisableChannelInterrupt </pre>	
Parameters	Parameter	Description
	Channel	ICU channel.
Returns	N/A	

2.7.1.22 Icu_ValueType Icu_GetTimeElapsed (Icu_ChannelType Channel)

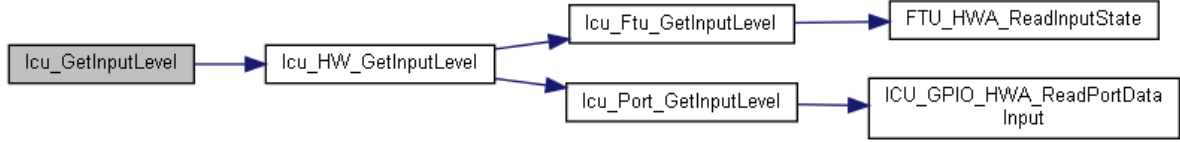
Function	Icu_ValueType Icu_GetTimeElapsed (Icu_ChannelType Channel)	
Description	Read the elapsed Signal Low, High, Period Time for the given channel.	
Diagram	<pre> graph LR Icu_GetTimeElapsed --> Icu_ClearBitChState Icu_GetTimeElapsed --> Icu_ValidateMeasurement </pre>	
Parameters	Parameter	Description
	Channel	ICU channel.
Returns	The elapsed Signal Low, High, Period Time for the given channel.	

2.7.1.23 void Icu_GetDutyCycleValues (Icu_ChannelType Channel, Icu_DutyCycleType * DutyCycleValues)

Function	void Icu_GetDutyCycleValues (Icu_ChannelType Channel, Icu_DutyCycleType * DutyCycleValues)	
Description	Reads the coherent active time and period time for the given ICU Channel.	
Diagram	<pre> graph LR Icu_GetDutyCycleValues --> Icu_ValidateMeasurement Icu_GetDutyCycleValues --> Icu_ValidateParamPtr </pre>	
Parameters	Parameter	Description
	Channel	ICU channel.
	DutyCycleValues	Pointer to a buffer where the results (high time and period time) shall be placed.
Returns	N/A	

2.7.1.24 Icu_InputLevelType Icu_GetInputLevel (Icu_ChannelType Channel)

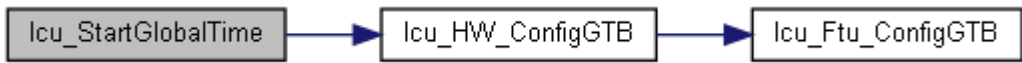
Function	Icu_InputLevelType Icu_GetInputLevel (Icu_ChannelType Channel)	
Description	Read the current input level of channel.	

Diagram	 <pre> graph LR A[Icu_GetInputLevel] --> B[Icu_HW_GetInputLevel] B --> C[Icu_Ftu_GetInputLevel] B --> D[Icu_Port_GetInputLevel] C --> E[FTU_HWA_ReadInputState] D --> F[ICU_GPIO_HWA_ReadPortData Input] </pre>	
Parameters	Parameter	Description
	Channel	ICU channel.
Returns	Input level of channel	

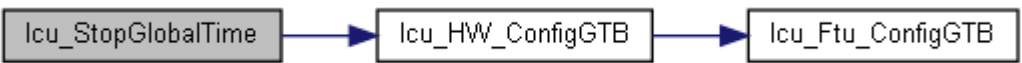
2.7.1.25 void Icu_GetVersionInfo (Std_VersionInfoType * versioninfo)

Function	void Icu_GetVersionInfo (Std_VersionInfoType * versioninfo)	
Description	Return the version information of this module.	
Diagram	N/A	
Parameters	Parameter	Description
	versioninfo	Pointer to where to store the version information of this module.
Returns	N/A	

2.7.1.26 void Icu_StartGlobalTime(const uint32 u32Group)

Function	void Icu_StartGlobalTime(const uint32 u32Group)	
Description	Enable the Ftu Gtb function.	
Diagram	 <pre> graph LR A[Icu_StartGlobalTime] --> B[Icu_HW_ConfigGTB] B --> C[Icu_Ftu_ConfigGTB] </pre>	
Parameters	Parameter	Description
	u32Group	Ftu Gtb Group index.
Returns	N/A	

2.7.1.27 void Icu_StopGlobalTime(const uint32 u32Group)

Function	void Icu_StopGlobalTime(const uint32 u32Group)	
Description	Disable the Ftu Gtb function.	
Diagram	 <pre> graph LR A[Icu_StopGlobalTime] --> B[Icu_HW_ConfigGTB] B --> C[Icu_Ftu_ConfigGTB] </pre>	
Parameters	Parameter	Description
	u32Group	Ftu Gtb Group index.
Returns	N/A	

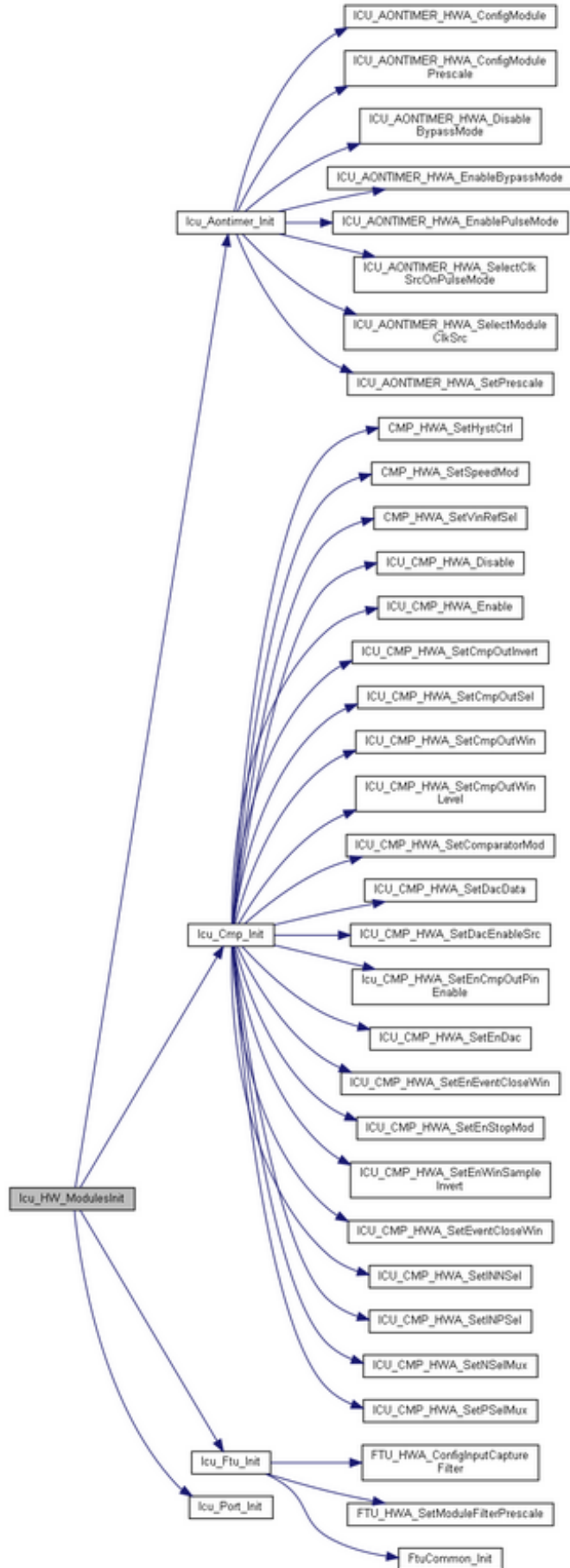
2.8 Hardware Functions

2.8.1 Functions in Icu_Hw.h

2.8.1.1 void Icu_HW_ModulesInit (const Icu_HwConfigType * pHwConfig)

Function	void Icu_HW_ModulesInit (const Icu_HwConfigType * pHwConfig)	
Description	Initialize hardware ICU Modules.	

Diagram



Parameters

Parameter

Description

pHwConfig

Pointer to module configuration structure.

Returns

N/A

Referenced By	void Icu_Init()
----------------------	-----------------

2.8.1.2 void Icu_HW_ChannelDeInit (Icu_HwType eHwType, uint32 u32ChannelOfHw)

Function	void Icu_HW_ChannelDeInit (Icu_HwType eHwType, uint32 u32ChannelOfHw)	
Description	De-initialize a hardware ICU channel.	
Diagram	<pre> graph LR A[Icu_HW_ChannelDeInit] --> B[Icu_Ftu_StopChannel] A --> C[Icu_Port_ChannelStop] B --> D[FTU_HWA_ClearChannelEdgeConfig] B --> E[FTU_HWA_DisableChannelInterrupt] C --> F[ICU_PORT_HWA_ClearPinInterruptFlag] C --> G[ICU_PORT_HWA_SetPinInterruptMode] </pre>	
Parameters	Parameter	Description
	pHwConfig	Module type of the channel.
	u32ChannelOfHw	Channel index of module.
Returns	N/A	
Referenced By	void Icu_DeInit()	

2.8.1.3 void Icu_HW_ModulesDeInit(const Icu_HwConfigType *pHwConfig)

Function	void Icu_HW_ModulesDeInit(const Icu_HwConfigType *pHwConfig)
Description	De-initialize hardware of ICU Modules.

Diagram	<pre> graph LR Icu_HW_ModulesDelnit --> Icu_Aontimer_Delnit Icu_HW_ModulesDelnit --> Icu_Cmp_Delnit Icu_HW_ModulesDelnit --> Icu_Ftu_Delnit Icu_HW_ModulesDelnit --> Icu_Port_Delnit Icu_Aontimer_Delnit --> ICU_AONTIMER_HWA_ConfigModule Icu_Aontimer_Delnit --> ICU_AONTIMER_HWA_ConfigModule_Prescale[ICU_AONTIMER_HWA_ConfigModule Prescale] Icu_Aontimer_Delnit --> ICU_AONTIMER_HWA_SetModule_CompareValue[ICU_AONTIMER_HWA_SetModule CompareValue] Icu_Cmp_Delnit --> ICU_CMP_HWA_ClearCCR1 Icu_Cmp_Delnit --> ICU_CMP_HWA_ClearCCR2 Icu_Cmp_Delnit --> ICU_CMP_HWA_ClearCmpout Icu_Cmp_Delnit --> ICU_CMP_HWA_ClearDCR Icu_Cmp_Delnit --> ICU_CMP_HWA_ClearIntFallingFlag Icu_Cmp_Delnit --> ICU_CMP_HWA_ClearIntRisingFlag Icu_Cmp_Delnit --> ICU_CMP_HWA_Disable Icu_Cmp_Delnit --> ICU_CMP_HWA_SetComparatorMod Icu_Cmp_Delnit --> ICU_CMP_HWA_SetDacEnableSrc Icu_Cmp_Delnit --> ICU_CMP_HWA_SetEnStopMod Icu_Cmp_Delnit --> ICU_CMP_HWA_SetIntEn_Falling Icu_Cmp_Delnit --> ICU_CMP_HWA_SetIntEn_Rising Icu_Cmp_Delnit --> FtuCommon_Delnit </pre>				
Parameters	<table border="1"> <thead> <tr> <th>Parameter</th><th>Description</th></tr> </thead> <tbody> <tr> <td>pHwConfig</td><td>Pointer to module configuration structure.</td></tr> </tbody> </table>	Parameter	Description	pHwConfig	Pointer to module configuration structure.
Parameter	Description				
pHwConfig	Pointer to module configuration structure.				
Returns	N/A				
Referenced By	void Icu_Delnit()				

2.8.1.4 void Icu_HW_ChannelInit(Icu_HwType eHwType, uint32 u32ChannelOfHw, Icu_ActivationType eActivationEdge)

Function	void Icu_HW_ChannelInit(Icu_HwType eHwType, uint32 u32ChannelOfHw, Icu_ActivationType eActivationEdge)				
Description	Initialize hardware of an ICU channel.				
Diagram	<pre> graph LR Icu_HW_ChannelInit --> Icu_Ftu_ChannelInit Icu_HW_ChannelInit --> Icu_Hw_GetFtuInputCapture Icu_HW_ChannelInit --> Icu_Port_ChannelInit Icu_Ftu_ChannelInit --> FTU_HWA_ClearChannelInterruptFlag Icu_Ftu_ChannelInit --> FTU_HWA_DisableChannelInterrupt Icu_Ftu_ChannelInit --> FTU_HWA_SetChannelInputCapture_OnBoth[FTU_HWA_SetChannelInputCapture OnBoth] Icu_Hw_GetFtuInputCapture --> Icu_Ftu_SetInputCapture Icu_Ftu_SetInputCapture --> FTU_HWA_SetChannelInputCapture_OnFalling[FTU_HWA_SetChannelInputCapture OnFalling] Icu_Ftu_SetInputCapture --> FTU_HWA_SetChannelInputCapture_OnRising[FTU_HWA_SetChannelInputCapture OnRising] Icu_Port_ChannelInit --> ICU_PORT_HWA_ClearPinInterruptFlag Icu_Port_ChannelInit --> ICU_PORT_HWA_SetPinInterruptMode </pre>				
Parameters	<table border="1"> <thead> <tr> <th>Parameter</th><th>Description</th></tr> </thead> <tbody> <tr> <td>eHwType</td><td>Module type of the channel.</td></tr> </tbody> </table>	Parameter	Description	eHwType	Module type of the channel.
Parameter	Description				
eHwType	Module type of the channel.				

	u32ChannelOfHw	Channel index of module.
	eActivationEdge	Activation edge of the channel.
Returns	N/A	
Referenced By	void Icu_Init()	

2.8.1.5 void Icu_HW_SetSleepMode(Icu_HwType eHwType, uint32 u32ChannelOfHw)

Function	void Icu_HW_SetSleepMode(Icu_HwType eHwType, uint32 u32ChannelOfHw)	
Description	Set hardware to sleep mode.	
Diagram	<pre> graph LR Icu_HW_SetSleepMode --> Icu_Aontimer_SetSleepMode Icu_HW_SetSleepMode --> Icu_Cmp_SetSleepMode Icu_HW_SetSleepMode --> Icu_Ftu_SetSleepMode Icu_HW_SetSleepMode --> Icu_Port_SetSleepMode Icu_Aontimer_SetSleepMode --> ICU_AONTIMER_HWA_ClearInterruptFlag Icu_Aontimer_SetSleepMode --> ICU_AONTIMER_HWA_DisableModuleInterrupt Icu_Cmp_SetSleepMode --> ICU_CMP_HWA_ClearIntFallingFlag Icu_Cmp_SetSleepMode --> ICU_CMP_HWA_ClearIntRisingFlag Icu_Cmp_SetSleepMode --> ICU_CMP_HWA_SetIntEn_Falling Icu_Cmp_SetSleepMode --> ICU_CMP_HWA_SetIntEn_Rising Icu_Ftu_SetSleepMode --> FTU_HWA_ClearChannelEdgeConfig Icu_Port_SetSleepMode --> ICU_PORT_HWA_ClearPinInterruptFlag Icu_Port_SetSleepMode --> ICU_PORT_HWA_SetPinInterruptMode </pre>	
Parameters	Parameter	Description
	eHwType	Module type of the channel.
	u32ChannelOfHw	Channel index of module.
Returns	N/A	
Referenced By	void Icu_SetMode()	

2.8.1.6 void Icu_HW_SetNormalMode(Icu_HwType eHwType, uint32 u32ChannelOfHw, Icu_ActivationType eActivationEdge)

Function	void Icu_HW_SetNormalMode(Icu_HwType eHwType, uint32 u32ChannelOfHw, Icu_ActivationType eActivationEdge)	
Description	Set hardware to normal mode.	

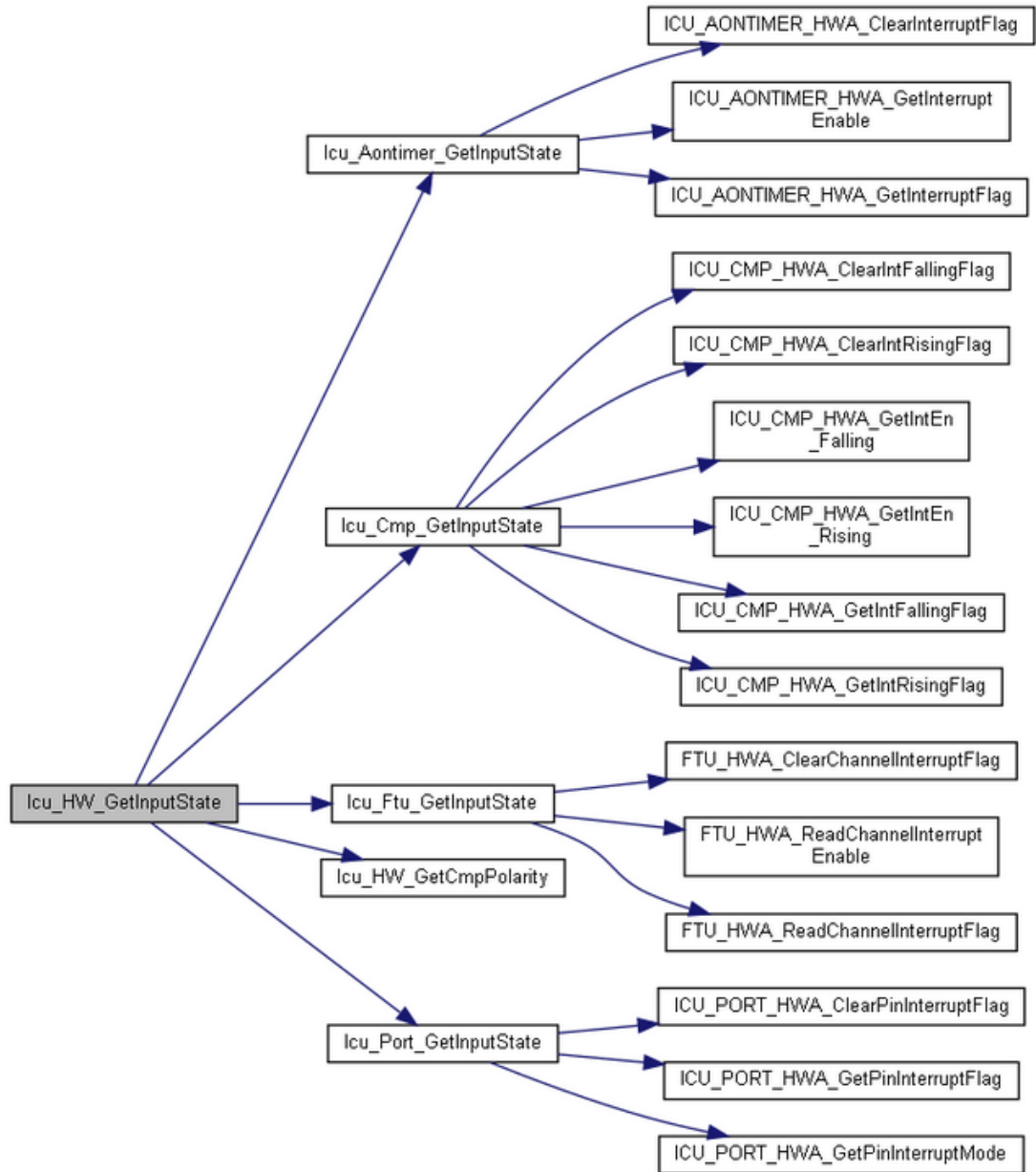
Diagram	<pre>graph LR; Icu_HW_SetNormalMode --> Icu_Aontimer_SetNormalMode; Icu_HW_SetNormalMode --> Icu_Cmp_SetNormalMode; Icu_HW_SetNormalMode --> Icu_Ftu_SetNormalMode; Icu_HW_SetNormalMode --> Icu_HW_GetAontimerPolarity; Icu_HW_SetNormalMode --> Icu_HW_GetCmpPolarity; Icu_HW_SetNormalMode --> Icu_HW_GetFtuInputCapture; Icu_HW_SetNormalMode --> Icu_HW_GetPortIntConfig; Icu_HW_SetNormalMode --> Icu_Port_SetNormalMode; Icu_Aontimer_SetNormalMode --> ICU_AONTIMER_HWA_ClearInterruptFlag; Icu_Aontimer_SetNormalMode --> ICU_AONTIMER_HWA_ConfigModulePolarity; Icu_Aontimer_SetNormalMode --> ICU_AONTIMER_HWA_EnableModuleInterrupt; Icu_Cmp_SetNormalMode --> ICU_CMP_HWA_ClearIntFallingFlag; Icu_Cmp_SetNormalMode --> ICU_CMP_HWA_ClearIntRisingFlag; Icu_Cmp_SetNormalMode --> ICU_CMP_HWA_SetIntEn_Falling; Icu_Ftu_SetNormalMode --> ICU_CMP_HWA_SetIntEn_Rising; Icu_Ftu_SetNormalMode --> Icu_Ftu_SetInputCapture; Icu_Ftu_SetInputCapture --> FTU_HWA_SetChannellInputCaptureOnBoth; Icu_Ftu_SetInputCapture --> FTU_HWA_SetChannellInputCaptureOnFalling; Icu_Ftu_SetInputCapture --> FTU_HWA_SetChannellInputCaptureOnRising; Icu_Port_SetNormalMode --> ICU_PORT_HWA_SetPinInterruptMode;</pre>									
Parameters	<table><tr><th>Parameter</th><th>Description</th></tr><tr><td>eHwType</td><td>Module type of the channel.</td></tr><tr><td>u32ChannelOfHw</td><td>Channel index of module.</td></tr><tr><td>eActivationEdge</td><td>Activation edge of the channel.</td></tr></table>	Parameter	Description	eHwType	Module type of the channel.	u32ChannelOfHw	Channel index of module.	eActivationEdge	Activation edge of the channel.	
Parameter	Description									
eHwType	Module type of the channel.									
u32ChannelOfHw	Channel index of module.									
eActivationEdge	Activation edge of the channel.									
Returns	N/A									
Referenced By	void Icu_SetMode()									

2.8.1.7 void Icu_HW_SetActivationCondition(Icu_HwType eHwType, uint32 u32ChannelOfHw, Icu_ActivationType eActivationEdge)

Function	void Icu_HW_SetActivationCondition(Icu_HwType eHwType, uint32 u32ChannelOfHw, Icu_ActivationType eActivationEdge)									
Description	Set activation for a hardware channel.									
Diagram										
Parameters	<table><tr><th>Parameter</th><th>Description</th></tr><tr><td>eHwType</td><td>Module type of the channel.</td></tr><tr><td>u32ChannelOfHw</td><td>Channel index of module.</td></tr><tr><td>eActivationEdge</td><td>Activation edge of the channel.</td></tr></table>	Parameter	Description	eHwType	Module type of the channel.	u32ChannelOfHw	Channel index of module.	eActivationEdge	Activation edge of the channel.	
Parameter	Description									
eHwType	Module type of the channel.									
u32ChannelOfHw	Channel index of module.									
eActivationEdge	Activation edge of the channel.									
Returns	N/A									
Referenced By	void Icu_SetActivationCondition()									

2.8.1.8 Icu_InputStateType Icu_HW_GetInputState(Icu_HwType eHwType, uint32 u32ChannelOfHw)

Function	Icu_InputStateType Icu_HW_GetInputState(Icu_HwType eHwType, uint32 u32ChannelOfHw)
Description	This function returns the state of the hardware channel.
Diagram	<pre> graph LR Icu_HW_GetInputState --> Icu_Aontimer_GetInputState Icu_HW_GetInputState --> Icu_Ftu_GetInputState Icu_HW_GetInputState --> Icu_Port_GetInputState Icu_Aontimer_GetInputState --> ICU_AONTIMER_HWA_ClearInterruptFlag Icu_Aontimer_GetInputState --> ICU_AONTIMER_HWA_GetInterruptEnable Icu_Aontimer_GetInputState --> ICU_AONTIMER_HWA_GetInterruptFlag Icu_Ftu_GetInputState --> FTU_HWA_ClearChannelInterruptFlag Icu_Ftu_GetInputState --> FTU_HWA_ReadChannelInterruptEnable Icu_Ftu_GetInputState --> FTU_HWA_ReadChannelInterruptFlag Icu_Port_GetInputState --> ICU_PORT_HWA_ClearPinInterruptFlag Icu_Port_GetInputState --> ICU_PORT_HWA_GetPinInterruptFlag Icu_Port_GetInputState --> ICU_PORT_HWA_GetPinInterruptMode </pre>



Parameters	Parameter	Description
	eHwType	Module type of the channel.
	u32ChannelOfHw	Channel index of module.
Returns	State of the channel ICU_ACTIVE An active edge has been detected. ICU_IDLE No active edge has been detected.	
Referenced By	Icu_InputStateType Icu_GetInputState()	

2.8.1.9 void Icu_HW_StartTimestamp(Icu_HwType eHwType, uint32 u32ChannelOfHw, Icu_ActivationType eActivationEdge)

Function	void Icu_HW_StartTimestamp(Icu_HwType eHwType, uint32 u32ChannelOfHw, Icu_ActivationType eActivationEdge)
Description	Start time stamp of a hardware channel.

Diagram	<pre> graph LR Icu_HW_StartTimestamp --> Icu_Ftu_EnableOverflow Icu_Ftu_EnableOverflow --> FtUCommon_EnableOverflow FtUCommon_EnableOverflow --> FTU_HWA_EnableModuleInterrupt Icu_HW_StartTimestamp --> Icu_Ftu_StartChannel Icu_Ftu_StartChannel --> FTU_HWA_ClearChannelInterruptFlag Icu_Ftu_StartChannel --> FTU_HWA_EnableChannelInterrupt Icu_HW_StartTimestamp --> Icu_HW_GetFtuInputCapture Icu_HW_GetFtuInputCapture --> Icu_Ftu_SetInputCapture Icu_Ftu_SetInputCapture --> FTU_HWA_SetChannelInputCaptureOnBoth Icu_Ftu_SetInputCapture --> FTU_HWA_SetChannelInputCaptureOnFalling Icu_Ftu_SetInputCapture --> FTU_HWA_SetChannelInputCaptureOnRising </pre>	
	Parameters	Description
	eHwType	Module type of the channel.
	u32ChannelOfHw	Channel index of module.
	eActivationEdge	Activation edge of the channel.
Returns	N/A	
Referenced By	void Icu_StartTimestamp()	

2.8.1.10 void Icu_HW_StopTimeStamp(Icu_HwType eHwType, uint32 u32ChannelOfHw)

Function	void Icu_HW_StopTimeStamp(Icu_HwType eHwType, uint32 u32ChannelOfHw)	
Description	Stop time stamp of a hardware channel.	
Diagram	<pre> graph LR Icu_HW_StopTimeStamp --> Icu_Ftu_DisableOverflow Icu_Ftu_DisableOverflow --> FtUCommon_DisableOverflow FtUCommon_DisableOverflow --> FTU_HWA_DisableOverflowInterrupt Icu_HW_StopTimeStamp --> Icu_Ftu_StopChannel Icu_Ftu_StopChannel --> FTU_HWA_ClearChannelEdgeConfig Icu_Ftu_StopChannel --> FTU_HWA_DisableChannelInterrupt </pre>	
Parameters	Parameter	Description
	eHwType	Module type of the channel.
	u32ChannelOfHw	Channel index of module.
Returns	N/A	
Referenced By	void Icu_StopTimestamp()	

2.8.1.11 boolean Icu_HW_GetEdgeNumbers(Icu_HwType eHwType, uint32 u32ChannelOfHw, Icu_EdgeNumberType *pEdgeNum)

Function	boolean Icu_HW_GetEdgeNumbers(Icu_HwType eHwType, uint32 u32ChannelOfHw, Icu_EdgeNumberType *pEdgeNum)	
Description	Get edge numbers of a channel from hardware.	
Diagram	<pre> graph LR Icu_HW_GetEdgeNumbers --> Icu_Aontimer_GetEdgeNumbers Icu_Aontimer_GetEdgeNumbers --> ICU_AONTIMER_HWA_GetModuleCounterValue Icu_Aontimer_GetEdgeNumbers --> ICU_AONTIMER_HWA_SetModuleCounterValue Icu_HW_GetEdgeNumbers --> Icu_Aontimer_IsEnable Icu_Aontimer_IsEnable --> ICU_AONTIMER_HWA_GetModuleEnable </pre>	
Parameters	Parameter	Description
	eHwType	Module type of the channel
	u32ChannelOfHw	Channel index of module
	pEdgeNum	Point to store the edge numbers

Returns	TRUE The channel supports getting edge numbers from hardware. FALSE The channel does not support getting edge numbers from hardware.
Referenced By	Icu_EdgeNumberType Icu_GetEdgeNumbers()

2.8.1.12 void Icu_HW_ResetEdgeNumbers(Icu_HwType eHwType, uint32 u32ChannelOfHw)

Function	void Icu_HW_ResetEdgeNumbers(Icu_HwType eHwType, uint32 u32ChannelOfHw)	
Description	Reset hardware edge numbers of the channel.	
Diagram	<pre> graph LR A[Icu_HW_ResetEdgeNumbers] --> B[Icu_Aontimer_ResetEdgeNumbers] B --> C[ICU_AONTIMER_HWA_DisableTimer] B --> D[ICU_AONTIMER_HWA_EnableTimer] </pre>	
Parameters	Parameter	Description
	eHwType	Module type of the channel.
	u32ChannelOfHw	Channel index of module.
Returns	N/A	
Referenced By	void Icu_ResetEdgeCount()	

2.8.1.13 void Icu_HW_EnableEdgeCount(Icu_HwType eHwType, uint32 u32ChannelOfHw, Icu_ActivationType eActivationEdge)

Function	void Icu_HW_EnableEdgeCount(Icu_HwType eHwType, uint32 u32ChannelOfHw, Icu_ActivationType eActivationEdge)	
Description	Enable hardware edge numbers of the channel.	
Diagram	<pre> graph LR A[Icu_HW_EnableEdgeCount] --> B[Icu_Aontimer_EnableEdgeCount] A --> C[Icu_Cmp_EnableEdgeCount] A --> D[Icu_Ftu_StartChannel] A --> E[Icu_HW_GetAontimerPolarity] A --> F[Icu_HW_GetCmpPolarity] A --> G[Icu_HW_GetFtuInputCapture] A --> H[Icu_HW_GetPortIntConfig] A --> I[Icu_Port_ChannelStart] B --> J[ICU_AONTIMER_HWA_ClearInterruptFlag] B --> K[ICU_AONTIMER_HWA_ConfigModulePolarity] B --> L[ICU_AONTIMER_HWA_EnableTimer] B --> M[ICU_AONTIMER_HWA_SetModuleCompareValue] C --> N[ICU_CMP_HWA_ClearIntFallingFlag] C --> O[ICU_CMP_HWA_ClearIntRisingFlag] C --> P[ICU_CMP_HWA_SetIntEn_Falling] C --> Q[ICU_CMP_HWA_SetIntEn_Rising] D --> R[FTU_HWA_ClearChannelInterruptFlag] D --> S[FTU_HWA_EnableChannelInterrupt] D --> T[Icu_Ftu_SetInputCapture] T --> U[FTU_HWA_SetChannelInputCaptureOnBoth] T --> V[FTU_HWA_SetChannelInputCaptureOnFalling] T --> W[FTU_HWA_SetChannelInputCaptureOnRising] I --> X[ICU_PORT_HWA_SetPinInterruptMode] </pre>	
Parameters	Parameter	Description
	eHwType	Module type of the channel.
	u32ChannelOfHw	Channel index of module.

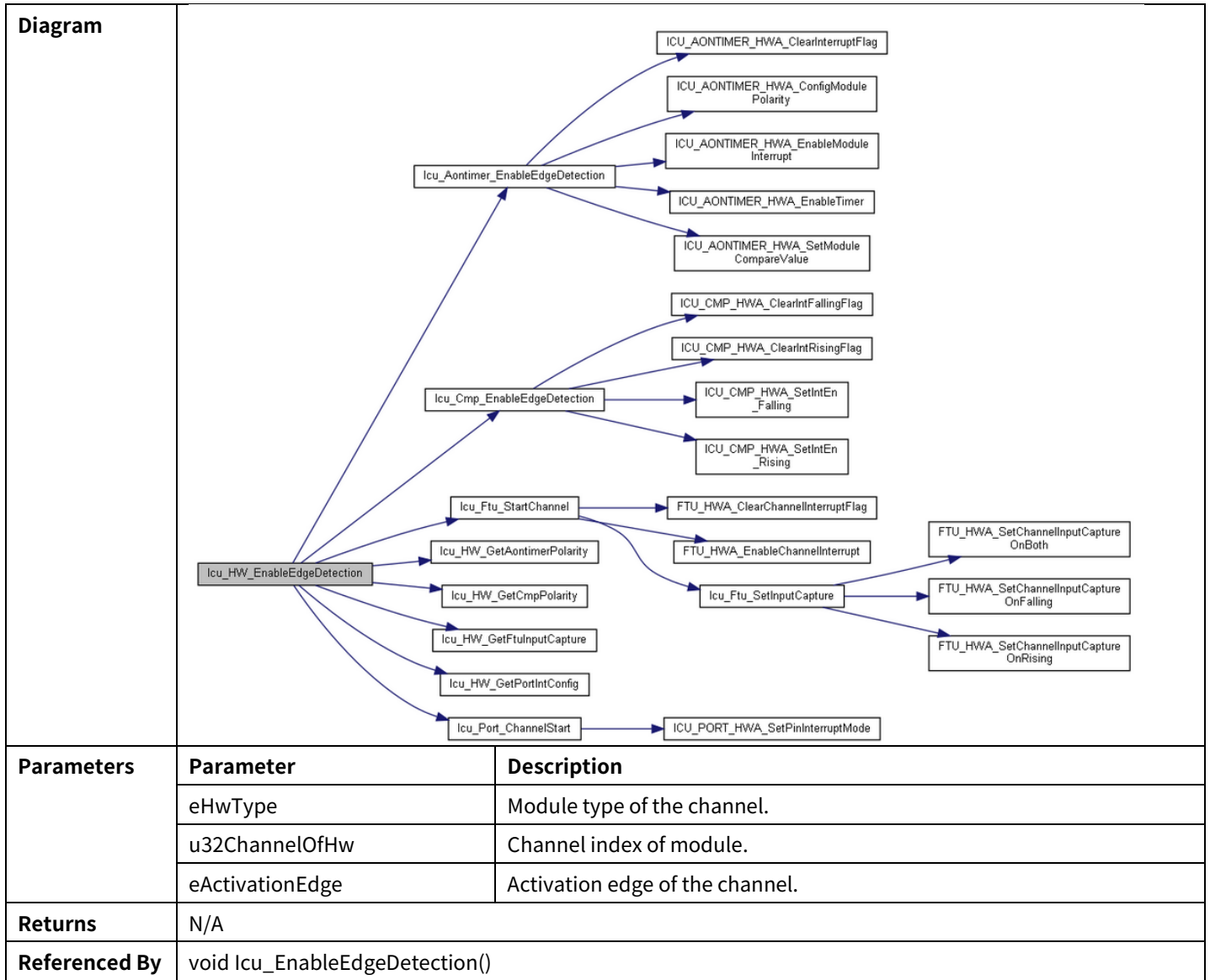
	eActivationEdge	Activation edge of the channel.
Returns	N/A	
Referenced By	void Icu_EnableEdgeCount()	

2.8.1.14 void Icu_HW_DisableEdgeCount(Icu_HwType eHwType, uint32 u32ChannelOfHw)

Function	void Icu_HW_DisableEdgeCount(Icu_HwType eHwType, uint32 u32ChannelOfHw)	
Description	Disable hardware edge numbers of the channel.	
Diagram	<pre> graph LR Icu_HW_DisableEdgeCount --> Icu_Aontimer_DisableEdgeCount Icu_HW_DisableEdgeCount --> Icu_Cmp_DisableEdgeCount Icu_HW_DisableEdgeCount --> Icu_Ftu_StopChannel Icu_HW_DisableEdgeCount --> Icu_Port_ChannelStop Icu_Aontimer_DisableEdgeCount --> ICU_AONTIMER_HWA_Disable_Timer Icu_Cmp_DisableEdgeCount --> ICU_CMP_HWA_ClearIntFallingFlag Icu_Cmp_DisableEdgeCount --> ICU_CMP_HWA_ClearIntRisingFlag Icu_Cmp_DisableEdgeCount --> ICU_CMP_HWA_SetIntEn_Falling Icu_Cmp_DisableEdgeCount --> ICU_CMP_HWA_SetIntEn_Rising Icu_Ftu_StopChannel --> FTU_HWA_ClearChannelEdge_Config Icu_Ftu_StopChannel --> FTU_HWA_DisableChannelInterrupt Icu_Port_ChannelStop --> ICU_PORT_HWA_ClearPinInterruptFlag Icu_Port_ChannelStop --> ICU_PORT_HWA_SetPinInterruptMode </pre>	
Parameters	Parameter	Description
	eHwType	Module type of the channel.
	u32ChannelOfHw	Channel index of module.
Returns	N/A	
Referenced By	void Icu_DisableEdgeCount()	

2.8.1.15 void Icu_HW_EnableEdgeDetection(Icu_HwType eHwType, uint32 u32ChannelOfHw, Icu_ActivationType eActivationEdge)

Function	void Icu_HW_EnableEdgeDetection(Icu_HwType eHwType, uint32 u32ChannelOfHw, Icu_ActivationType eActivationEdge)	
Description	Enable hardware edge detection of the channel.	



2.8.1.16 void Icu_HW_DisableEdgeDetection(Icu_HwType eHwType, uint32 u32ChannelOfHw)

Function	void Icu_HW_DisableEdgeDetection(Icu_HwType eHwType, uint32 u32ChannelOfHw)
Description	Disable hardware edge detection of the channel

Diagram	<pre>graph LR; Icu_HW_DisableEdgeDetection --> Icu_Aontimer_DisableEdgeDetection; Icu_HW_DisableEdgeDetection --> Icu_Cmp_DisableEdgeDetection; Icu_HW_DisableEdgeDetection --> Icu_Ftu_StopChannel; Icu_HW_DisableEdgeDetection --> Icu_Port_ChannelStop; Icu_Aontimer_DisableEdgeDetection --> ICU_AONTIMER_HWA_DisableModuleInterrupt; Icu_Aontimer_DisableEdgeDetection --> ICU_AONTIMER_HWA_DisableTimer; Icu_Cmp_DisableEdgeDetection --> ICU_CMP_HWA_Disable; Icu_Cmp_DisableEdgeDetection --> ICU_CMP_HWA_SetIntEn_Falling; Icu_Cmp_DisableEdgeDetection --> ICU_CMP_HWA_SetIntEn_Rising; Icu_Ftu_StopChannel --> FTU_HWA_ClearChannelEdgeConfig; Icu_Ftu_StopChannel --> FTU_HWA_DisableChannelInterrupt; Icu_Port_ChannelStop --> ICU_PORT_HWA_ClearPinInterruptFlag; Icu_Port_ChannelStop --> ICU_PORT_HWA_SetPinInterruptMode;</pre>							
Parameters	<table><tr><th>Parameter</th><th>Description</th></tr><tr><td>eHwType</td><td>Module type of the channel.</td></tr><tr><td>u32ChannelOfHw</td><td>Channel index of module.</td></tr></table>	Parameter	Description	eHwType	Module type of the channel.	u32ChannelOfHw	Channel index of module.	
Parameter	Description							
eHwType	Module type of the channel.							
u32ChannelOfHw	Channel index of module.							
Returns	N/A							
Referenced By	void Icu_DisableEdgeDetection()							

2.8.1.17 void Icu_HW_StartSignalMeasurement(Icu_HwType eHwType, uint32 u32ChannelOfHw)

Function	void Icu_HW_StartSignalMeasurement(Icu_HwType eHwType, uint32 u32ChannelOfHw)	
Description	Start hardware signal measurement of the channel.	
Diagram		
Parameters	Parameter	Description
	eHwType	Module type of the channel.
	u32ChannelOfHw	Channel index of module.
Returns	N/A	
Referenced By	void Icu_StartSignalMeasurement()	

2.8.1.18 void Icu_HW_StopSignalMeasurement(Icu_HwType eHwType, uint32 u32ChannelOfHw)

Function	void Icu_HW_StopSignalMeasurement(Icu_HwType eHwType, uint32 u32ChannelOfHw)	
Description	Stop hardware signal measurement of the channel.	

Diagram	<pre>graph LR; A[Icu_HW_StopSignalMeasurement] --> B[Icu_Ftu_DisableOverflow]; A --> C[Icu_Ftu_StopChannel]; B --> D[FtuCommon_DisableOverflow]; D --> E[FTU_HWA_DisableOverflowInterrupt]; C --> F[FTU_HWA_ClearChannelEdge Config]; F --> G[FTU_HWA_DisableChannelInterrupt];</pre>	
Parameters	Parameter	Description
	eHwType	Module type of the channel.
	u32ChannelOfHw	Channel index of module.
Returns	N/A	
Referenced By	void Icu_StopSignalMeasurement()	

2.8.1.19 boolean Icu_HW_GetInputLevel (Icu_HwType eHwType,uint32 u32ChannelOfHw, Icu_InputLevelType *pLevel)

Function	boolean Icu_HW_GetInputLevel (Icu_HwType eHwType,uint32 u32ChannelOfHw, Icu_InputLevelType *pLevel)	
Description	Get hardware input level of the channel.	
Diagram	<pre>graph LR; A[Icu_HW_GetInputLevel] --> B[Icu_Cmp_GetInputLevel]; A --> C[Icu_Ftu_GetInputLevel]; A --> D[Icu_Port_GetInputLevel]; B --> E[ICU_CMP_HWA_GetCmpOut]; C --> F[FTU_HWA_ReadInputState]; D --> G[ICU_GPIO_HWA_ReadPortData Input];</pre>	
Parameters	Parameter	Description
	eHwType	Module type of the channel.
	u32ChannelOfHw	Channel index of module.
	pLevel	Point to store the input level
Returns	TRUE. The channel supports getting input level from hardware. FALSE. The channel does not support get input level from hardware.	
Referenced By	Icu_GetInputLevel()	

2.8.1.20 Icu_ValueType Icu_HW_GetOverflowValue (Icu_HwType eHwType)

Function	Icu_ValueType Icu_HW_GetOverflowValue (Icu_HwType eHwType)	
Description	Get overflow value of the channel.	
Diagram	<pre>graph LR; A[Icu_HW_GetOverflowValue] --> B[Icu_Ftu_GetInstanceMaxCount]; B --> C[FTU_HWA_ReadMaxCount];</pre>	
Parameters	Parameter	Description
	eHwType	Module type of the channel.
Returns	Overflow count	
Referenced By	Icu_StartSignalMeasurement().	

2.8.1.21 Icu_HW_ConfigGTB(const uint32 u32Group,boolean benable)

Function	Icu_HW_ConfigGTB(const uint32 u32Group,boolean benable)	
Description	Config the FTU GTB Function	
Diagram	<pre> graph LR A[Icu_HW_ConfigGTB] --> B[Icu_Ftu_ConfigGTB] </pre>	

Parameters	Parameter	Description
	u32Group	FTU GTB Group
	benable	enable/disable Ftu Gtb function.
Returns	Overflow count	
Referenced By	Icu_HW_FtuOverflowIntrProcess().	

2.8.1.22 void Icu_ChannelInterruptCallback(Icu_HwChannelType tHwChannel,uint32 u32Value, Icu_InputLevelType eInputLevel)

Function	void Icu_ChannelInterruptCallback(Icu_HwChannelType tHwChannel,uint32 u32Value, Icu_InputLevelType eInputLevel)	
Description	The callback of Icu channel interrupt.	
Diagram	<pre> graph LR Icu_ChannelInterruptCallback --> Icu_ReportWakeup Icu_ChannelInterruptCallback --> Icu_SetBitChState Icu_ChannelInterruptCallback --> Icu_GetBitChState Icu_ChannelInterruptCallback --> Icu_CallNotification Icu_ChannelInterruptCallback --> Icu_TimeStampProcess Icu_ChannelInterruptCallback --> Icu_ClearBitChState Icu_ChannelInterruptCallback --> Icu_Hw_StopTimeStamp Icu_ChannelInterruptCallback --> Icu_SignalMeasurementLevel Icu_ChannelInterruptCallback --> Icu_SignalMeasurementPeriod Icu_ChannelInterruptCallback --> Icu_SignalMeasurementDutyCycle Icu_ChannelInterruptCallback --> Icu_Ftu_DisableOverflow Icu_ChannelInterruptCallback --> Icu_Ftu_StopChannel Icu_ChannelInterruptCallback --> Icu_GetSigalDuration Icu_ReportWakeup --> Icu_SetBitChState Icu_ReportWakeup --> Icu_GetBitChState Icu_SetBitChState --> Icu_GetBitChState Icu_GetBitChState --> Icu_CallNotification Icu_CallNotification --> Icu_TimeStampProcess Icu_TimeStampProcess --> Icu_ClearBitChState Icu_ClearBitChState --> Icu_Hw_StopTimeStamp Icu_Hw_StopTimeStamp --> Icu_Ftu_DisableOverflow Icu_Ftu_DisableOverflow --> FtUCommon_DisableOverflow FtUCommon_DisableOverflow --> FTU_HWA_DisableOverflowInterrupt Icu_Ftu_DisableOverflow --> Icu_Ftu_StopChannel Icu_Ftu_StopChannel --> FTU_HWA_DisableChannelInterrupt FTU_HWA_DisableChannelInterrupt --> FTU_HWA_ClearChannelEdgeConfig Icu_SignalMeasurementLevel --> Icu_SignalMeasurementPeriod Icu_SignalMeasurementPeriod --> Icu_SignalMeasurementDutyCycle Icu_SignalMeasurementDutyCycle --> Icu_GetSigalDuration Icu_GetSigalDuration --> Icu_Ftu_StopChannel </pre>	
Parameters	Parameter	Description
	tHwChannel	Hardware channel index
	u32Value	Tick count of the hardware
	eInputLevel	Input level
Returns	Overflow count	
Referenced By	Icu_StartSignalMeasurement().	

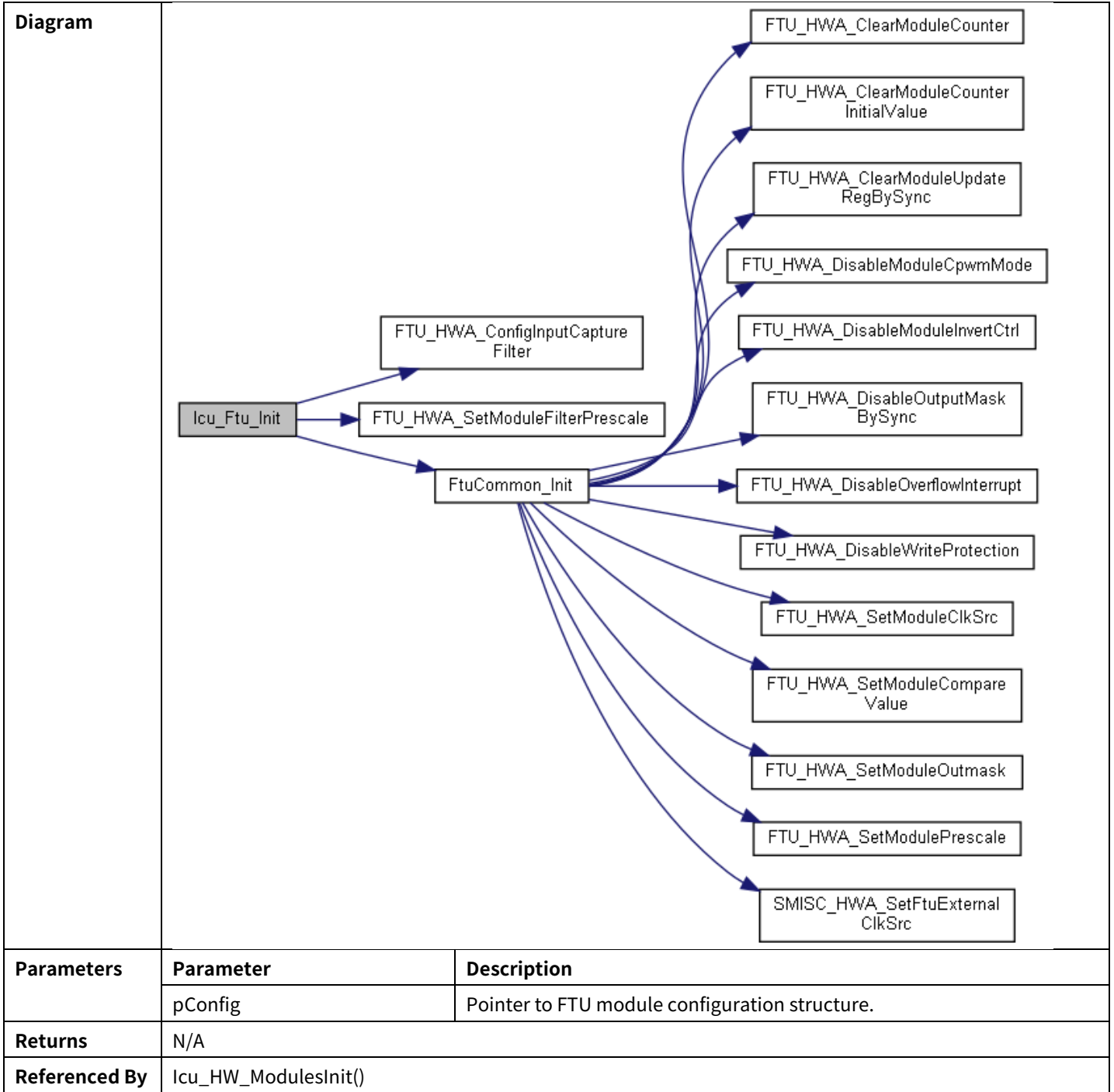
2.8.1.23 void Icu_OverflowCallback (Icu_HwChannelType tHwChannel)

Function	void Icu_OverflowCallback(Icu_HwChannelType tHwChannel)	
Description	The callback of Icu channel overflow	
Diagram	<pre> graph LR Icu_OverflowCallback --> Icu_GetBitChState </pre>	
Parameters	Parameter	Description
	tHwChannel	Hardware channel index
Returns	Overflow count	
Referenced By	Icu_HW_FtuOverflowIntrProcess().	

2.8.2 Functions in Icu_Ftu.h

2.8.2.1 void Icu_Ftu_Init (const Icu_FtuModuleConfigType * pConfig)

Function	void Icu_Ftu_Init (const Icu_FtuModuleConfigType * pConfig)
Description	Initialize a FTU Module.



2.8.2.2 void Icu_Ftu_ChannelInit (const Icu_FtuInstanceType eFtuInstance, const uint8 u8Channel, const Icu_FtuInputCaptureType eActivationEdge)

Function	void Icu_Ftu_ChannelInit (const Icu_FtuInstanceType eFtuInstance, const uint8 u8Channel, const Icu_FtuInputCaptureType eActivationEdge)
Description	Initialize a FTU channel.

Diagram	<pre>graph LR; Icu_Ftu_ChannelInit --> FTU_HWA_ClearChannelInterruptFlag; Icu_Ftu_ChannelInit --> FTU_HWA_DisableChannelInterrupt; Icu_Ftu_ChannelInit --> Icu_Ftu_SetInputCapture; Icu_Ftu_SetInputCapture --> FTU_HWA_SetChannelInputCaptureOnBoth; Icu_Ftu_SetInputCapture --> FTU_HWA_SetChannelInputCaptureOnFalling; Icu_Ftu_SetInputCapture --> FTU_HWA_SetChannelInputCaptureOnRising;</pre>								
	Parameters	<table><thead><tr><th>Parameter</th><th>Description</th></tr></thead><tbody><tr><td>eFtuInstance</td><td>FTU instance ID</td></tr><tr><td>u8Channel</td><td>FTU channel index.</td></tr><tr><td>eActivationEdge</td><td>Activation edge of the channel.</td></tr></tbody></table>	Parameter	Description	eFtuInstance	FTU instance ID	u8Channel	FTU channel index.	eActivationEdge
Parameter	Description								
eFtuInstance	FTU instance ID								
u8Channel	FTU channel index.								
eActivationEdge	Activation edge of the channel.								
Returns	N/A								
Referenced By	Icu_HW_ChannelInit()								

2.8.2.3 void Icu_Ftu_DelInit (const Icu_FtuInstanceType eFtuInstance)

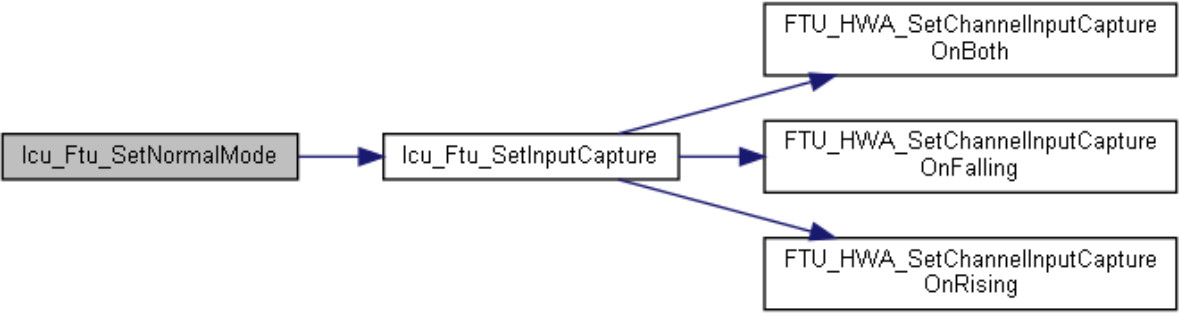
Function	void Icu_Ftu_DelInit (const Icu_FtuInstanceType eFtuInstance)	
Description	De-initialize a FTU Module.	
Diagram	<pre>graph LR; Icu_Ftu_Delnit[Icu_Ftu_Delnit] --> FtCommon_Delnit[FtuCommon_Delnit]; FtCommon_Delnit --> FTU_HWA_ClearModuleCounter[FTU_HWA_ClearModuleCounter]; FtCommon_Delnit --> FTU_HWA_ClearModuleCounter_InitialValue[FTU_HWA_ClearModuleCounter InitialValue]; FtCommon_Delnit --> FTU_HWA_ClearModuleUpdate_RegBySync[FTU_HWA_ClearModuleUpdate RegBySync]; FtCommon_Delnit --> FTU_HWA_ClearOverflowFlag[FTU_HWA_ClearOverflowFlag]; FtCommon_Delnit --> FTU_HWA_DeinitModuleCommon[FTU_HWA_DeinitModuleCommon]; FtCommon_Delnit --> FTU_HWA_DisableOverflowInterrupt[FTU_HWA_DisableOverflowInterrupt]; FtCommon_Delnit --> FTU_HWA_SetModuleCompare_Value[FTU_HWA_SetModuleCompare Value]; FtCommon_Delnit --> FTU_HWA_SetModuleSYNCRreg[FTU_HWA_SetModuleSYNCRreg];</pre>	
Parameters	Parameter	Description
	eFtuInstance	FTU instance ID.
Returns	N/A	
Referenced By	Icu_HW_ModulesDelnit()	

2.8.2.4 void Icu_Ftu_SetSleepMode (const Icu_FtuInstanceType eFtuInstance, const uint8 u8Channel)

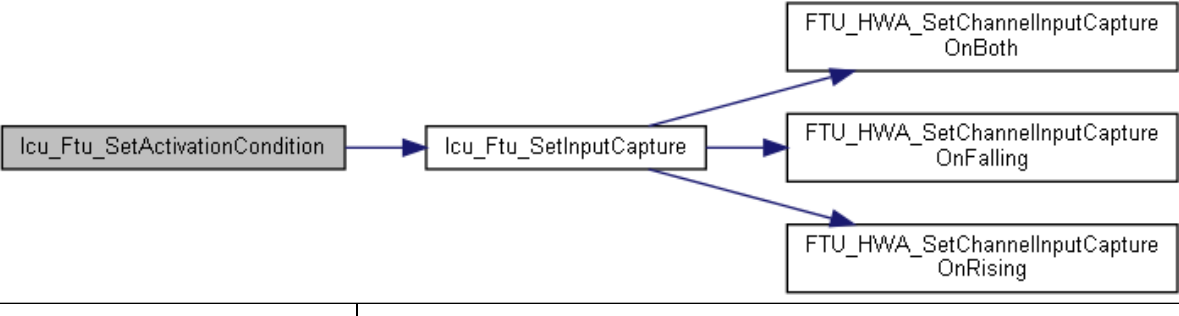
Function	void Icu_Ftu_SetSleepMode (const Icu_FtuInstanceType eFtuInstance, const uint8 u8Channel)	
Description	Set a FTU channel to sleep mode.	

Diagram	 <pre> graph LR A[Icu_Ftu_SetSleepMode] --> B[FTU_HWA_ClearChannelEdge Config] </pre>	
Parameters	Parameter	Description
	eFtuInstance	FTU instance ID.
	u8Channel	FTU channel index.
Returns	N/A	
Referenced By	Icu_HW_SetSleepMode()	

2.8.2.5 `void Icu_Ftu_SetNormalMode ( const Icu_FtuInstanceType eFtuInstance, const uint8 u8Channel, const Icu_FtuInputCaptureType eActivationEdge )`

Function	void Icu_Ftu_SetNormalMode (const Icu_FtuInstanceType eFtuInstance, const uint8 u8Channel, const Icu_FtuInputCaptureType eActivationEdge)	
Description	Set a FTU channel to normal mode.	
Diagram	 <pre> graph LR A[Icu_Ftu_SetNormalMode] --> B[Icu_Ftu_SetInputCapture] B --> C[FTU_HWA_SetChannelInputCapture OnBoth] B --> D[FTU_HWA_SetChannelInputCapture OnFalling] B --> E[FTU_HWA_SetChannelInputCapture OnRising] </pre>	
Parameters	Parameter	Description
	eFtuInstance	FTU instance ID.
	u8Channel	FTU channel index.
	eActivationEdge	Activation edge of the channel.
Returns	N/A	
Referenced By	Icu_HW_SetNormalMode()	

2.8.2.6 `void Icu_Ftu_SetActivationCondition ( const Icu_FtuInstanceType eFtuInstance, const uint8 u8Channel, const Icu_FtuInputCaptureType eActivationEdge )`

Function	void Icu_Ftu_SetActivationCondition (const Icu_FtuInstanceType eFtuInstance, const uint8 u8Channel, const Icu_FtuInputCaptureType eActivationEdge)	
Description	Set activation for a FTU channel.	
Diagram	 <pre> graph LR A[Icu_Ftu_SetActivationCondition] --> B[Icu_Ftu_SetInputCapture] B --> C[FTU_HWA_SetChannelInputCapture OnBoth] B --> D[FTU_HWA_SetChannelInputCapture OnFalling] B --> E[FTU_HWA_SetChannelInputCapture OnRising] </pre>	
Parameters	Parameter	Description
	eFtuInstance	FTU instance ID
	u8Channel	FTU channel index.

	eActivationEdge	Activation edge of the channel.
Returns	N/A	
Referenced By	Icu_HW_SetActivationCondition()	

2.8.2.7 boolean Icu_Ftu_GetInputState (const Icu_FtuInstanceType eFtuInstance, const uint8 u8Channel)

Function	boolean Icu_Ftu_GetInputState (const Icu_FtuInstanceType eFtuInstance, const uint8 u8Channel)	
Description	This function returns the state of the FTU channel.	
Diagram	<pre> graph LR A[Icu_Ftu_GetInputState] --> B[FTU_HWA_ClearChannelInterruptFlag] A --> C[FTU_HWA_ReadChannelInterrupt Enable] A --> D[FTU_HWA_ReadChannelInterruptFlag] </pre>	
Parameters	Parameter	Description
	eFtuInstance	FTU instance ID.
	u8Channel	FTU channel index.
Returns	State of the channel.	
Referenced By	Icu_HW_GetInputState()	

2.8.2.8 void Icu_Ftu_StartChannel (const Icu_FtuInstanceType eFtuInstance, const uint8 u8Channel, const Icu_FtuInputCaptureType eActivationEdge)

Function	void Icu_Ftu_StartChannel (const Icu_FtuInstanceType eFtuInstance, const uint8 u8Channel, const Icu_FtuInputCaptureType eActivationEdge)	
Description	Start a FTU channel.	
Diagram	<pre> graph LR A[Icu_Ftu_StartChannel] --> B[FTU_HWA_ClearChannelInterruptFlag] A --> C[FTU_HWA_EnableChannelInterrupt] A --> D[Icu_Ftu_SetInputCapture] D --> E[FTU_HWA_SetChannelInputCapture OnBoth] D --> F[FTU_HWA_SetChannelInputCapture OnFalling] D --> G[FTU_HWA_SetChannelInputCapture OnRising] </pre>	
Parameters	Parameter	Description
	eFtuInstance	FTU instance ID.
	u8Channel	FTU channel index.
	eActivationEdge	Activation edge of the channel.
Returns	N/A	
Referenced By	Icu_HW_EnableEdgeCount(), Icu_HW_EnableEdgeDetection(), Icu_HW_StartSignalMeasurement(), Icu_HW_StartTimestamp()	

2.8.2.9 void Icu_Ftu_StopChannel (const Icu_FtuInstanceType eFtuInstance, const uint8 u8Channel)

Function	void Icu_Ftu_StopChannel (const Icu_FtuInstanceType eFtuInstance, const uint8 u8Channel)
-----------------	--

Description	Stop a FTU channel.	
Diagram	<pre> graph LR A[Icu_Ftu_StopChannel] --> B[FTU_HWA_ClearChannelEdge Config] A --> C[FTU_HWA_DisableChannelInterrupt] </pre>	
Parameters	Parameter	Description
	eFtuInstance	FTU instance ID
	u8Channel	FTU channel index.
Returns	N/A	
Referenced By	Icu_HW_ChannelDeInit(), Icu_HW_DisableEdgeCount(), Icu_HW_DisableEdgeDetection(), Icu_HW_StopSignalMeasurement(), Icu_HW_StopTimeStamp()	

2.8.2.10 Icu_ValueType Icu_Ftu_GetInstanceMaxCount(const Icu_FtuInstanceType eFtuInstance)

Function	Icu_ValueType Icu_Ftu_GetInstanceMaxCount(const Icu_FtuInstanceType eFtuInstance)	
Description	Get the Maximum count value of the Ftu instance.	
Diagram	<pre> graph LR A[Icu_Ftu_GetInstanceMaxCount] --> B[FTU_HWA_ReadMaxCount] </pre>	
Parameters	Parameter	Description
	eFtuInstance	FTU instance ID
Returns	Maximum count value of the Ftu instance.	
Referenced By	Icu_HW_GetOverflowValue()	

2.8.2.11 void Icu_Ftu_EnableOverflow(const Icu_FtuInstanceType eFtuInstance, const uint8 u8Channel)

Function	void Icu_Ftu_EnableOverflow(const Icu_FtuInstanceType eFtuInstance, const uint8 u8Channel)	
Description	Enable the overflow interrupt of the channel	
Diagram	<pre> graph LR A[Icu_Ftu_EnableOverflow] --> B[FtuCommon_EnableOverflow] B --> C[FTU_HWA_EnableModuleInterrupt] </pre>	
Parameters	Parameter	Description
	eFtuInstance	FTU instance.
	u8Channel	FTU channel index
Returns	N/A	
Referenced By	Icu_HW_StartSignalMeasurement(), Icu_HW_StartTimeStamp().	

2.8.2.12 void Icu_Ftu_DisableOverflow(const Icu_FtuInstanceType eFtuInstance, const uint8 u8Channel)

Function	void Icu_Ftu_DisableOverflow(const Icu_FtuInstanceType eFtuInstance, const uint8 u8Channel)	
Description	Disable the overflow interrupt of the channel	
Diagram	<pre> graph LR A[Icu_Ftu_DisableOverflow] --> B[FtuCommon_DisableOverflow] B --> C[FTU_HWA_DisableOverflowInterrupt] </pre>	
Parameters	Parameter	Description
	eFtuInstance	FTU instance.
	u8Channel	FTU channel index
Returns	N/A	
Referenced By	Icu_HW_StopSignalMeasurement(), Icu_HW_StopTimeStamp().	

2.8.2.13 void Icu_Ftu_ConfigGTB(const uint32 u32Group, boolean benable)

Function	void Icu_Ftu_ConfigGTB(const uint32 u32Group, boolean benable)	
Description	Config the FTU GTB Function	
Diagram	N/A	
Parameters	Parameter	Description
	u32Group	FTU GTB Group
	benable	enable/disable Ftu Gtb function
Returns	N/A	
Referenced By		

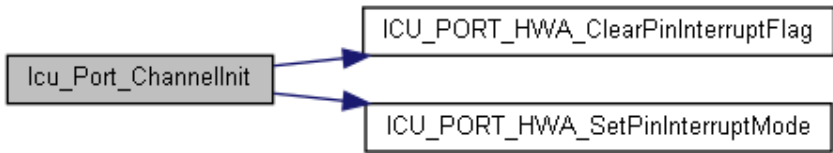
2.8.3 Functions in Icu_Port.h**2.8.3.1 void Icu_Port_Init(const Icu_PortInstanceType ePortInstance)**

Function	void Icu_Port_Init(const Icu_PortInstanceType ePortInstance)	
Description	Initialize a port module of ICU.	
Diagram	N/A	
Parameters	Parameter	Description
	ePortInstance	Port instance.
Returns	N/A	
Referenced By	Icu_HW_ModulesInit()	

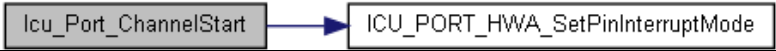
2.8.3.2 void Icu_Port_DeInit(const Icu_PortInstanceType ePortInstance)

Function	void Icu_Port_DeInit(const Icu_PortInstanceType ePortInstance)	
Description	Deinitialize a port module of ICU.	
Diagram	N/A	
Parameters	Parameter	Description
	ePortInstance	Port instance.
Returns	N/A	
Referenced By	Icu_HW_ModulesDeInit()	

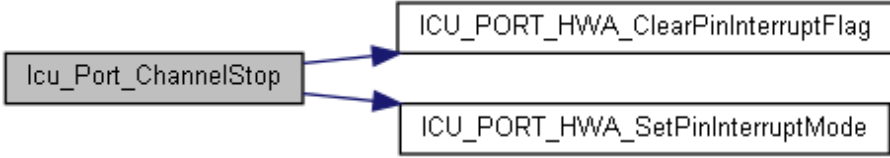
2.8.3.3 void Icu_Port_Channellnit (const Icu_PortInstanceType ePortInstance, const uint8 u8Channel)

Function	void Icu_Port_Channellnit (const Icu_PortInstanceType ePortInstance, const uint8 u8Channel)	
Description	Initialize a port channel of ICU.	
Diagram	 <pre> graph LR A[Icu_Port_Channellnit] --> B[ICU_PORT_HWA_ClearPinInterruptFlag] A --> C[ICU_PORT_HWA_SetPinInterruptMode] </pre>	
Parameters	Parameter	Description
	ePortInstance	Port instance.
	u8Channel	Pin number.
Returns	N/A	
Referenced By	Icu_HW_Channellnit()	

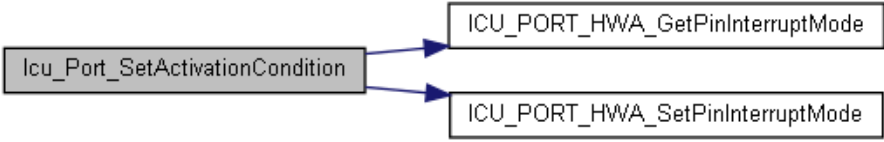
2.8.3.4 void Icu_Port_ChannelStart (const Icu_PortInstanceType ePortInstance, const uint8 u8Channel, const Icu_PortIntConfigType eIntConfig)

Function	void Icu_Port_ChannelStart (const Icu_PortInstanceType ePortInstance, const uint8 u8Channel, const Icu_PortIntConfigType eIntConfig)	
Description	Start a port channel of ICU.	
Diagram	 <pre> graph LR A[Icu_Port_ChannelStart] --> B[ICU_PORT_HWA_SetPinInterruptMode] </pre>	
Parameters	Parameter	Description
	ePortInstance	Port instance.
	u8Channel	Pin number.
	eIntConfig	Pin IRQC configuration value.
Returns	N/A	
Referenced By	Icu_HW_EnableEdgeCount(), Icu_HW_EnableEdgeDetection(), Icu_HW_SetNormalMode()	

2.8.3.5 void Icu_Port_ChannelStop (const Icu_PortInstanceType ePortInstance, const uint8 u8Channel)

Function	void Icu_Port_ChannelStop (const Icu_PortInstanceType ePortInstance, const uint8 u8Channel)	
Description	Stop a port channel of ICU.	
Diagram	 <pre> graph LR A[Icu_Port_ChannelStop] --> B[ICU_PORT_HWA_ClearPinInterruptFlag] A --> C[ICU_PORT_HWA_SetPinInterruptMode] </pre>	
Parameters	Parameter	Description
	ePortInstance	Port instance.
	u8Channel	Pin number.
Returns	N/A	
Referenced By	Icu_HW_ChannelDeInit(), Icu_HW_DisableEdgeCount(), Icu_HW_DisableEdgeDetection(), Icu_HW_SetSleepMode()	

2.8.3.6 void Icu_Port_SetActivationCondition(const Icu_PortInstanceType eFtuInstance, const uint8 u8Channel, const Icu_PortIntConfigType eActivationEdge)

Function	void Icu_Port_SetActivationCondition(const Icu_PortInstanceType eFtuInstance, const uint8 u8Channel, const Icu_PortIntConfigType eActivationEdge)	
Description	Set activation edge of a port channel.	
Diagram	 <pre> graph LR A[Icu_Port_SetActivationCondition] --> B[ICU_PORT_HWA_GetPinInterruptMode] A --> C[ICU_PORT_HWA_SetPinInterruptMode] </pre>	
Parameters	Parameter	Description
	ePortInstance	Port instance.
	u8Channel	Pin number.
	eActivationEdge	Pin IRQC configuration value.
Returns	N/A	
Referenced By	Icu_HW_SetActivationCondition()	

2.8.3.7 `boolean Icu_Port_GetInputState(const Icu_PortInstanceType ePortInstance, const uint8 u8Channel)`

Function	<code>boolean Icu_Port_GetInputState(const Icu_PortInstanceType ePortInstance, const uint8 u8Channel)</code>	
Description	This function returns the state of the port channel.	
Diagram	<pre> graph LR A[Icu_Port_GetInputState] --> B[ICU_PORT_HWA_ClearPinInterruptFlag] A --> C[ICU_PORT_HWA_GetPinInterruptFlag] A --> D[ICU_PORT_HWA_GetPinInterruptMode] </pre>	
Parameters	Parameter	Description
	ePortInstance	Port instance.
	u8Channel	Pin number.
Returns	State of the channel.	
Referenced By	Icu_HW_GetInputState()	

2.8.3.8 `boolean Icu_Port_GetInputLevel (const Icu_PortInstanceType ePortInstance,const uint8 u8Channel)`

Function	<code>boolean Icu_Port_GetInputLevel (const Icu_PortInstanceType ePortInstance,const uint8 u8Channel)</code>	
Description	This function returns the input level of the port channel	
Diagram	<pre> graph LR A[Icu_Port_GetInputLevel] --> B[ICU_GPIO_HWA_ReadPortData Input] </pre>	
Parameters	Parameter	Description
	ePortInstance	Port instance.
	u8Channel	Pin number.
Returns	input level of the channel	
Referenced By	Icu_HW_GetInputLevel().	

2.8.4 Functions in Icu_Aontimer.h**2.8.4.1 `void Icu_Aontimer_Init(const Icu_AontimerModuleConfigType *pConfig)`**

Function	<code>void Icu_Aontimer_Init(const Icu_AontimerModuleConfigType *pConfig)</code>
Description	Initialize an AONTIMER Module.

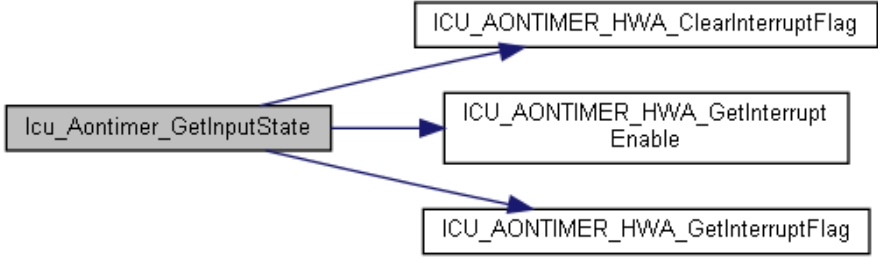
Diagram	<pre> graph LR Icu_Aontimer_Init --> ICU_AONTIMER_HWA_ConfigModule Icu_Aontimer_Init --> ICU_AONTIMER_HWA_ConfigModule_Prescale[ICU_AONTIMER_HWA_ConfigModule Prescale] Icu_Aontimer_Init --> ICU_AONTIMER_HWA_Disable_BypassMode[ICU_AONTIMER_HWA_Disable BypassMode] Icu_Aontimer_Init --> ICU_AONTIMER_HWA_EnableBypassMode Icu_Aontimer_Init --> ICU_AONTIMER_HWA_EnablePulseMode Icu_Aontimer_Init --> ICU_AONTIMER_HWA_SelectClk_SrcOnPulseMode[ICU_AONTIMER_HWA_SelectClk SrcOnPulseMode] Icu_Aontimer_Init --> ICU_AONTIMER_HWA_SelectModule_ClkSrc[ICU_AONTIMER_HWA_SelectModule ClkSrc] Icu_Aontimer_Init --> ICU_AONTIMER_HWA_SetPrescale </pre>				
Parameters	<table border="1"> <thead> <tr> <th>Parameter</th><th>Description</th></tr> </thead> <tbody> <tr> <td>pConfig</td><td>Pointer to AONTIMER module configuration structure.</td></tr> </tbody> </table>	Parameter	Description	pConfig	Pointer to AONTIMER module configuration structure.
Parameter	Description				
pConfig	Pointer to AONTIMER module configuration structure.				
Returns	N/A				
Referenced By	Icu_HW_ModulesInit()				

2.8.4.2 `void Icu_Aontimer_SetActivationCondition(const Icu_AontimerInstanceType eInstance, const Icu_AontimerPulsePolarityType ePolarity)`

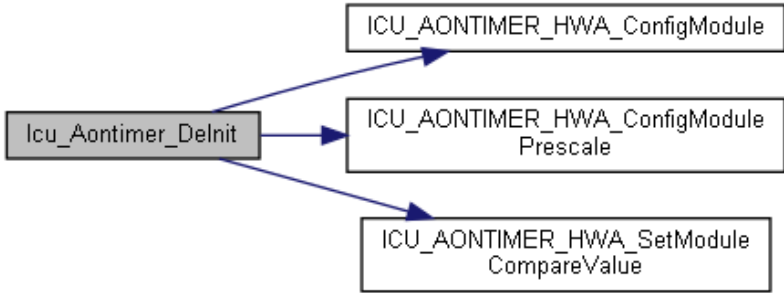
Function	void Icu_Aontimer_SetActivationCondition(const Icu_AontimerInstanceType eInstance, const Icu_AontimerPulsePolarityType ePolarity)						
Description	Set activation for AONTIMER.						
Diagram	<pre> graph LR Icu_Aontimer_SetActivationCondition --> ICU_AONTIMER_HWA_ConfigModule_Polarity[ICU_AONTIMER_HWA_ConfigModule Polarity] </pre>						
Parameters	<table border="1"> <thead> <tr> <th>Parameter</th><th>Description</th></tr> </thead> <tbody> <tr> <td>eInstance</td><td>AONTIMER instance ID.</td></tr> <tr> <td>ePolarity</td><td>The polarity of pulse mode.</td></tr> </tbody> </table>	Parameter	Description	eInstance	AONTIMER instance ID.	ePolarity	The polarity of pulse mode.
Parameter	Description						
eInstance	AONTIMER instance ID.						
ePolarity	The polarity of pulse mode.						
Returns	N/A						
Referenced By	Icu_HW_SetActivationCondition()						

2.8.4.3 `boolean Icu_Aontimer_GetInputState(const Icu_AontimerInstanceType eInstance)`

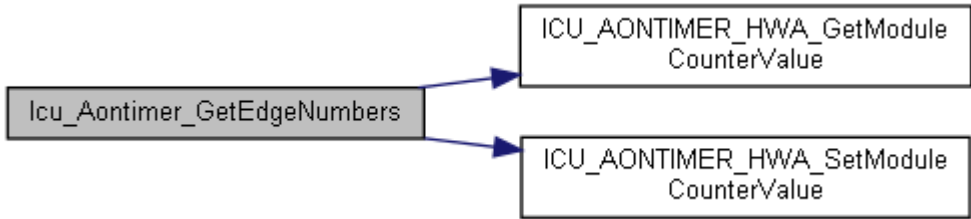
Function	boolean Icu_Aontimer_GetInputState(const Icu_AontimerInstanceType eInstance)
Description	This function returns the state of AONTIMER.

Diagram	 <pre> graph LR A[Icu_Aontimer_GetInputState] --> B[ICU_AONTIMER_HWA_ClearInterruptFlag] A --> C[ICU_AONTIMER_HWA_GetInterruptEnable] A --> D[ICU_AONTIMER_HWA_GetInterruptFlag] </pre>	
Parameters	Parameter	Description
	eInstance	AONTIMER instance ID.
Returns	state of the Aontimer	
Referenced By	Icu_HW_GetInputState()	

2.8.4.4 void Icu_Aontimer_DeInit(const Icu_AontimerInstanceType eInstance)

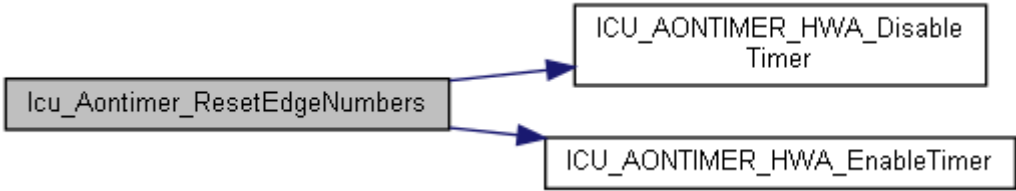
Function	void Icu_Aontimer_DeInit(const Icu_AontimerInstanceType eInstance)	
Description	De-initialize AONTIMER module.	
Diagram	 <pre> graph LR A[Icu_Aontimer_DeInit] --> B[ICU_AONTIMER_HWA_ConfigModule] A --> C[ICU_AONTIMER_HWA_ConfigModulePrescale] A --> D[ICU_AONTIMER_HWA_SetModuleCompareValue] </pre>	
Parameters	Parameter	Description
	eInstance	AONTIMER instance ID.
Returns	N/A	
Referenced By	Icu_HW_ModulesDeInit()	

2.8.4.5 Icu_EdgeNumberType Icu_Aontimer_GetEdgeNumbers(const Icu_AontimerInstanceType eInstance)

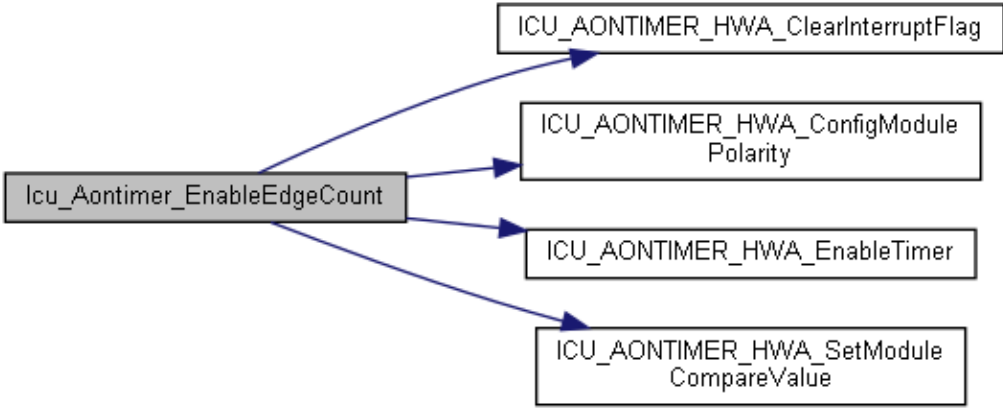
Function	Icu_EdgeNumberType Icu_Aontimer_GetEdgeNumbers(const Icu_AontimerInstanceType eInstance)	
Description	Get AONTIMER edge numbers.	
Diagram	 <pre> graph LR A[Icu_Aontimer_GetEdgeNumbers] --> B[ICU_AONTIMER_HWA_GetModuleCounterValue] A --> C[ICU_AONTIMER_HWA_SetModuleCounterValue] </pre>	
Parameters	Parameter	Description
	eInstance	AONTIMER instance ID.
Returns	AONTIMER edge numbers.	
Referenced By	Icu_HW_GetEdgeNumbers()	

2.8.4.6 void Icu_Aontimer_ResetEdgeNumbers(const Icu_AontimerInstanceType eInstance)

Function	void Icu_Aontimer_ResetEdgeNumbers(const Icu_AontimerInstanceType eInstance)	
Description	Reset AONTIMER edge numbers.	

Diagram	 <pre> graph LR A[Icu_Aontimer_ResetEdgeNumbers] --> B[ICU_AONTIMER_HWA_DisableTimer] A --> C[ICU_AONTIMER_HWA_EnableTimer] </pre>	
Parameters	Parameter	Description
	eInstance	AONTIMER instance ID.
Returns	N/A	
Referenced By	Icu_HW_ResetEdgeNumbers()	

2.8.4.7 `void Icu_Aontimer_EnableEdgeCount(const Icu_AontimerInstanceType eInstance, const Icu_AontimerPulsePolarityType ePolarity)`

Function	void Icu_Aontimer_EnableEdgeCount(const Icu_AontimerInstanceType eInstance, const Icu_AontimerPulsePolarityType ePolarity)	
Description	Enable AONTIMER edge count.	
Diagram	 <pre> graph LR A[Icu_Aontimer_EnableEdgeCount] --> B[ICU_AONTIMER_HWA_ClearInterruptFlag] A --> C[ICU_AONTIMER_HWA_ConfigModulePolarity] A --> D[ICU_AONTIMER_HWA_EnableTimer] A --> E[ICU_AONTIMER_HWA_SetModuleCompareValue] </pre>	
Parameters	Parameter	Description
	eInstance	AONTIMER instance ID.
	ePolarity	The polarity of pulse mode.
Returns	N/A	
Referenced By	Icu_HW_EnableEdgeCount()	

2.8.4.8 `void Icu_Aontimer_DisableEdgeCount(const Icu_AontimerInstanceType eInstance)`

Function	void Icu_Aontimer_DisableEdgeCount(const Icu_AontimerInstanceType eInstance)	
Description	Disable AONTIMER edge count.	
Diagram	 <pre> graph LR A[Icu_Aontimer_DisableEdgeCount] --> B[ICU_AONTIMER_HWA_DisableTimer] </pre>	
Parameters	Parameter	Description
	eInstance	AONTIMER instance ID.
Returns	N/A	
Referenced By	Icu_HW_DisableEdgeCount()	

2.8.4.9 `void` `Icu_Aontimer_EnableEdgeDetection(const Icu_AontimerInstanceType eInstance, const Icu_AontimerPulsePolarityType ePolarity)`

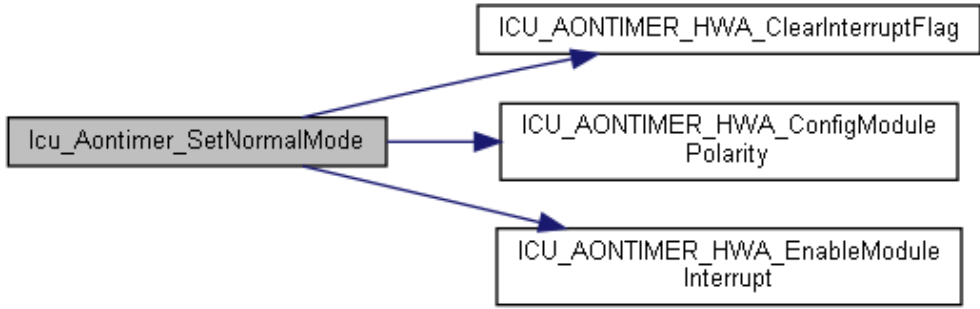
Function	<code>void Icu_Aontimer_EnableEdgeDetection(const Icu_AontimerInstanceType eInstance, const Icu_AontimerPulsePolarityType ePolarity)</code>	
Description	Enable AONTIMER edge detection.	
Diagram	<pre> graph LR A[Icu_Aontimer_EnableEdgeDetection] --> B[ICU_AONTIMER_HWA_ClearInterruptFlag] A --> C[ICU_AONTIMER_HWA_ConfigModulePolarity] A --> D[ICU_AONTIMER_HWA_EnableModuleInterrupt] A --> E[ICU_AONTIMER_HWA_EnableTimer] A --> F[ICU_AONTIMER_HWA_SetModuleCompareValue] </pre>	
Parameters	Parameter	Description
	eInstance	AONTIMER instance ID.
	ePolarity	The polarity of pulse mode.
Returns	N/A	
Referenced By	Icu_HW_EnableEdgeDetection()	

2.8.4.10 `void` `Icu_Aontimer_DisableEdgeDetection(const Icu_AontimerInstanceType eInstance)`

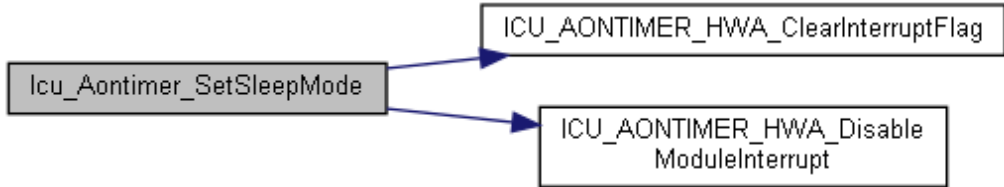
Function	<code>void Icu_Aontimer_DisableEdgeDetection(const Icu_AontimerInstanceType eInstance)</code>	
Description	Disable AONTIMER edge detection	
Diagram	<pre> graph LR A[Icu_Aontimer_DisableEdgeDetection] --> B[ICU_AONTIMER_HWA_DisableModuleInterrupt] A --> C[ICU_AONTIMER_HWA_DisableTimer] </pre>	
Parameters	Parameter	Description
	eInstance	AONTIMER instance ID.
Returns	N/A	
Referenced By	Icu_HW_DisableEdgeDetection()	

2.8.4.11 `void` `Icu_Aontimer_SetNormalMode(const Icu_AontimerInstanceType eInstance, const Icu_AontimerPulsePolarityType ePolarity)`

Function	<code>void Icu_Aontimer_SetNormalMode(const Icu_AontimerInstanceType eInstance, const Icu_AontimerPulsePolarityType ePolarity)</code>	
Description	Set AONTIMER to normal mode.	

Diagram	 <pre>graph LR; A[Icu_Aontimer_SetNormalMode] --> B[ICU_AONTIMER_HWA_ClearInterruptFlag]; A --> C[ICU_AONTIMER_HWA_ConfigModulePolarity]; A --> D[ICU_AONTIMER_HWA_EnableModuleInterrupt];</pre>							
Parameters	<table><tr><th>Parameter</th><th>Description</th></tr><tr><td>eInstance</td><td>AONTIMER instance ID.</td></tr><tr><td>ePolarity</td><td>The polarity of pulse mode.</td></tr></table>	Parameter	Description	eInstance	AONTIMER instance ID.	ePolarity	The polarity of pulse mode.	
Parameter	Description							
eInstance	AONTIMER instance ID.							
ePolarity	The polarity of pulse mode.							
Returns	N/A							
Referenced By	Icu_HW_SetNormalMode()							

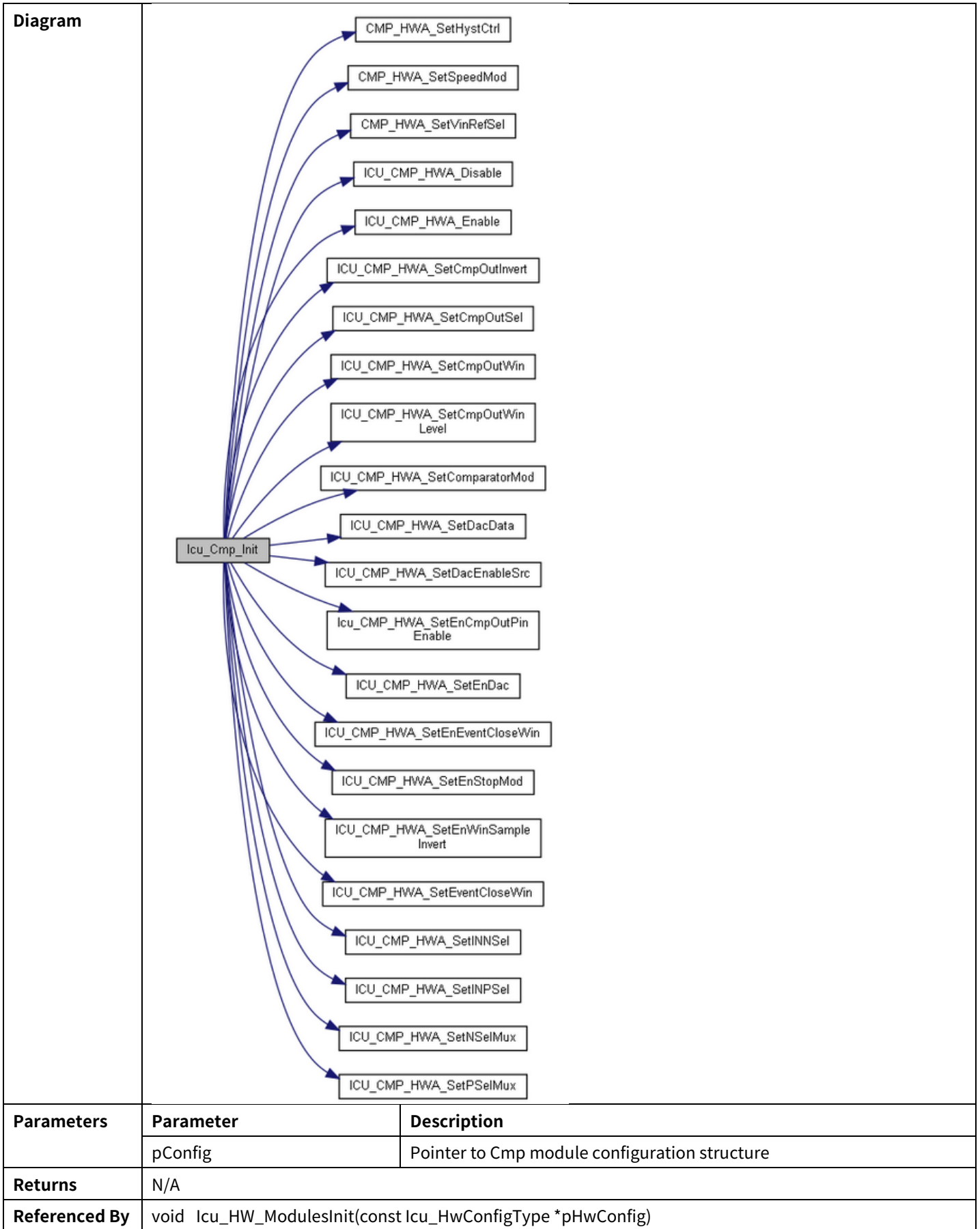
2.8.4.12 void Icu_Aontimer_SetSleepMode(const Icu_AontimerInstanceType eInstance)

Function	void Icu_Aontimer_SetSleepMode(const Icu_AontimerInstanceType eInstance)	
Description	Set AONTIMER to sleep mode.	
Diagram	 <pre>graph LR; A[Icu_Aontimer_SetSleepMode] --> B[ICU_AONTIMER_HWA_ClearInterruptFlag]; A --> C[ICU_AONTIMER_HWA_DisableModuleInterrupt];</pre>	
Parameters	Parameter	Description
	eInstance	AONTIMER instance ID.
Returns	N/A	
Referenced By	Icu_HW_SetSleepMode()	

2.8.5 Functions in Icu_Cmp.h

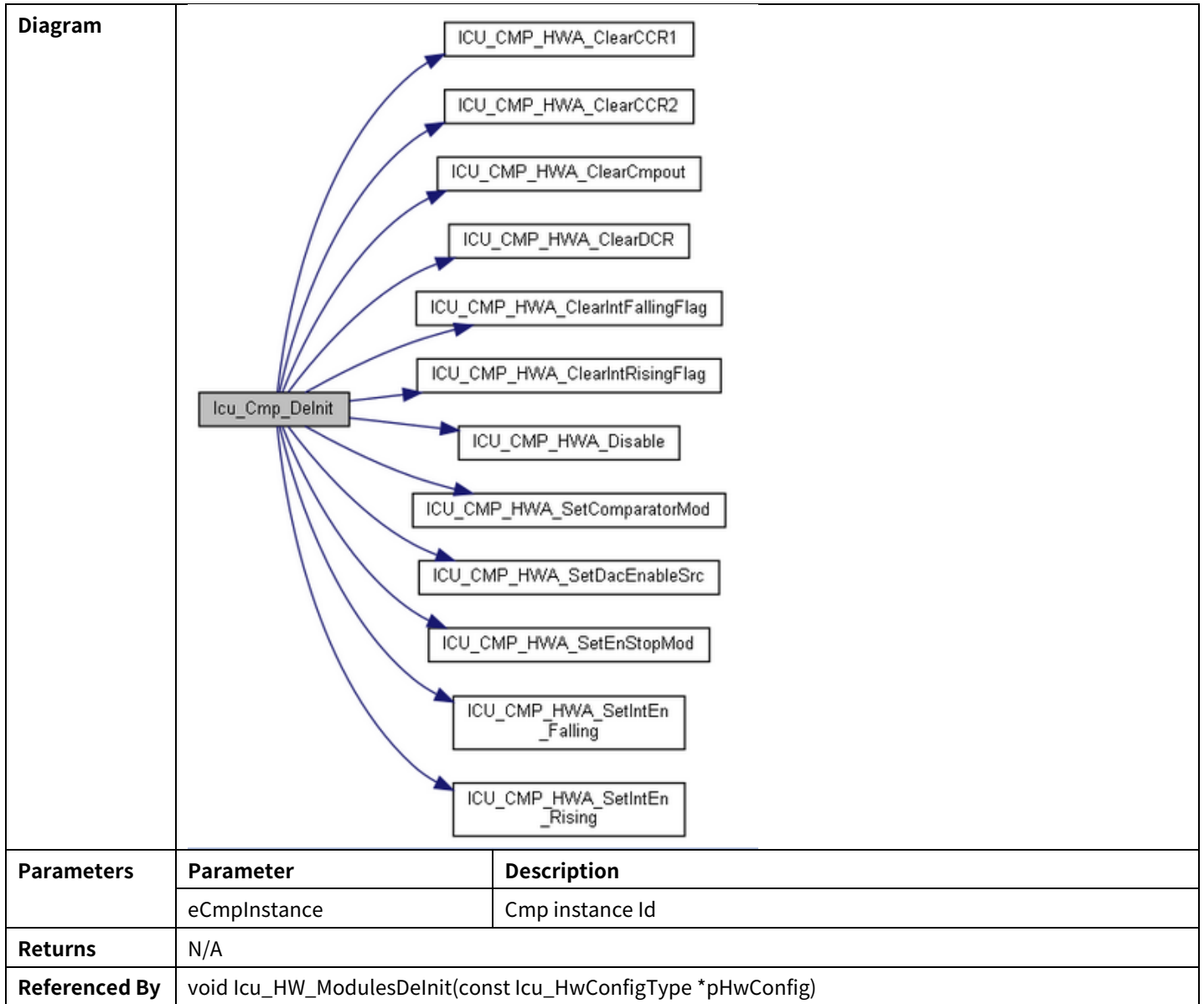
2.8.5.1 void Icu_Cmp_Init(const Icu_CmpmoduleConfigType *pConfig)

Function	void Icu_Cmp_Init(const Icu_CmpmoduleConfigType *pConfig)
Description	Initialize a Cmp module of ICU.



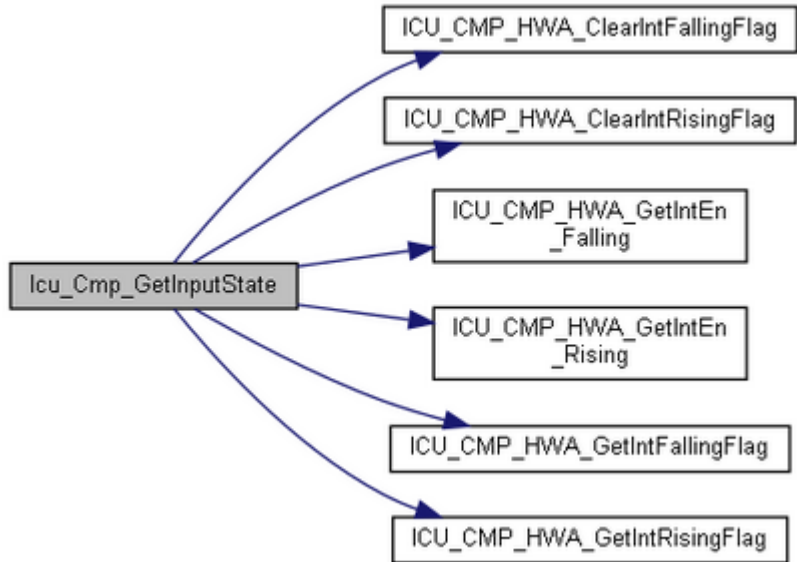
2.8.5.2 void Icu_Cmp_DeInit(const Icu_CmpInstanceType eCmpInstance)

Function	void Icu_Cmp_DeInit(const Icu_CmpInstanceType eCmpInstance)
Description	De-initialize a Cmp Module

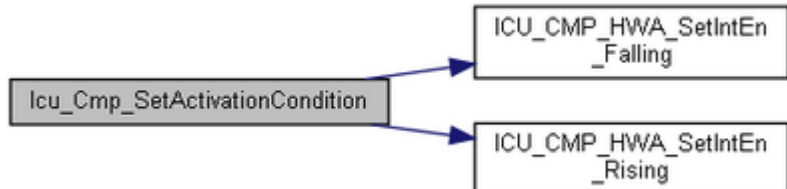


2.8.5.3 `boolean Icu_Cmp_GetInputState(const Icu_CmplInstanceType eCmplInstance, const ICU_CmpOutPolarityType ePolarity)`

Function	Boolean Icu_Cmp_GetInputState(const Icu_CmplInstanceType eCmplInstance, const ICU_CmpOutPolarityType ePolarity)
Description	Get the state of cmp

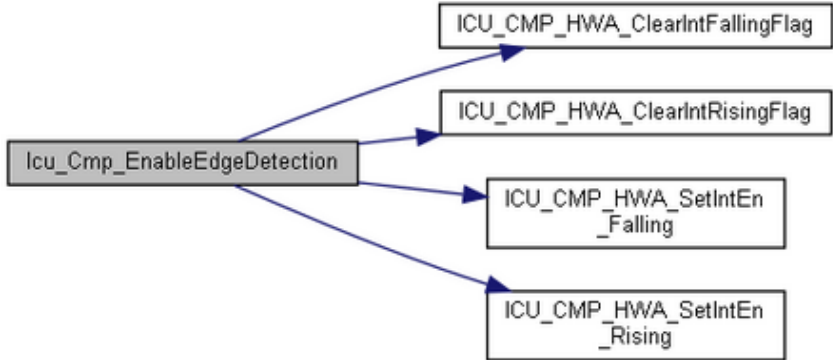
Diagram	 <pre>graph LR; A[Icu_Cmp_GetInputState] --> B[ICU_CMP_HWA_ClearIntFallingFlag]; A --> C[ICU_CMP_HWA_ClearIntRisingFlag]; A --> D[ICU_CMP_HWA_GetIntEn_Falling]; A --> E[ICU_CMP_HWA_GetIntEn_Rising]; A --> F[ICU_CMP_HWA_GetIntFallingFlag]; A --> G[ICU_CMP_HWA_GetIntRisingFlag];</pre>			
Parameters	Parameter	Description		
	eCmplInstance	Cmp instance Id		
	ePolarity	input state		
Returns	state of the cmp			
Referenced By	Icu_InputStateType	Icu_HW_GetInputState(Icu_HwType	eHwType,	uint32
	u32ChannelOfHw,Icu_ActivationType eActivationEdge)			

2.8.5.4 void Icu_Cmp_SetActivationCondition(const Icu_CmplInstanceType eCmplInstance,const ICU_CmpOutPolarityType ePolarity)

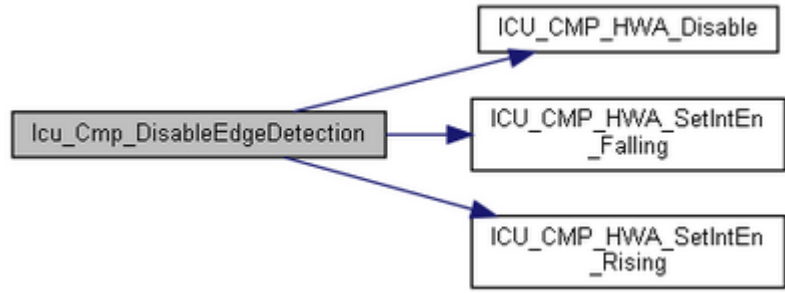
Function	void Icu_Cmp_SetActivationCondition(const Icu_CmpInstanceType eCmpInstance, const ICU_CmpOutPolarityType ePolarity)	
Description	Set activation for cmp	
Diagram	 <pre>graph LR; A[Icu_Cmp_SetActivationCondition] --> B[ICU_CMP_HWA_SetIntEn_Falling]; A --> C[ICU_CMP_HWA_SetIntEn_Rising];</pre>	
Parameters	Parameter	Description
	eCmpInstance	Cmp instance Id
	ePolarity	The polarity of pulse mode
Returns	N/A	
Referenced By	void Icu_HW_SetActivationCondition(Icu_HwType eHwType, uint32 u32ChannelOfHw, Icu_ActivationType eActivationEdge)	

2.8.5.5 void Icu_Cmp_EnableEdgeDetection(const Icu_CmplInstanceType eCmplInstance, const ICU_CmpOutPolarityType ePolarity)

Function	void Icu_Cmp_EnableEdgeDetection(const Icu_CmplInstanceType eCmplInstance, const ICU_CmpOutPolarityType ePolarity)		
Description	Enable cmp edge detection		

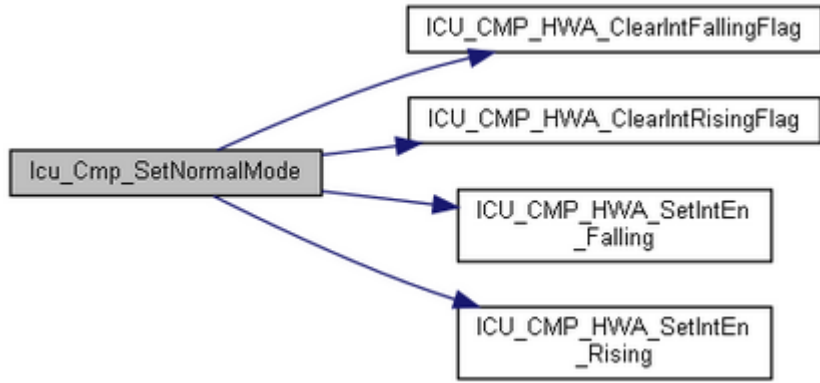
Diagram	 <pre> graph LR A[Icu_Cmp_EnableEdgeDetection] --> B[ICU_CMP_HWA_ClearIntFallingFlag] A --> C[ICU_CMP_HWA_ClearIntRisingFlag] A --> D[ICU_CMP_HWA_SetIntEn_Falling] A --> E[ICU_CMP_HWA_SetIntEn_Rising] </pre>	
Parameters	Parameter	Description
	eCmplInstance	Cmp instance Id
	ePolarity	The polarity of pulse mode
Returns	N/A	
Referenced By	void Icu_EnableEdgeDetection(Icu_ChannelType Channel)	

2.8.5.6 void Icu_Cmp_DisableEdgeDetection(const Icu_CmplInstanceType eCmplInstance)

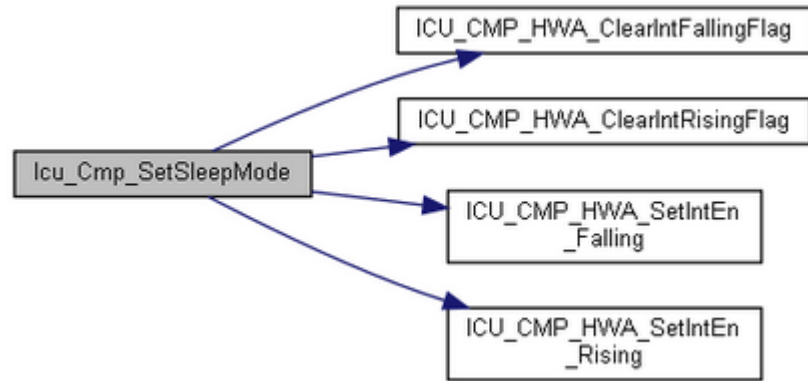
Function	void Icu_Cmp_DisableEdgeDetection(const Icu_CmplInstanceType eCmplInstance)	
Description	Disable cmp edge detection	
Diagram	 <pre> graph LR A[Icu_Cmp_DisableEdgeDetection] --> B[ICU_CMP_HWA_Disable] A --> C[ICU_CMP_HWA_SetIntEn_Falling] A --> D[ICU_CMP_HWA_SetIntEn_Rising] </pre>	
Parameters	Parameter	Description
	eCmplInstance	Cmp instance Id
	ePolarity	The polarity of pulse mode
Returns	N/A	
Referenced By	void Icu_DisableEdgeDetection(Icu_ChannelType Channel)	

2.8.5.7 void Icu_Cmp_SetNormalMode (const Icu_CmplInstanceType eCmplInstance, const ICU_CmpOutPolarityType ePolarity)

Function	void Icu_Cmp_SetNormalMode (const Icu_CmplInstanceType eCmplInstance, const ICU_CmpOutPolarityType ePolarity)	
Description	Set cmp to normal mode.	

Diagram	 <pre>graph LR; A[Icu_Cmp_SetNormalMode] --> B[ICU_CMP_HWA_ClearIntFallingFlag]; A --> C[ICU_CMP_HWA_ClearIntRisingFlag]; A --> D[ICU_CMP_HWA_SetIntEn_Falling]; A --> E[ICU_CMP_HWA_SetIntEn_Rising];</pre>	
Parameters	Parameter	Description
	eCmplInstance	Cmp instance Id
	ePolarity	The polarity of pulse mode
Returns	N/A	
Referenced By	void Icu_HW_SetNormalMode(Icu_HwType eHwType, uint32 u32ChannelOfHw, Icu_ActivationType eActivationEdge)	

2.8.5.8 void Icu_Cmp_SetSleepMode (const Icu_CmplInstanceType eCmplInstance)

Function	Icu_Cmp_SetSleepMode (const Icu_CmplInstanceType eCmplInstance)					
Description	Set cmp to sleep mode.					
Diagram	 <pre>graph LR; A[Icu_Cmp_SetSleepMode] --> B[ICU_CMP_HWA_ClearIntFallingFlag]; A --> C[ICU_CMP_HWA_ClearIntRisingFlag]; A --> D[ICU_CMP_HWA_SetIntEn_Falling]; A --> E[ICU_CMP_HWA_SetIntEn_Rising];</pre>					
Parameters	<table><tr><th>Parameter</th><th>Description</th></tr><tr><td>eCmplInstance</td><td>Cmp instance Id</td></tr></table>	Parameter	Description	eCmplInstance	Cmp instance Id	
Parameter	Description					
eCmplInstance	Cmp instance Id					
Returns	N/A					
Referenced By	void Icu_HW_SetNormalMode(Icu_HwType eHwType, uint32 u32ChannelOfHw, Icu_ActivationType eActivationEdge)					

2.8.5.9 boolean Icu_Cmp_GetInputLevel (const Icu_CmplInstanceType eCmplInstance)

Function	boolean Icu_Cmp_GetInputLevel (const Icu_CmplInstanceType eCmplInstance)		
Description	This function returns the input level of the port channel.		
Diagram	 <pre>graph LR; A[Icu_Cmp_GetInputLevel] --> B[ICU_CMP_HWA_GetCmpOut]</pre>		
Parameters	Parameter	Description	
	eCmplInstance	Cmp instance Id	
Returns	N/A		
Referenced By	boolean Icu_HW_GetInputLevel(Icu_HwType eHwType, uint32 u32ChannelOfHw, Icu_InputLevelType *pLevel)		

2.8.5.10 void Icu_Cmp_EnableEdgeCount (const Icu_CmplInstanceType eCmplInstance, const ICU_CmpOutPolarityType ePolarity)

Function	void Icu_Cmp_EnableEdgeCount (const Icu_CmplInstanceType eCmplInstance, const ICU_CmpOutPolarityType ePolarity)	
Description	Enable Cmp edge count	
Diagram	<pre> sequenceDiagram participant Func as Icu_Cmp_EnableEdgeCount participant HWA1 as ICU_CMP_HWA_ClearIntFallingFlag participant HWA2 as ICU_CMP_HWA_ClearIntRisingFlag participant HWA3 as ICU_CMP_HWA_SetIntEn_Falling participant HWA4 as ICU_CMP_HWA_SetIntEn_Rising Func->>HWA1 Func->>HWA2 Func->>HWA3 Func->>HWA4 </pre>	
Parameters	Parameter	Description
	eCmplInstance	Cmp instance Id
	ePolarity	The polarity of pulse mode
Returns	N/A	
Referenced By	void Icu_HW_EnableEdgeCount(Icu_HwType eHwType, uint32 u32ChannelOfHw, Icu_ActivationType eActivationEdge)	

2.8.5.11 void Icu_Cmp_DisableEdgeCount (const Icu_CmplInstanceType eCmplInstance)

Function	Icu_Cmp_DisableEdgeCount (const Icu_CmplInstanceType eCmplInstance)	
Description	Enable Cmp edge count	
Diagram	<pre> sequenceDiagram participant Func as Icu_Cmp_DisableEdgeCount participant HWA1 as ICU_CMP_HWA_ClearIntFallingFlag participant HWA2 as ICU_CMP_HWA_ClearIntRisingFlag participant HWA3 as ICU_CMP_HWA_SetIntEn_Falling participant HWA4 as ICU_CMP_HWA_SetIntEn_Rising Func->>HWA1 Func->>HWA2 Func->>HWA3 Func->>HWA4 </pre>	
Parameters	Parameter	Description
	eCmplInstance	Cmp instance Id
Returns	N/A	
Referenced By	void Icu_HW_DisableEdgeCount(Icu_HwType eHwType, uint32 u32ChannelOfHw)	

2.9 API Sequence Diagram

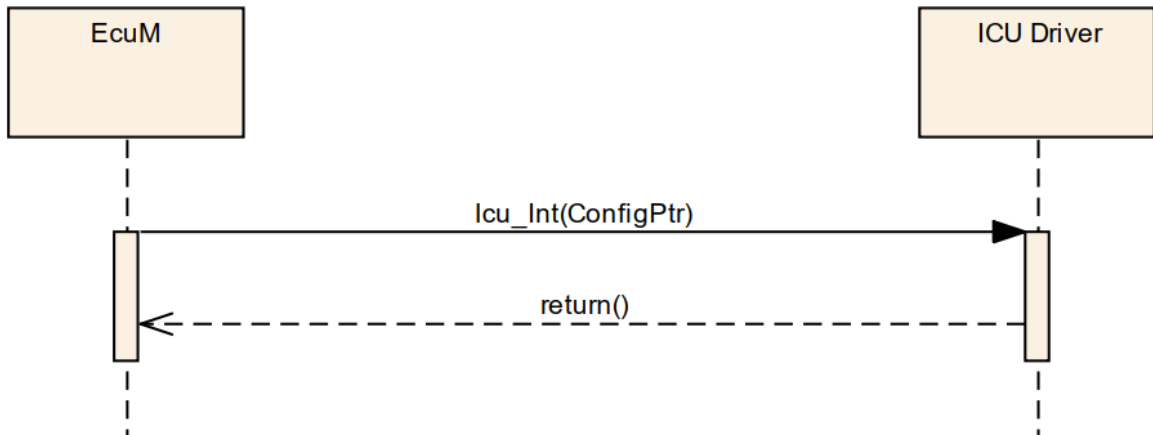
2.9.1 Initialization

ICU Driver Initialization

Module operates in ICU_MODE_NORMAL.

Used HW is configured according to referenced configuration structure.

- a) All notifications are disabled.
- b) All used interrupts are disabled.

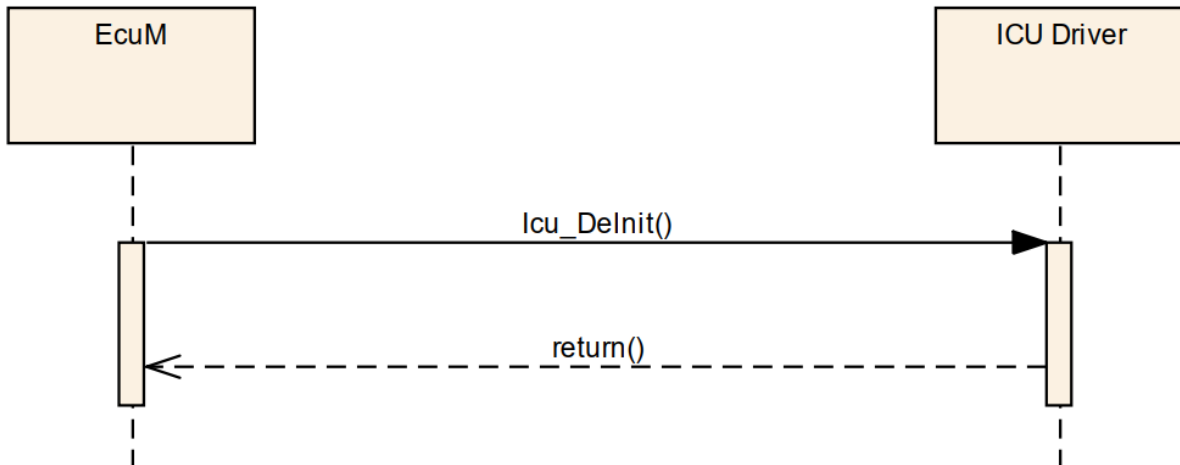


2.9.2 De-initialization

ICU Driver De-Initialization

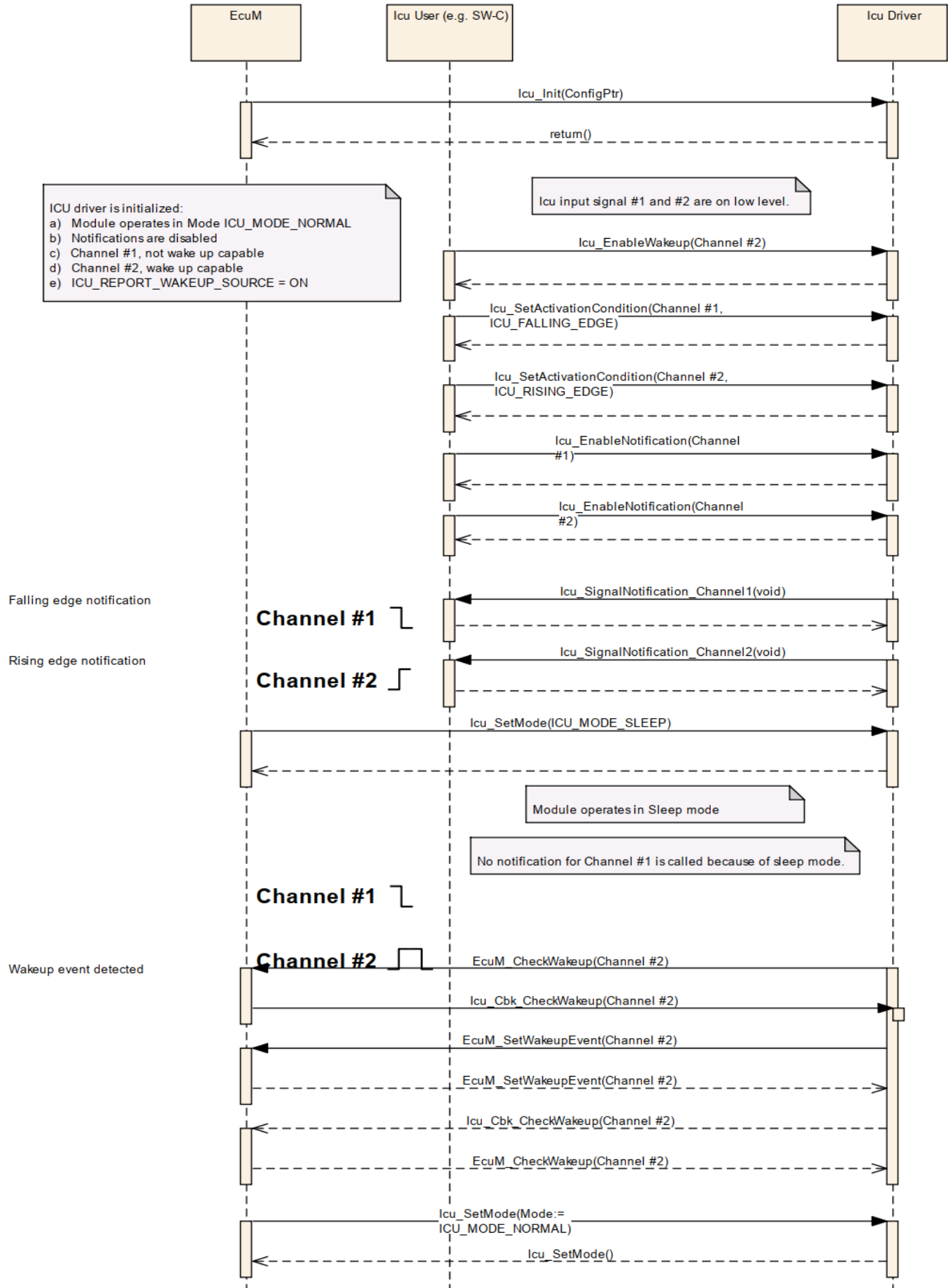
Used HW is configured according to referenced configuration structure.

- a) All notifications are disabled.
- b) All used interrupts are disabled.

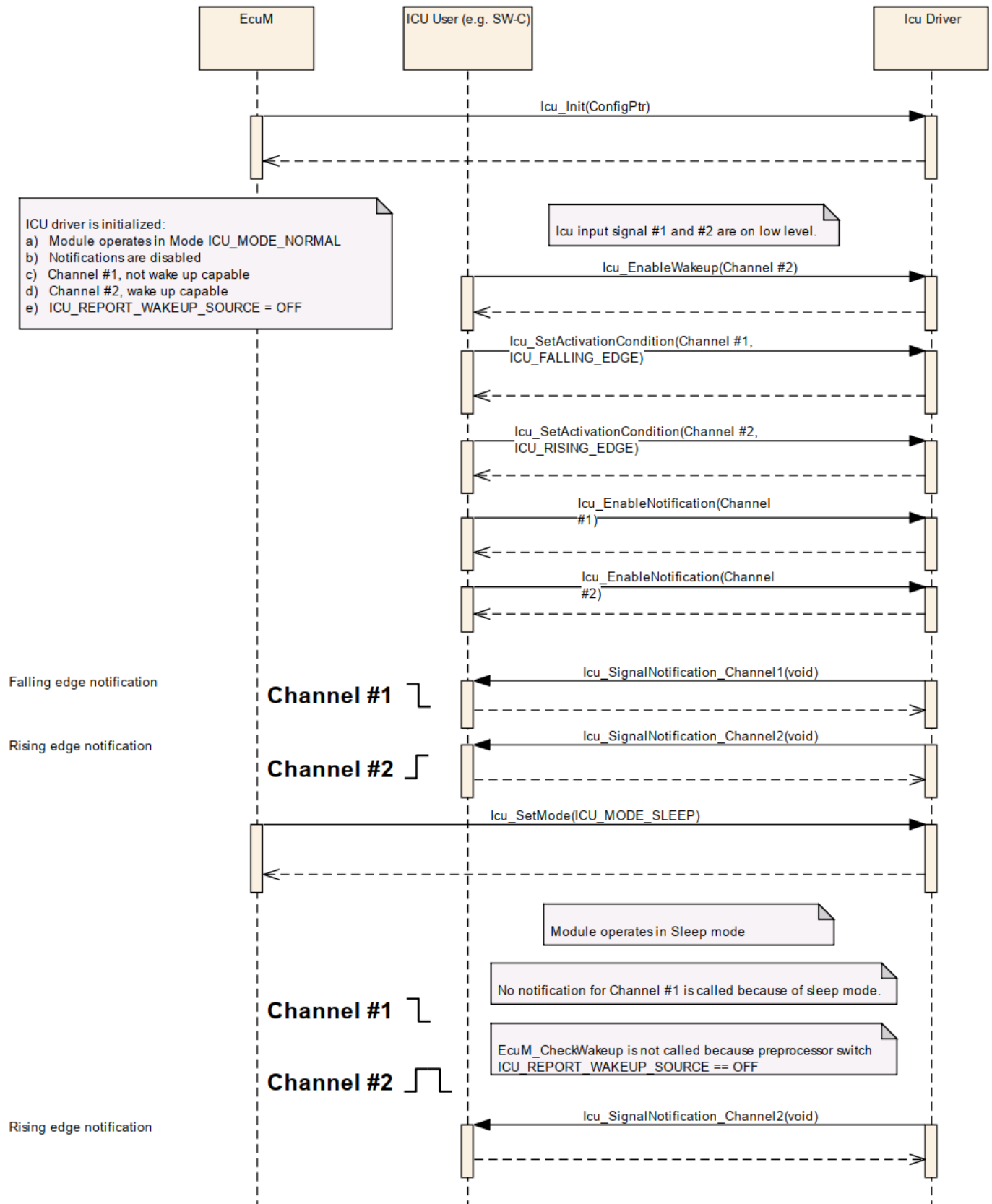


2.9.3 Setting Mode

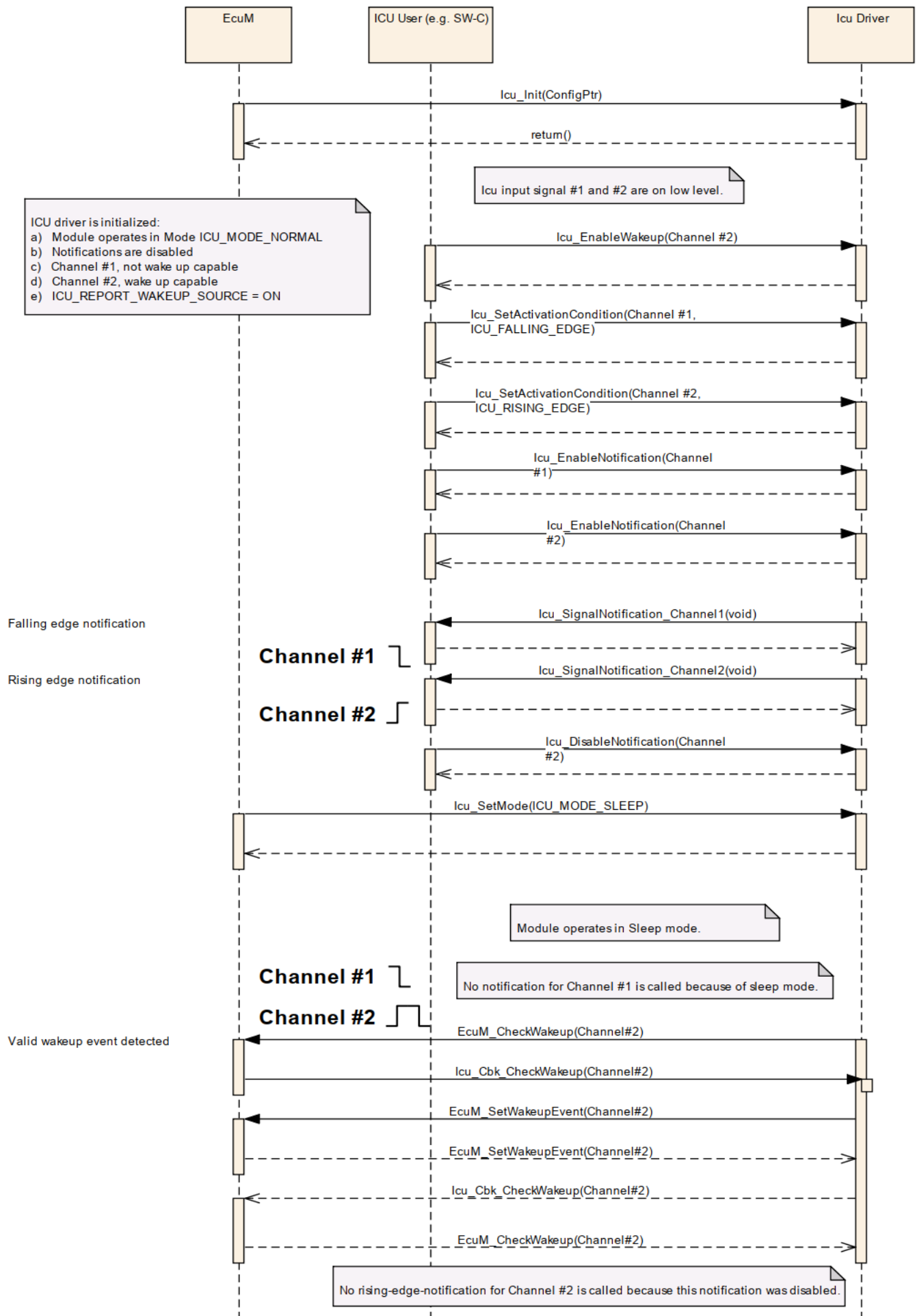
2.9.3.1 Enabled Notifications in SLEEP mode



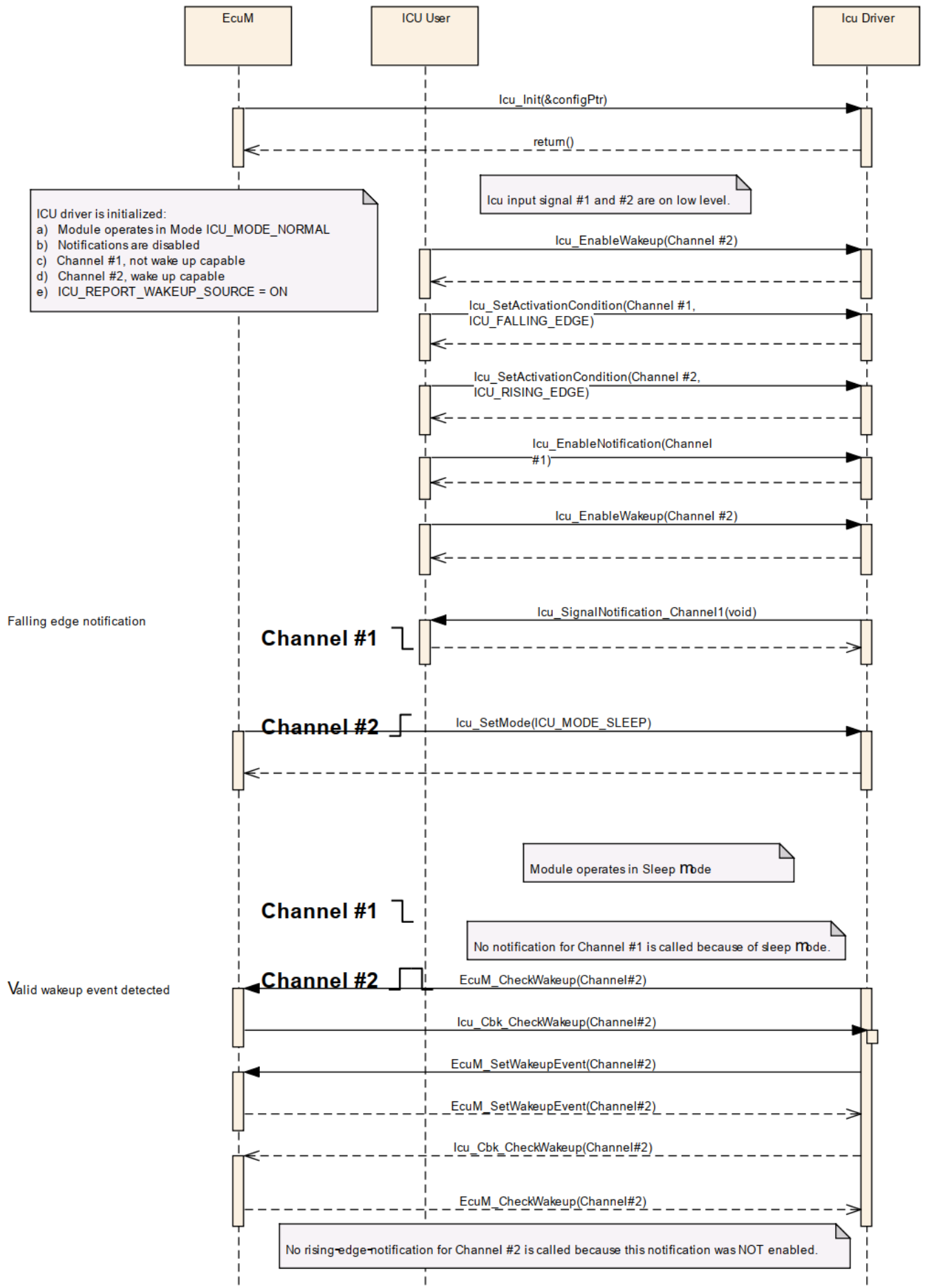
2.9.3.2 Disabled Reporting of Wakeup Sources in SLEEP Mode



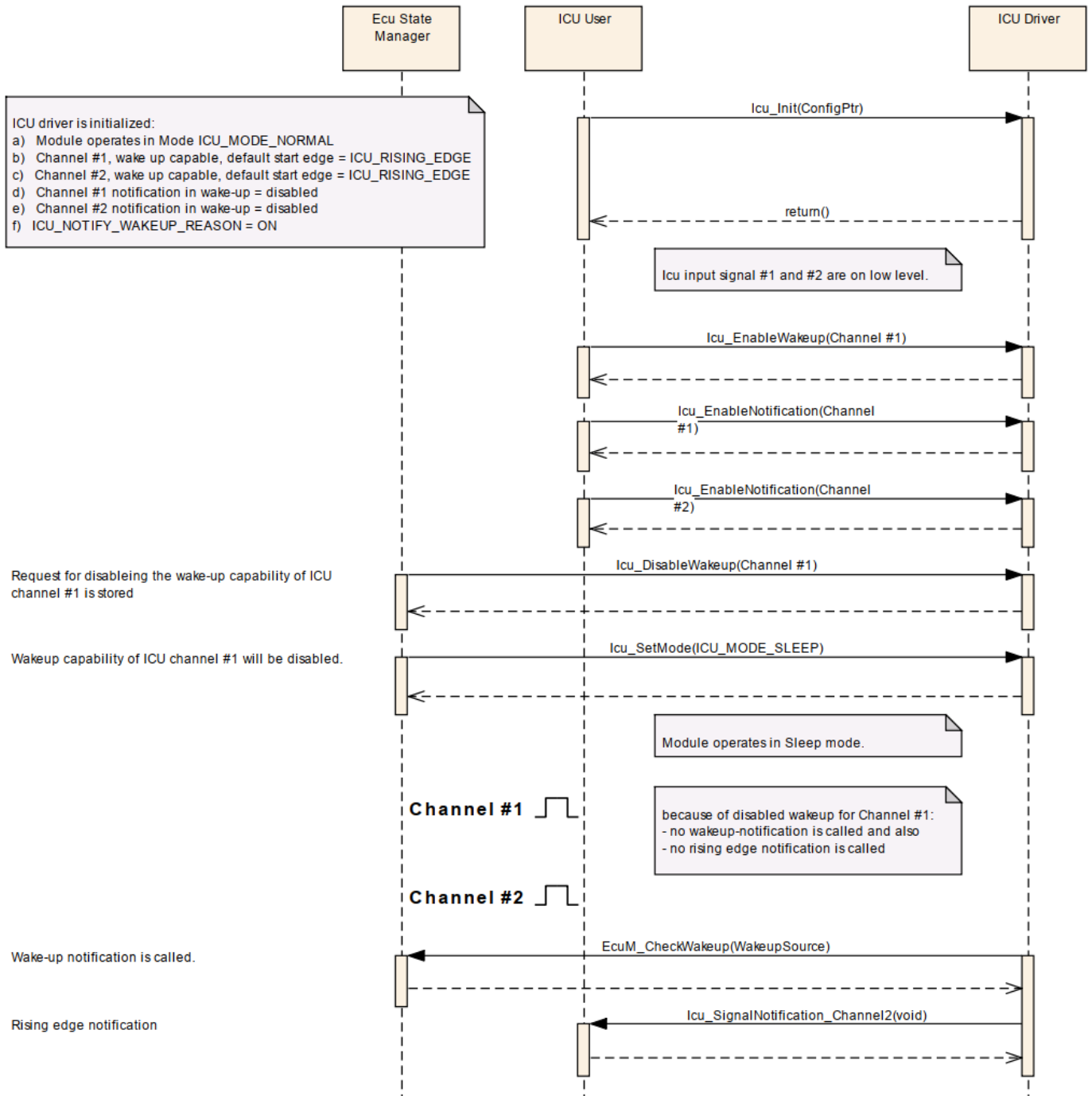
2.9.3.3 Disabled Edge Notification in SLEEP Mode



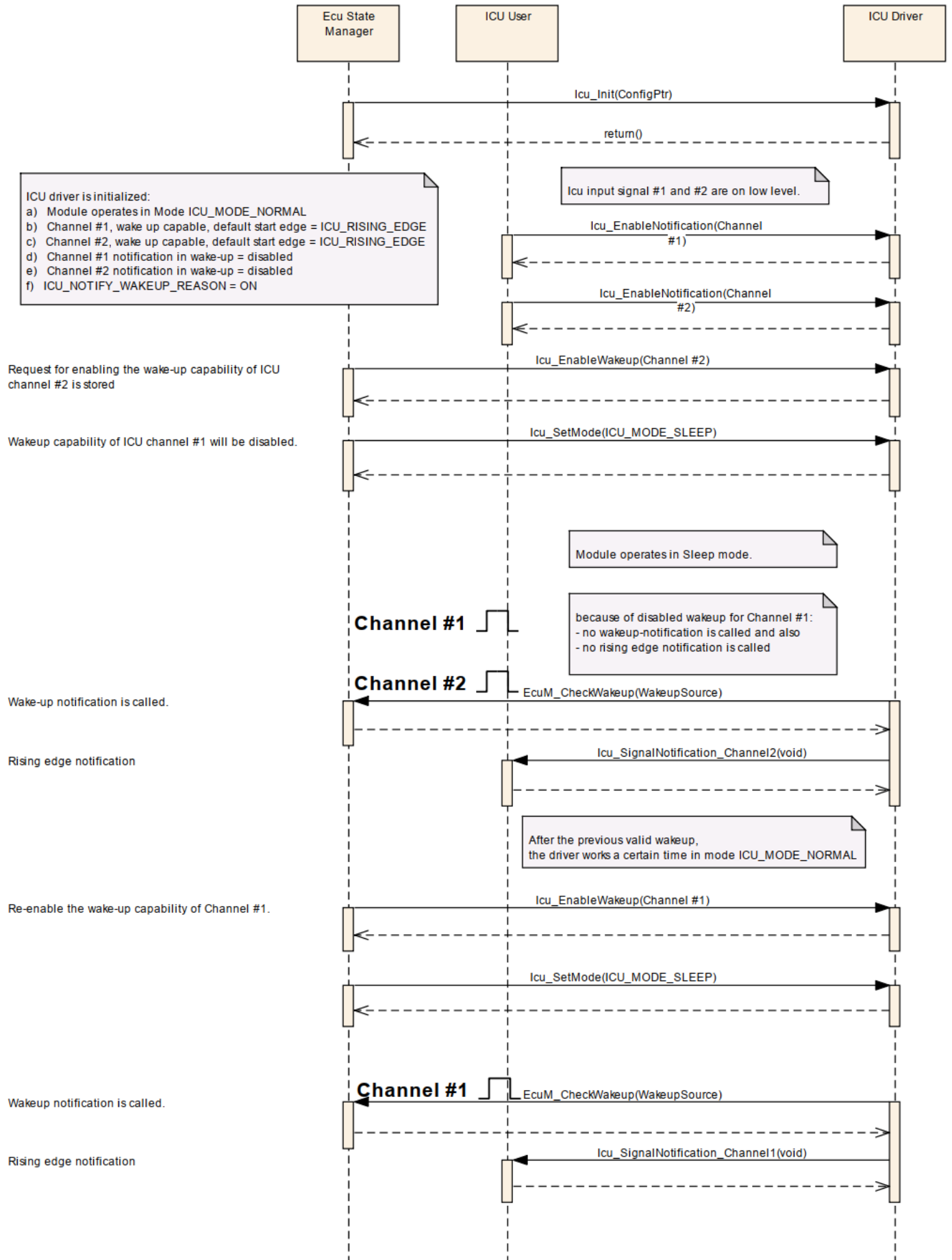
2.9.3.4 Un-Enabled Reporting of Notifications in SLEEP Mode



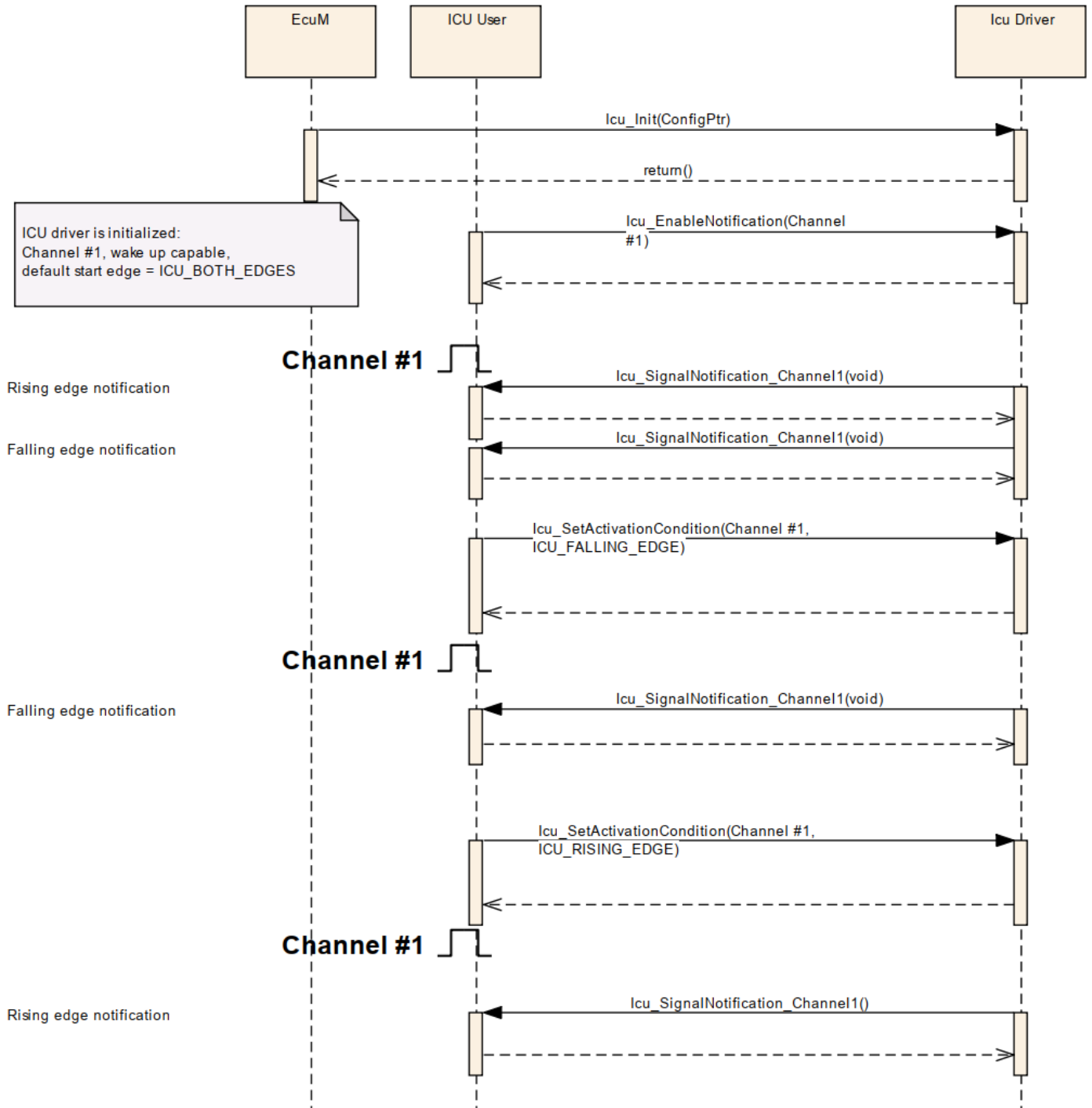
2.9.4 Disabling of Wakeup-Capabilities



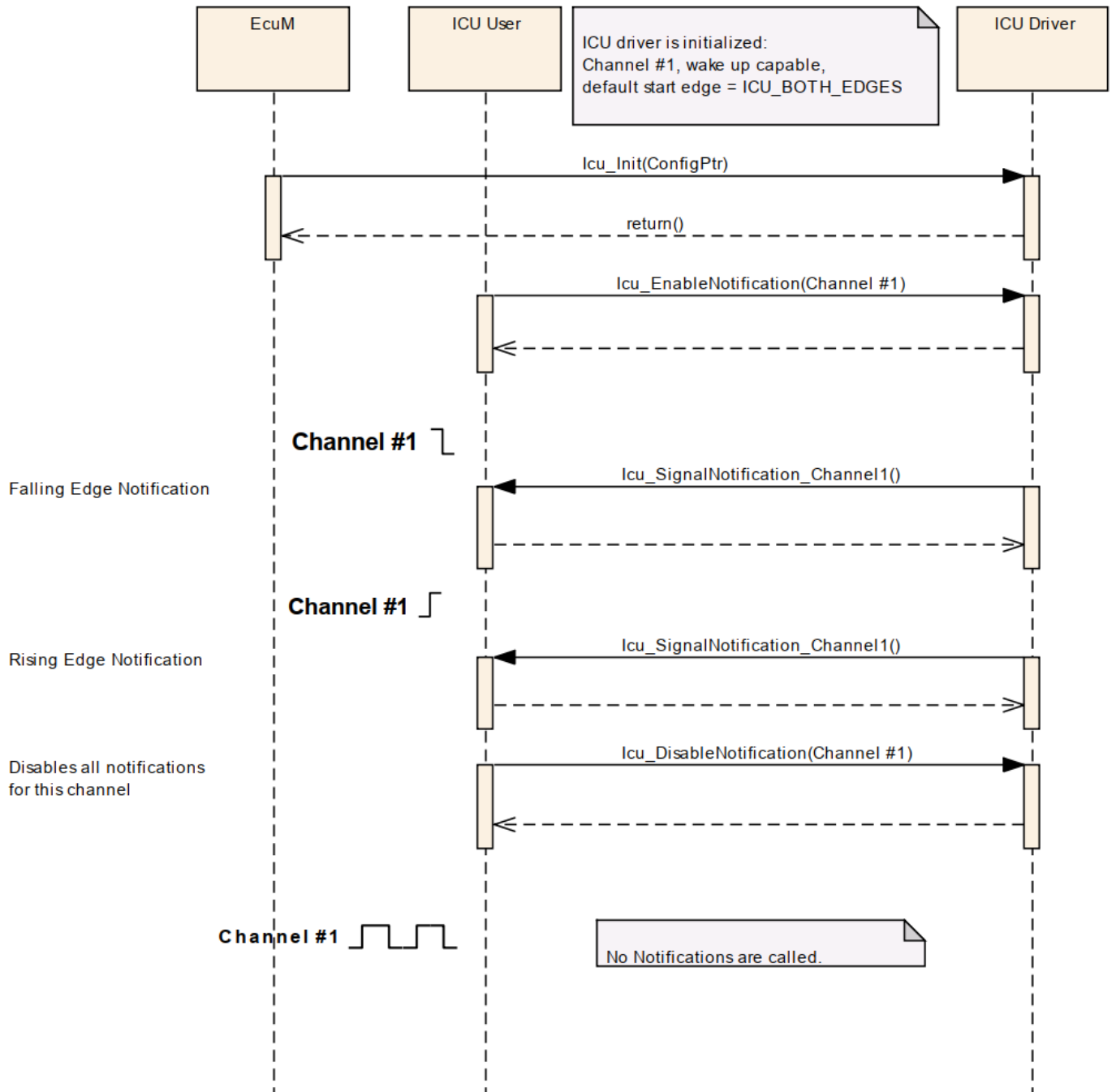
2.9.5 Enabling of Wakeup-Capabilities



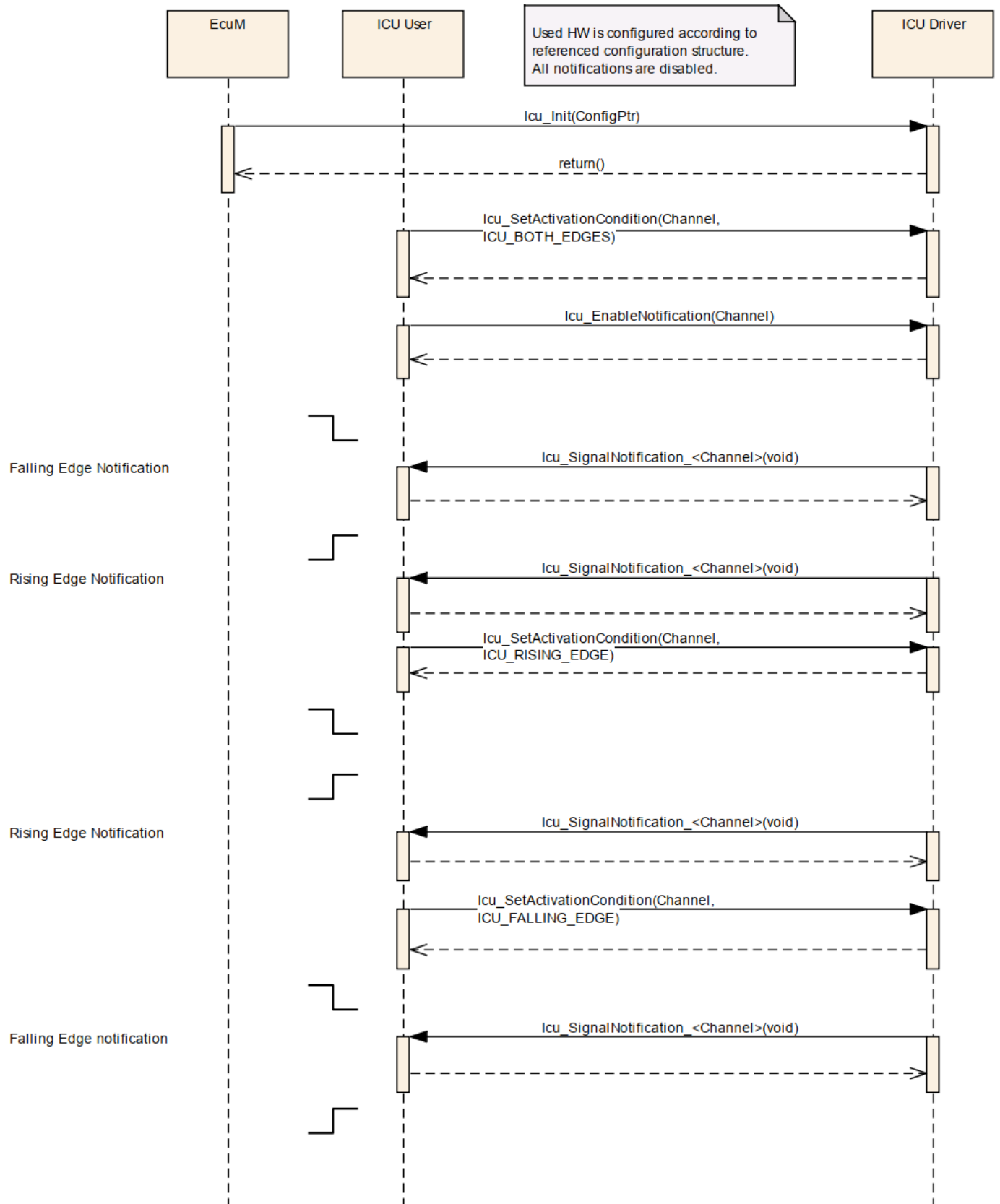
2.9.6 Setting up the Activation Condition for a Channel



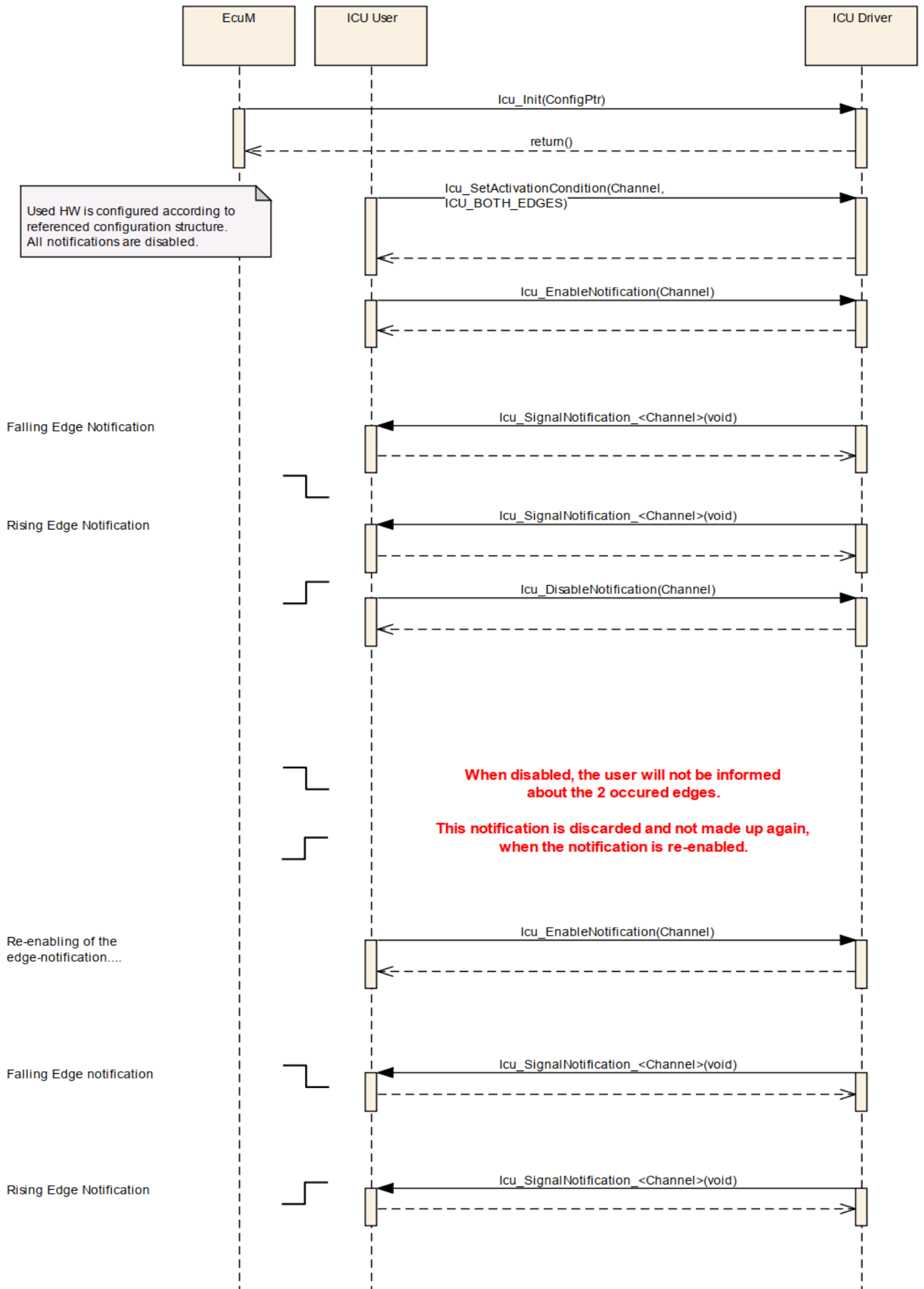
2.9.7 Disabling of the Notification for a Channel



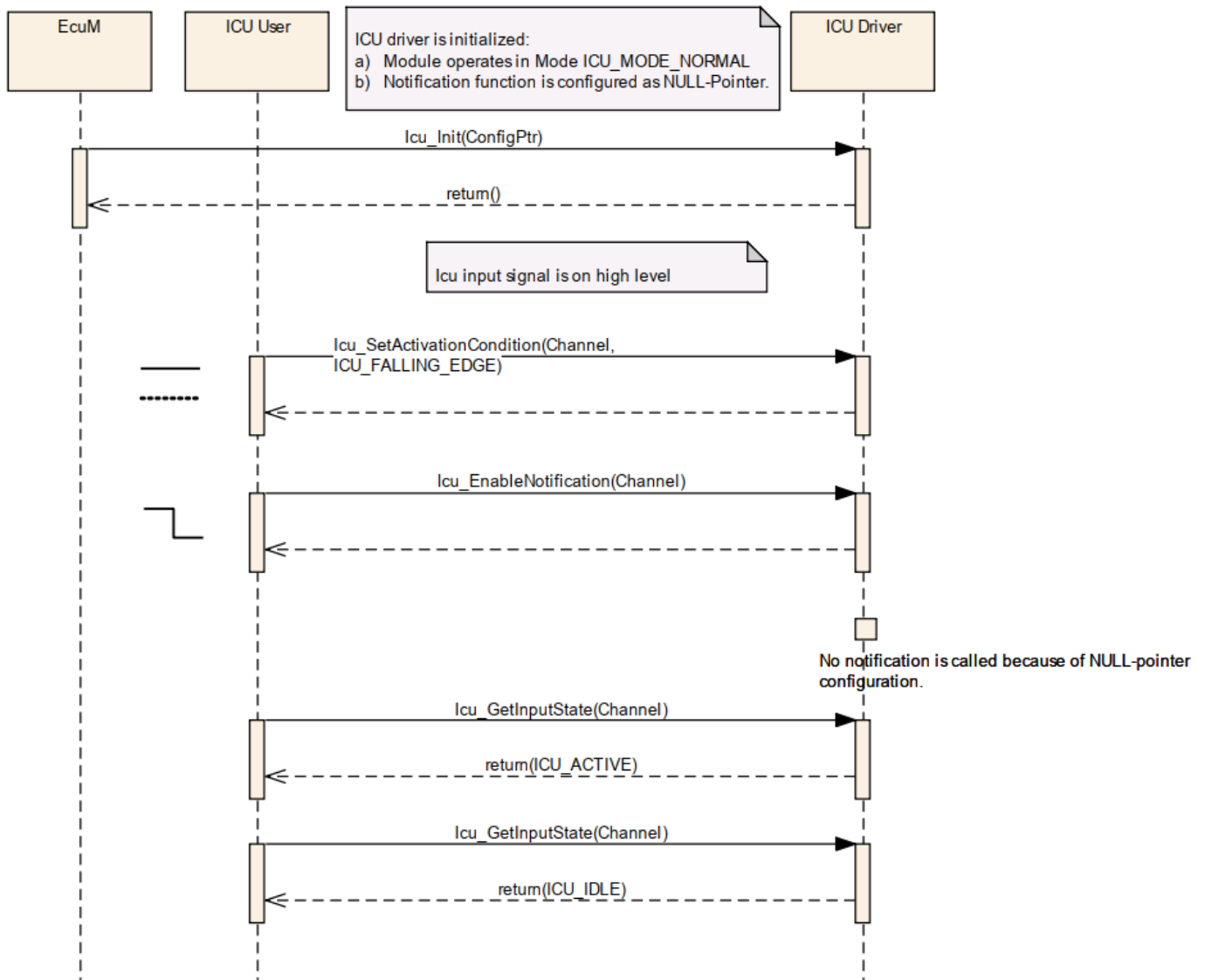
2.9.8 Enabling of the Edge-Notification for a Channel



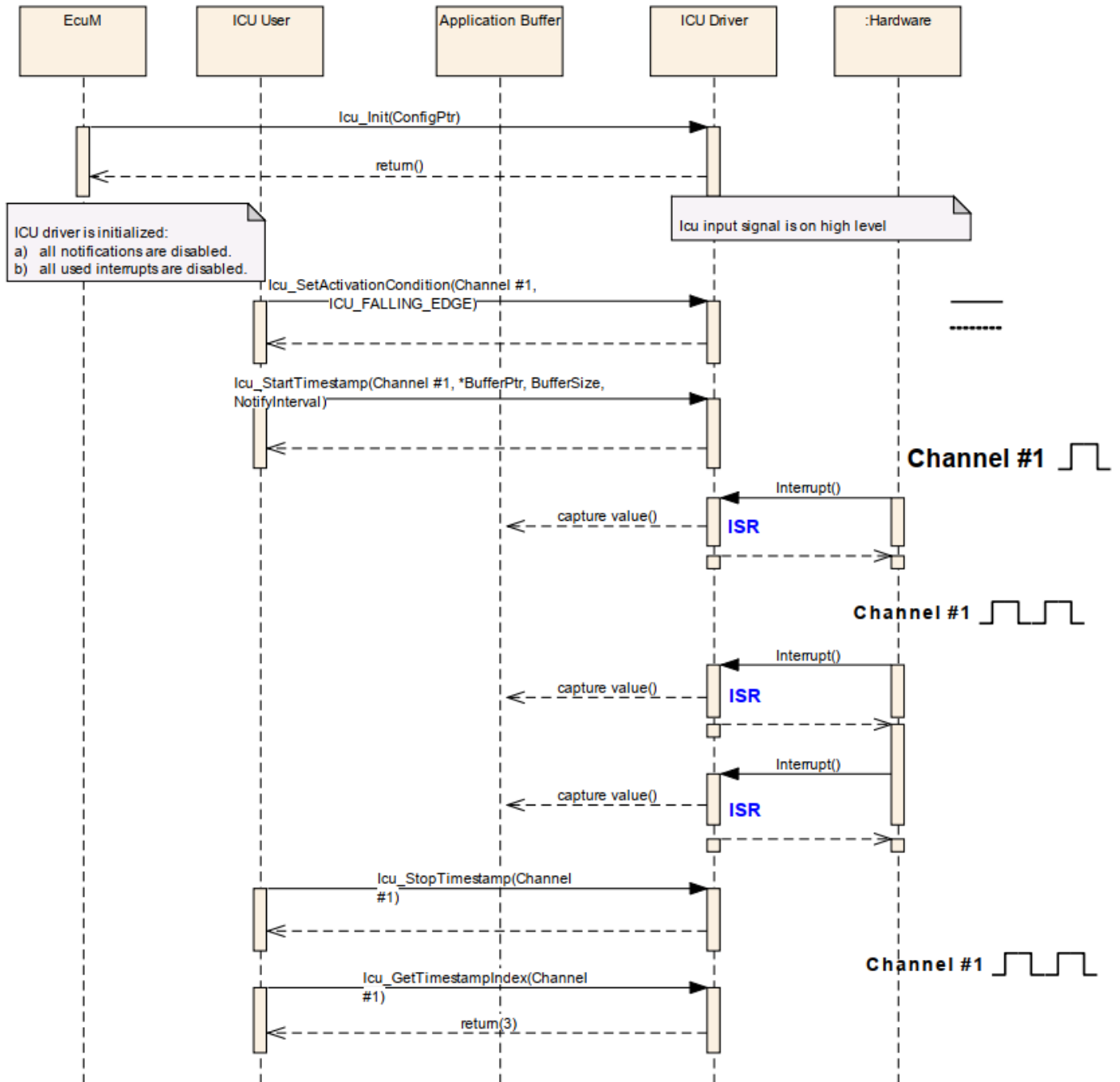
2.9.9 Re-enabling of the Notification for a Channel



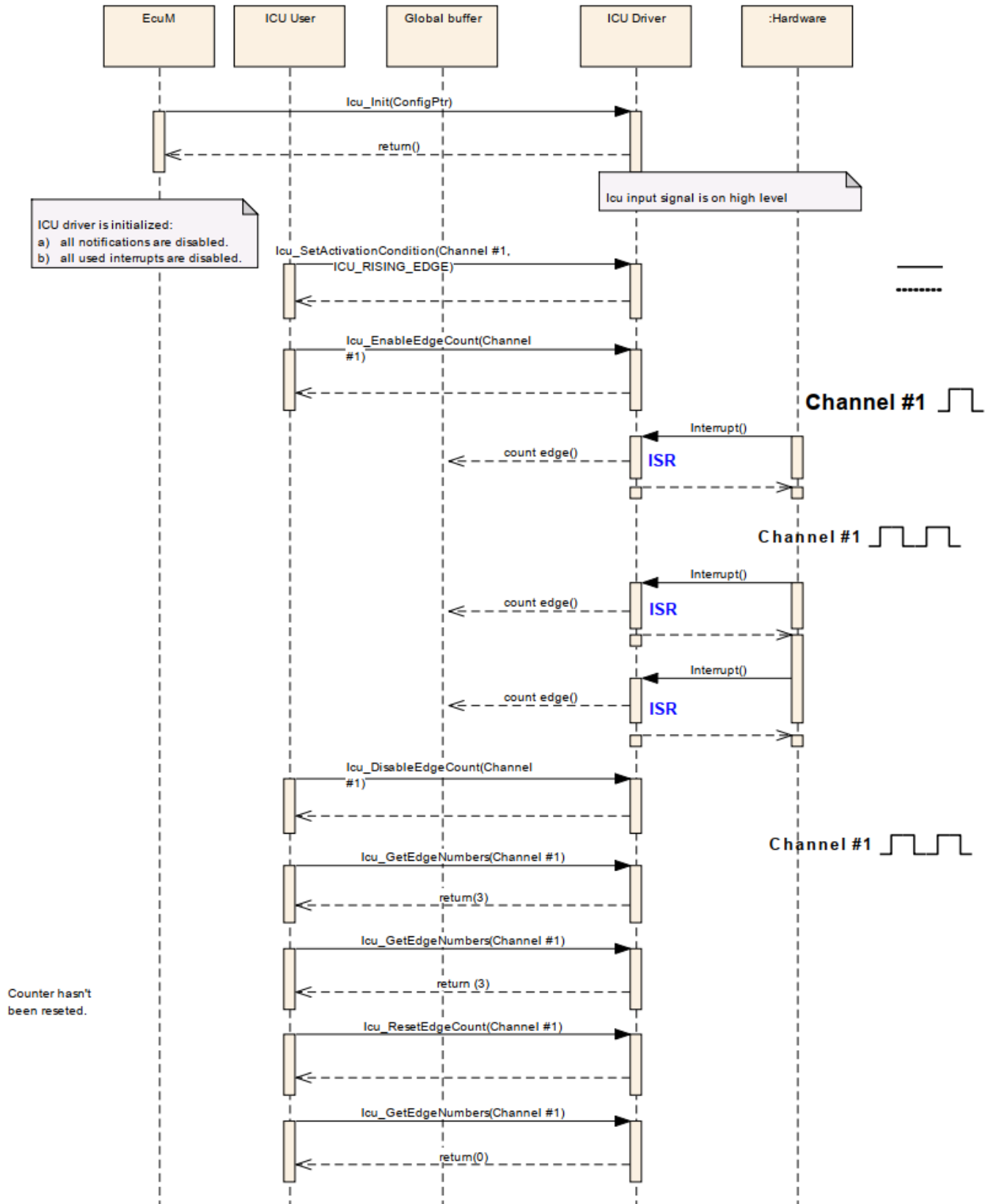
2.9.10 Polling of the Channel Status



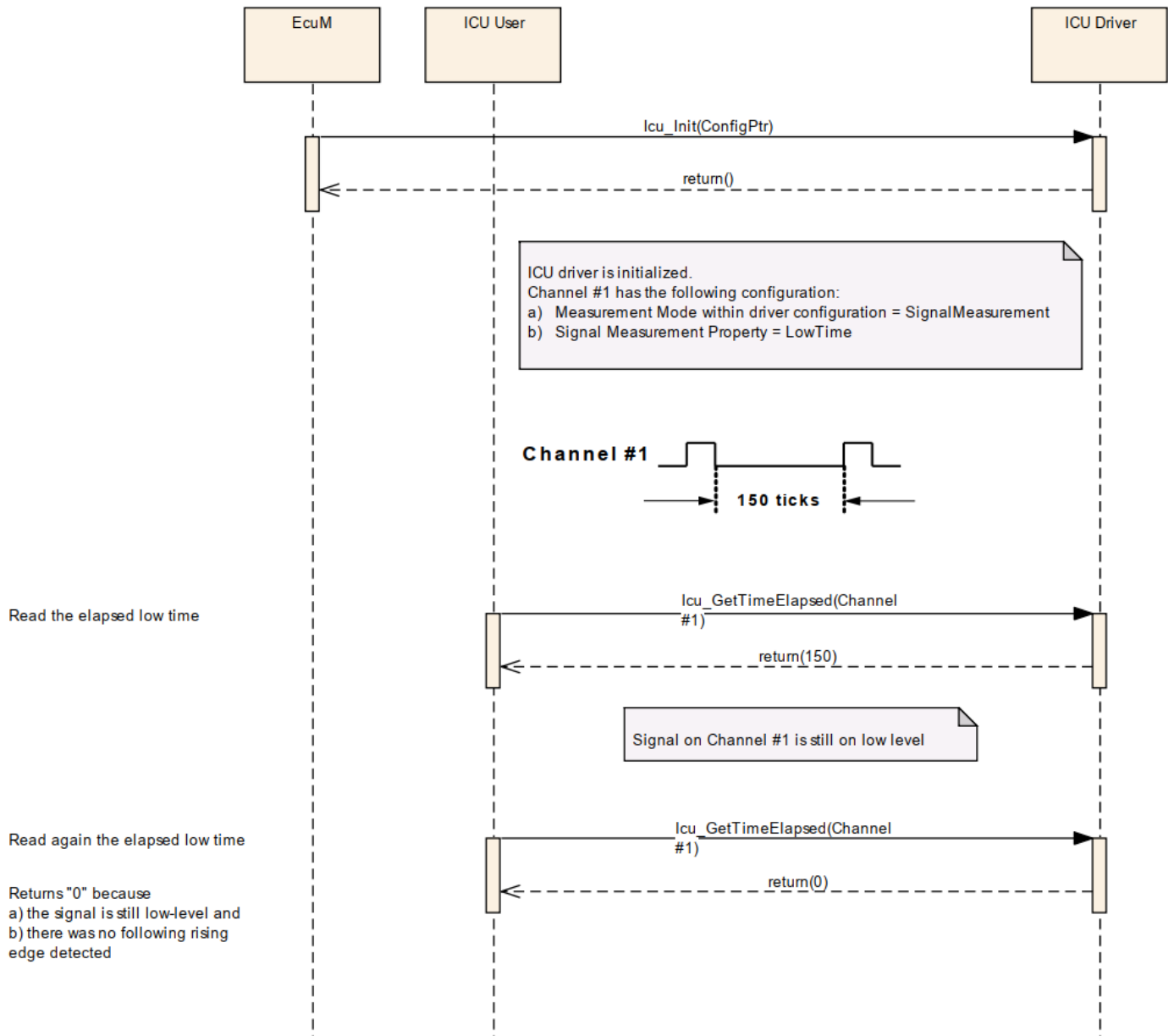
2.9.11 Overview of the Timestamping Functionality of the ICU Driver



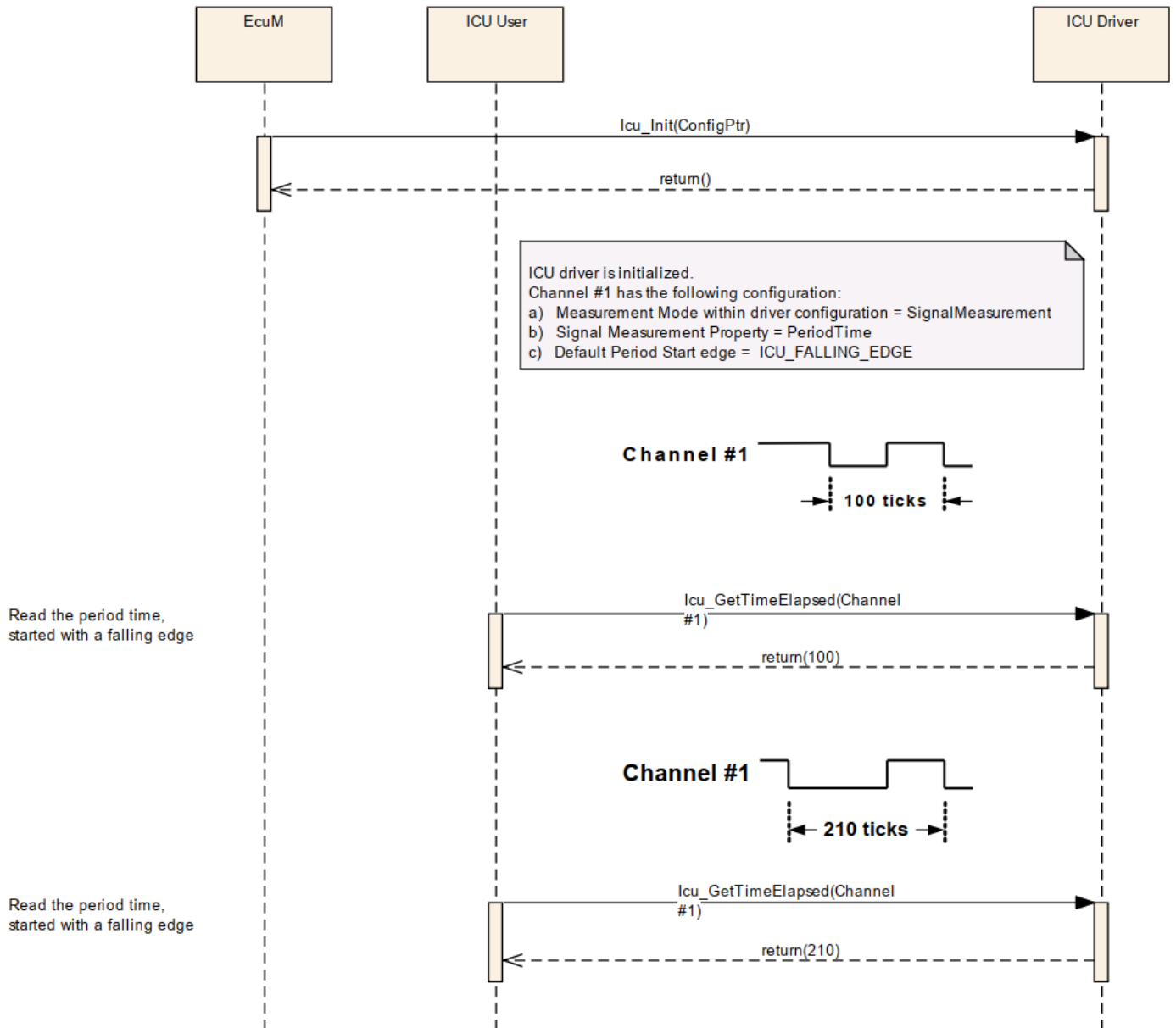
2.9.12 Inquire the Number of Counted Edges



2.9.13 Inquire the Elapsed Level-Time of a Channel



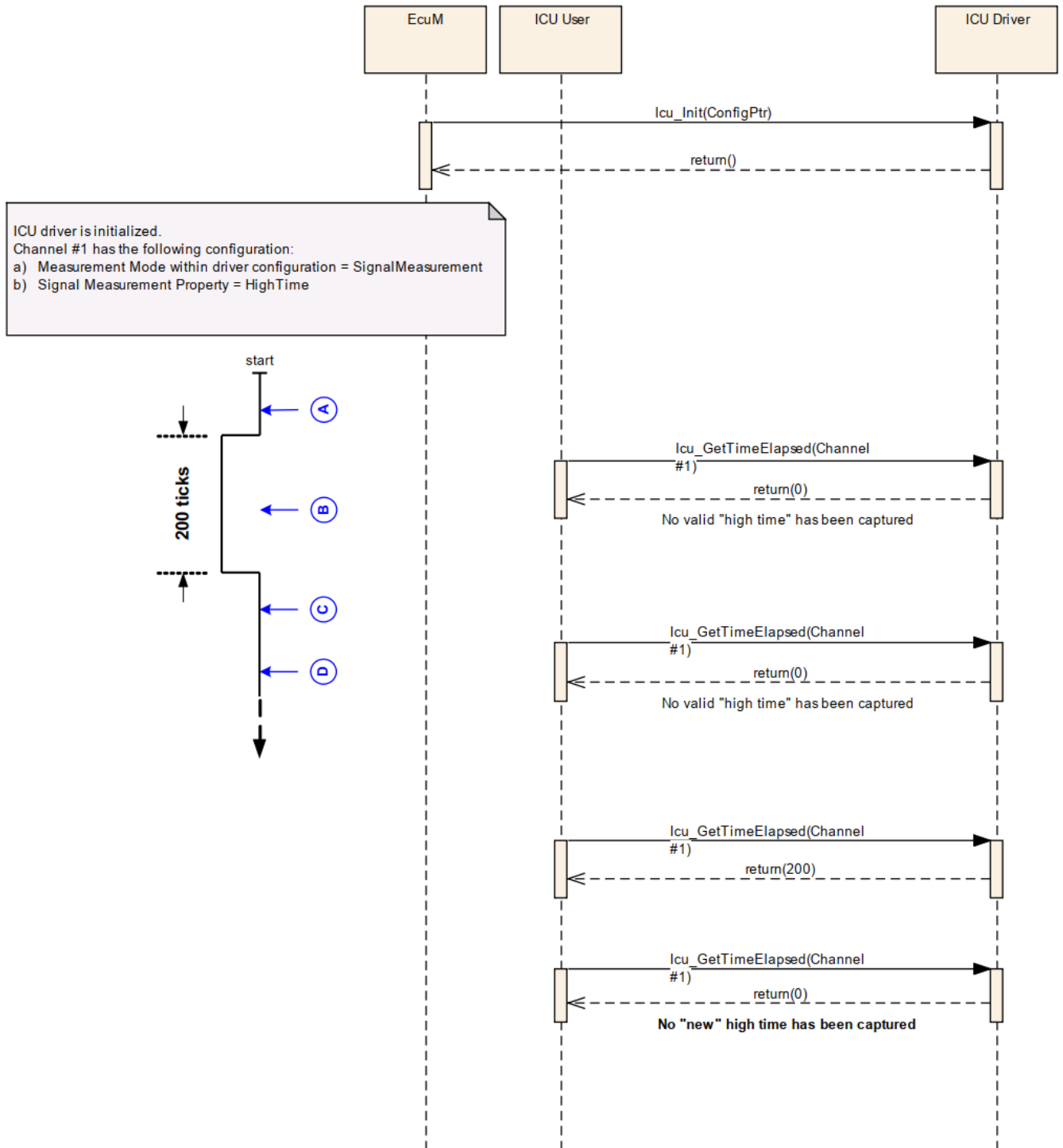
2.9.14 Inquire the Elapsed Period Time of a Channel



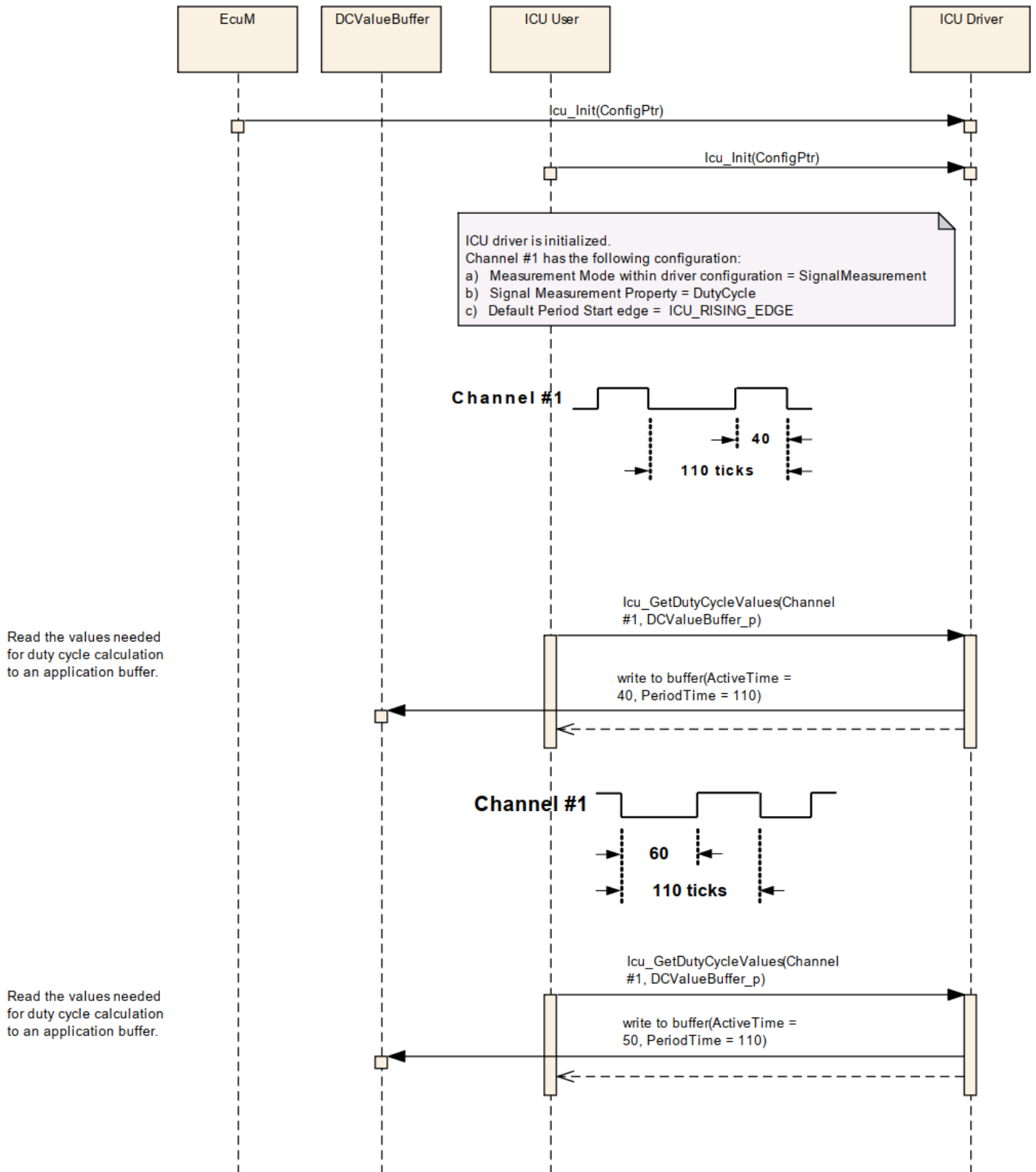
2.9.15 Inquire the Elapsed High Time of a Channel

The following example shows the exemplary behavior before, while and after capturing the "high time" of a signal.

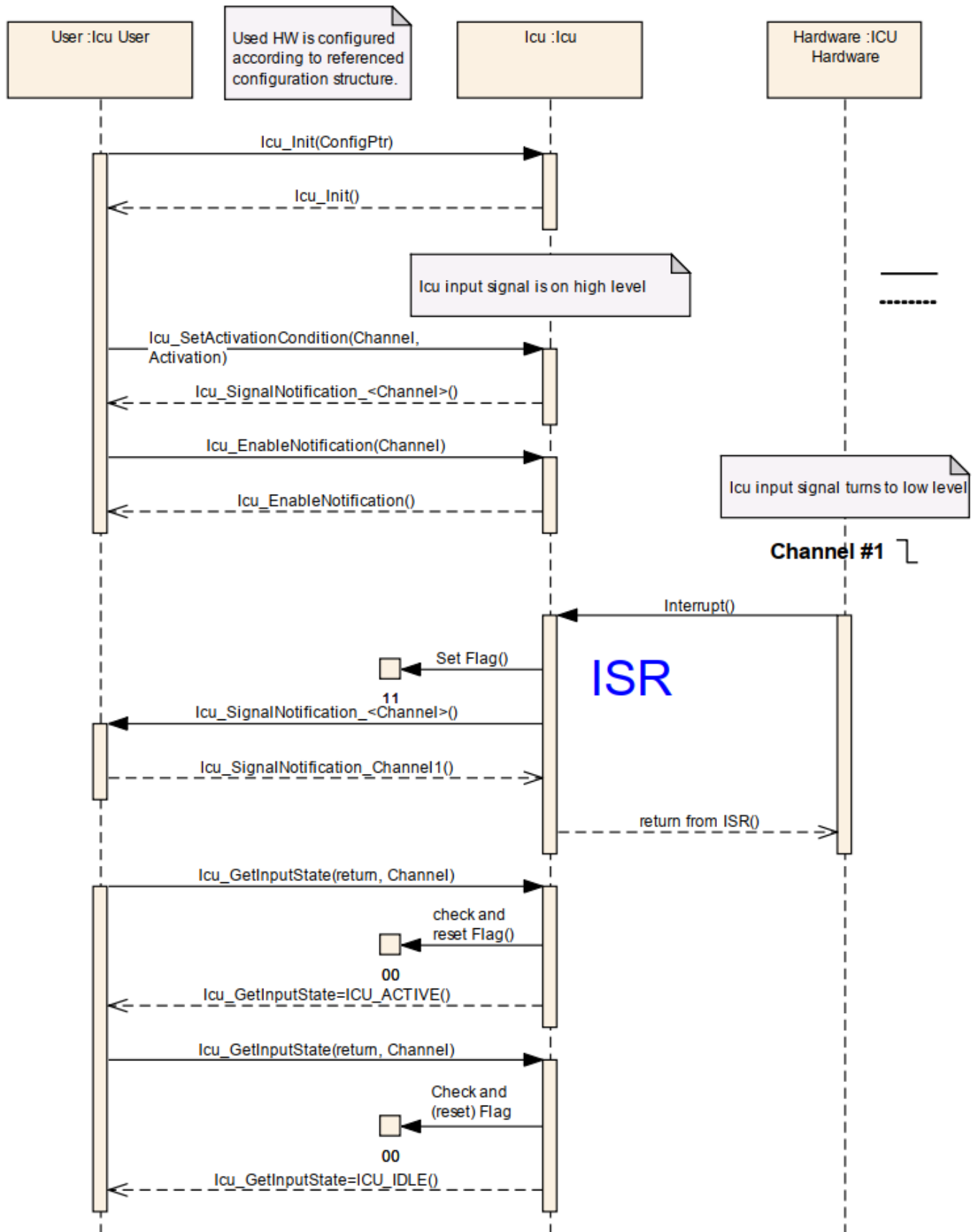
The shown behavior is also appropriate for the service `Icu_GetDutyCycleValues()`.



2.9.16 Measure the Values Needed for Calculation of Duty Cycles



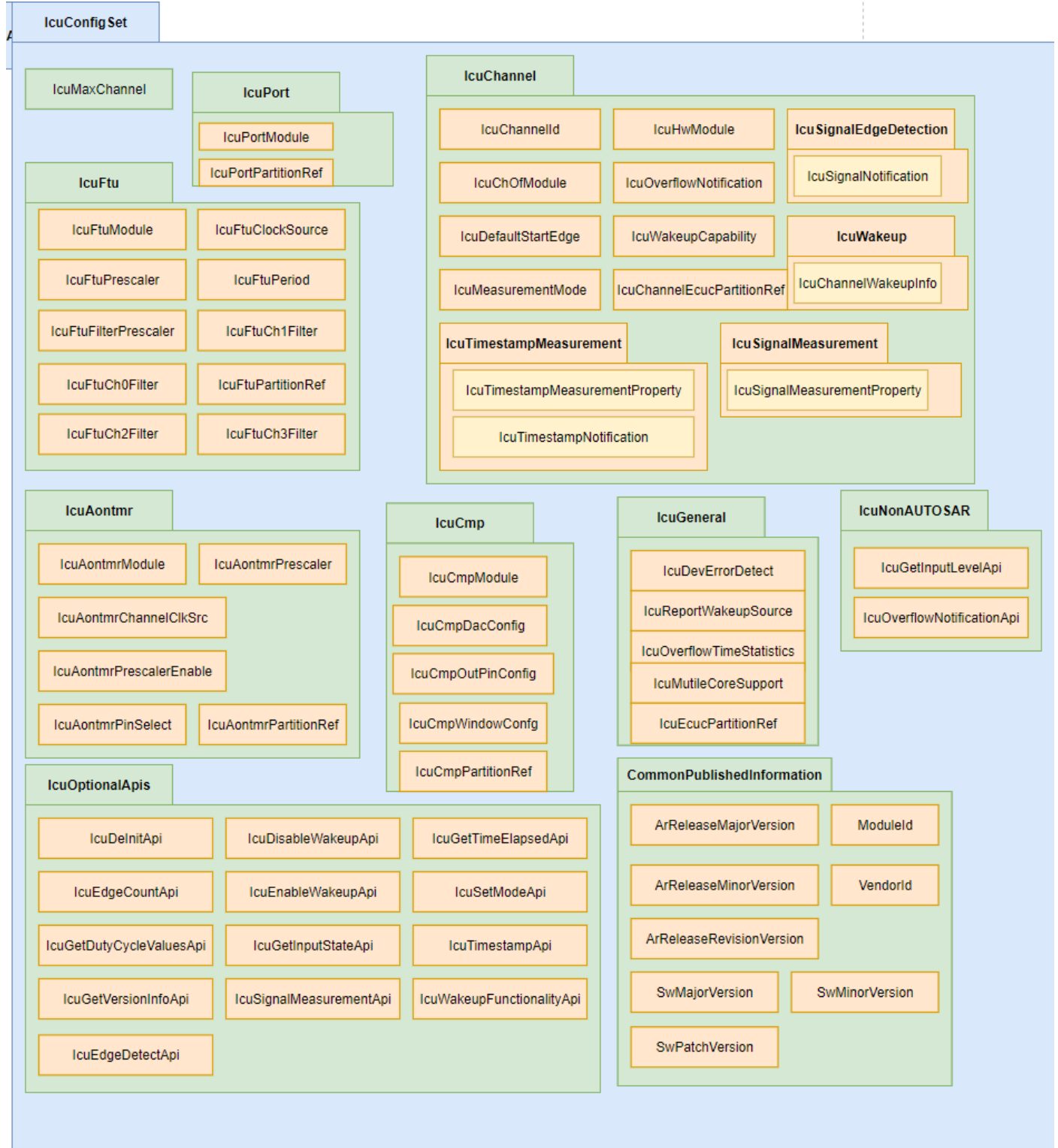
2.9.17 Cooperative Usage of Notification and Polling Mechanism

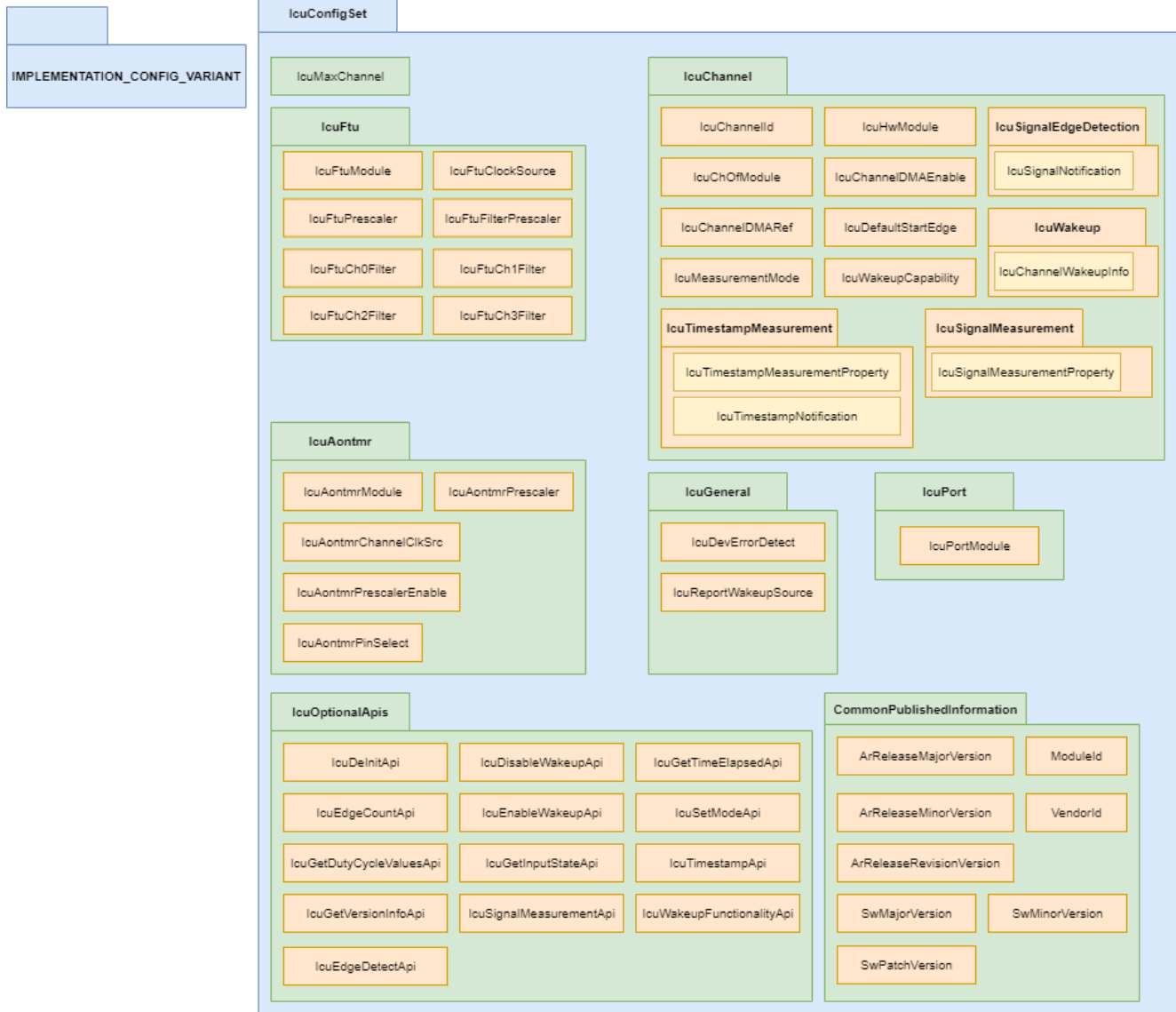


Chapter 3 Tresos Configuration Items

3.1 Container Inclusion Relation

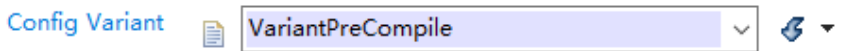
The container inclusion relation is shown as below:





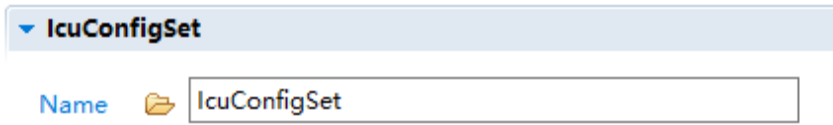
3.2 Containers and Variables

3.2.1 IMPLEMENTATION_CONFIG_VARIANT

Variable	IMPLEMENTATION_CONFIG_VARIANT	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	Variable: enumeration
	Label	Config Variant
	Default	VariantPostBuild
	Range	VariantPostBuild VariantPreCompile
Returns	N/A	

3.2.2 IcuConfigSet

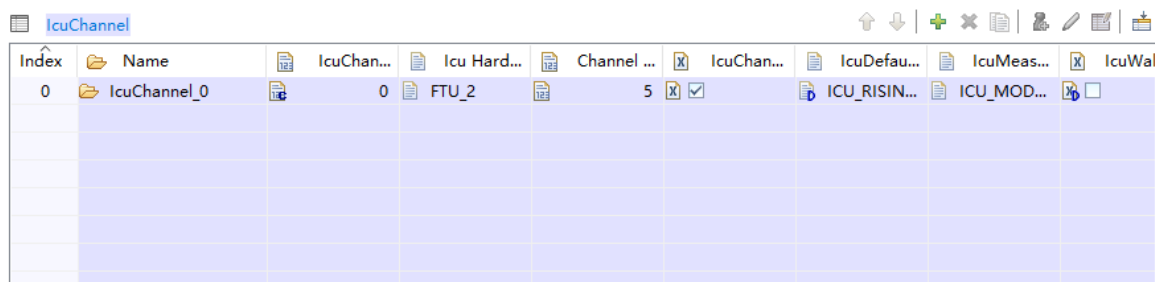
Container	IcuConfigSet
Description	This container is the base for a multiple configuration set.

Screenshot		
	Name  IcuConfigSet	
Properties	Property	Value
	Type	Container

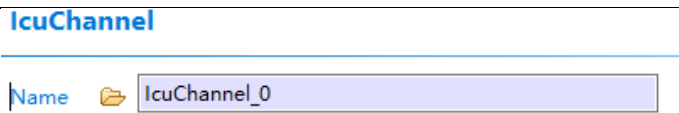
3.2.2.1 IcuMaxChannel

Variable	IcuMaxChannel	
Description	The value for the IcuMaxChannel must match with the number of IcuChannel configured.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.3 IcuChannel

Container	IcuChannel	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	Map
	MIN	1

3.2.3.1 IcuChannel

Container	IcuChannel	
Description	Configuration of an individual ICU channel.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.2 IcuChannelId

Variable	IcuChannelId	
Description	Channel ID of the ICU channel. This value will be assigned to the symbolic name derived of the IcuChannel container short name.	
Screenshot		

Properties	Property	Value
	Type	Variable: INTEGER
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	true
	Range	0~ 172

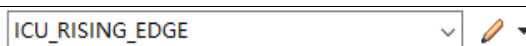
3.2.3.3 IcuHwModule

Variable	IcuHwModule	
Description	Select the hardware module of the ICU channel.	
Screenshot	Icu Hardware Module 	
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	FTU0~11,PORT_A, PORT_B, PORT_C, PORT_D, PORT_E,PORT_F,PORT_G,PORT_H,PORT_I,AONTIMER,CMP0~2

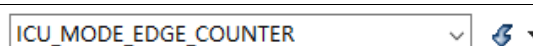
3.2.3.4 IcuChOfModule

Variable	IcuChOfModule	
Description	Hardware sub-channel number of the ICU channel.	
Screenshot	Channel Of Hardware Module 	
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	According to hardware module

3.2.3.5 IcuDefaultStartEdge

Variable	IcuDefaultStartEdge	
Description	Configures the default-activation-edge which shall be used for this channel.	
Screenshot	IcuDefaultStartEdge 	
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Range	ICU_BOTH_EDGES, ICU_FALLING_EDGE, ICU_RISING_EDGE

3.2.3.6 IcuMeasurementMode

Variable	IcuMeasurementMode	
Description	Configures the measurement mode of this channel.	
Screenshot	IcuMeasurementMode 	

Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Range	ICU_MODE_EDGE_COUNTER, ICU_MODE_SIGNAL_EDGE_DETECT, ICU_MODE_SIGNAL_MEASUREMENT, ICU_MODE_TIMESTAMP

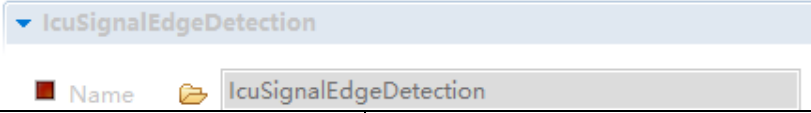
3.2.3.7 IcuOverflowNotification

Variable	IcuOverflowNotification	
Description	Icu Overflow Notification Handler	
Screenshot		
Properties	Property	Value
	Type	FUNCTION-NAME
	Origin	FLAGCHIP
	SymbolicNameValue	false

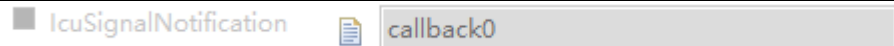
3.2.3.8 IcuWakeupCapability

Variable	IcuWakeupCapability	
Description	Information about the wakeup-capability of this channel.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.3.9 IcuSignalEdgeDetection

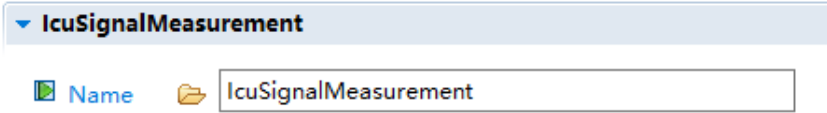
Container	IcuSignalEdgeDetection	
Description	This container contains the configuration (parameters) in case the measurement mode is IcuSignalEdgeDetection.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.10 IcuSignalNotification

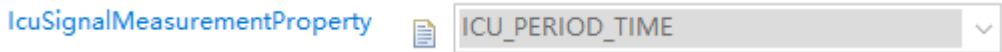
Variable	IcuSignalNotification	
Description	Notification function for signal notification.	
Screenshot		
Properties	Property	Value
	Type	Variable: FUNCTION-NAME

	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Optional	true

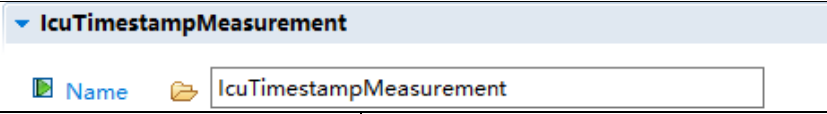
3.2.3.11 IcuSignalMeasurement

Container	IcuSignalMeasurement	
Description	This container contains the configuration (parameters) in case the measurement mode is IcuSignalMeasurement.	
Screenshot		
Properties	Property	Value
	Type	Container

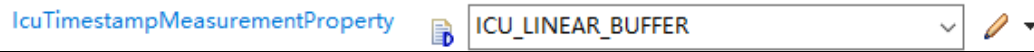
3.2.3.12 IcuSignalMeasurementProperty

Variable	IcuSignalMeasurementProperty	
Description	Configures the property that could be measured in case the mode is IcuSignalMeasurement.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Range	ICU_DUTY_CYCLE, ICU_HIGH_TIME, ICU_LOW_TIME, ICU_PERIOD_TIME

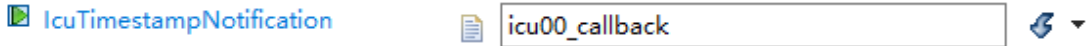
3.2.3.13 IcuTimestampMeasurement

Container	IcuTimestampMeasurement	
Description	This container contains the configuration (parameters) in case the measurement mode is IcuTimestamp.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.3.14 IcuTimestampMeasurementProperty

Variable	IcuTimestampMeasurementProperty	
Description	Configures the handling of the buffer in case the mode is Timestamp.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Range	ICU_CIRCULAR_BUFFER, ICU_LINEAR_BUFFER

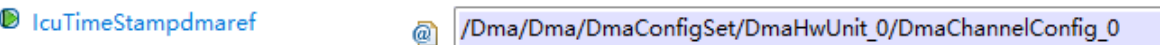
3.2.3.15 IcuTimestampNotification

Variable	IcuTimestampNotification	
Description	Notification function if the number of requested timestamps (Notification interval) are acquired.	
Screenshot		
Properties	Property	Value
	Type	Variable: FUNCTION-NAME
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Optional	true

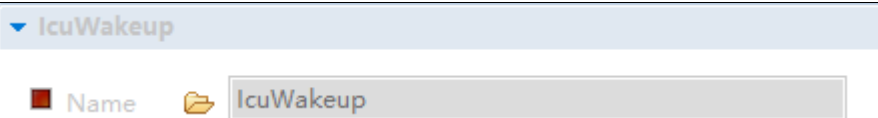
3.2.3.16 IcuTimeStampDmaSupport

Variable	IcuTimeStampDmaSupport	
Description	switch for the Icu TimeStamp transmit way.	
Screenshot		
Properties	Property	Value
	Type	Boolean(Range)
	Origin	Flagchip
	SymbolicNameValue	false
	Optional	true

3.2.3.17 IcuTimeStampdmaref

Variable	IcuTimeStampdmaref	
Description	Dma channel reference. Implementation Type: reference to DMA Hw channel.	
Screenshot		
Properties	Property	Value
	Type	Variable: SYMBOLIC-NAME-REFERENCE
	Origin	Flagchip
	Ref	/Dma/Dma/DmaConfigSet/DmaHwUnit_0/DmaChannelConfig_0

3.2.3.18 IcuWakeup

Container	IcuWakeup	
Description	This container contains the configuration (parameters) needed to configure a wakeup capable channel.	
Screenshot		
Properties	Property	Value
	Type	Container

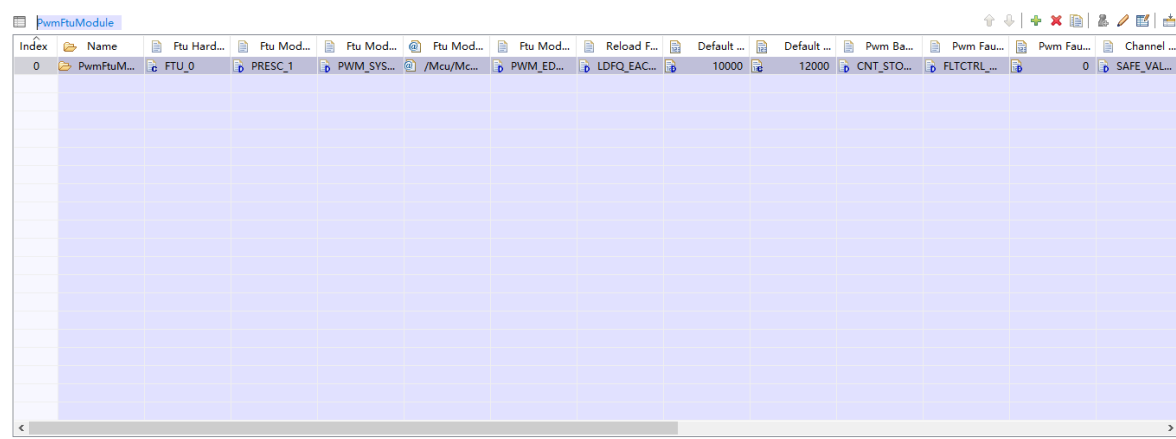
3.2.3.19 IcuChannelWakeupInfo

Variable	IcuChannelWakeupInfo	
Description	If the wakeup-capability is true the wakeup source referenced is transmitted to the ECU State Manager (EcuM).	
Screenshot		
Properties	Property	Value
	Type	Variable: SYMBOLIC-NAME-REFERENCE
	Origin	AUTOSAR_ECUC
	Ref	ASPathDataOfSchema:/AUTOSAR/EcucDefs/EcuM/EcuMConfiguration/EcuMCommonConfiguration/EcuMWakeupSource

3.2.3.20 IcuChannelEcucPartitionRef

Variable	IcuChannelEcucPartitionRef	
Description	If the wakeup-capability is true the wakeup source referenced is transmitted to the ECU State Manager (EcuM).	
Screenshot		
Properties	Property	Value
	Type	Variable: REFERENCE
	Origin	AUTOSAR_ECUC
	Ref	ASPathDataOfSchema:/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

3.2.4 IcuFtu

Container	IcuFtu	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	Map
	MIN	0

3.2.4.1 IcuFtuModule

Variable	IcuFtuModule
Description	Select the physical FTU Module.

Screenshot	Ftu Hardware Module 	
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	FTU_0, FTU_1, FTU_2, FTU_3, FTU_4, FTU_5, FTU_6, FTU_7, FTU_8, FTU_9, FTU_10, FTU_11 1 (For FC7240 only FTU_0~F TU_7)

3.2.4.2 IcuFtuClockSource

Variable	IcuFtuClockSource	
Description	Select the clock source of FTU module.	
Screenshot	Icu FlexTimer clock source 	
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	ICU_FTU_CORE_CLK, ICU_FTU_PCC_CLK, ICU_FTU_EXTERNAL_CLK0, ICU_FTU_EXTERNAL_CLK1, ICU_FTU_EXTERNAL_CLK2

3.2.4.3 IcuFtuPrescaler

Variable	IcuFtuPrescaler	
Description	Configures the prescaler value of FTU module.	
Screenshot	Ftu Module clock selection 	
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	ICU_FTU_PRESCALER_DIV_1, ICU_FTU_PRESCALER_DIV_2, ICU_FTU_PRESCALER_DIV_4, ICU_FTU_PRESCALER_DIV_8, ICU_FTU_PRESCALER_DIV_16, ICU_FTU_PRESCALER_DIV_32, ICU_FTU_PRESCALER_DIV_64, ICU_FTU_PRESCALER_DIV_128

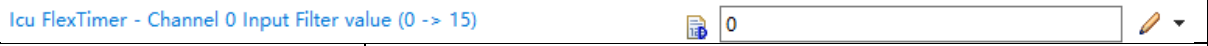
3.2.4.4 IcuFtuPeriod

Variable	IcuFtuPeriod	
Description	Default period value used at initialization.	
Screenshot	Icu FlexTimer Period Value (0 -> 65535) 	
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	0~65535

3.2.4.5 IcuFtuFilterPrescaler

Variable	IcuFtuFilterPrescaler		
Description	Selects the filter value for the channel input.		
Screenshot			
Properties	Property	Value	
	Type	Variable: ENUMERATION	
	Origin	FLAGCHIP	
	SymbolicNameValue	false	
	Range	ICU_FTU_FLT_DIV_1, ICU_FTU_FLT_DIV_2, ICU_FTU_FLT_DIV_3, ICU_FTU_FLT_DIV_4, ICU_FTU_FLT_DIV_5, ICU_FTU_FLT_DIV_6, ICU_FTU_FLT_DIV_7, ICU_FTU_FLT_DIV_8, ICU_FTU_FLT_DIV_9, ICU_FTU_FLT_DIV_10, ICU_FTU_FLT_DIV_11, ICU_FTU_FLT_DIV_12, ICU_FTU_FLT_DIV_13, ICU_FTU_FLT_DIV_14, ICU_FTU_FLT_DIV_15, ICU_FTU_FLT_DIV_16	

3.2.4.6 IcuFtuCh0Filter

Variable	IcuFtuCh0Filter		
Description	Selects the filter value for the channel 0 input. The filter is disabled when the value is zero.		
Screenshot			
Properties	Property	Value	
	Type	Variable: INTEGER	
	Origin	FLAGCHIP	
	SymbolicNameValue	false	
	Range	0~15	

3.2.4.7 IcuFtuCh1Filter

Variable	IcuFtuCh1Filter		
Description	Selects the filter value for the channel 1 input. The filter is disabled when the value is zero.		
Screenshot			
Properties	Property	Value	
	Type	Variable: INTEGER	
	Origin	FLAGCHIP	
	SymbolicNameValue	false	
	Range	0~15	

3.2.4.8 IcuFtuCh2Filter

Variable	IcuFtuCh2Filter		
Description	Selects the filter value for the channel 2 input. The filter is disabled when the value is zero.		
Screenshot			
Properties	Property	Value	
	Type	Variable: INTEGER	
	Origin	FLAGCHIP	

	SymbolicNameValue	false
	Range	0~15

3.2.4.9 IcuFtuCh3Filter

Variable	IcuFtuCh3Filter	
Description	Selects the filter value for the channel 3 input. The filter is disabled when the value is zero.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	0~15

3.2.4.10 IcuFtuGtb

Variable	IcuFtuGtb	
Description	Switch for enable FTU Global Time Base.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

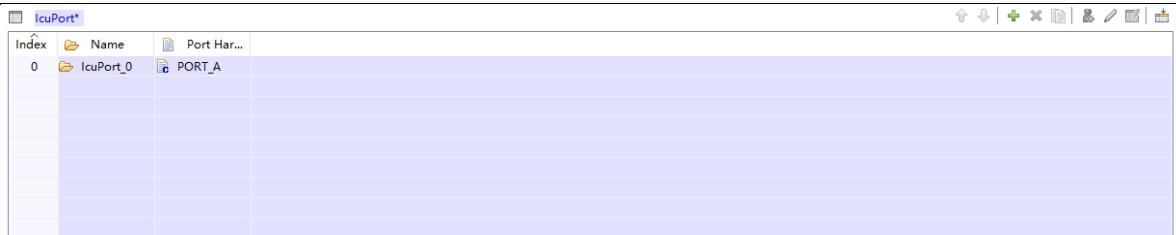
3.2.4.11 IcuFtuGtbRef

Variable	IcuFtuGtbRef	
Description	Maps the Icu ftu instance to Icu gtb group	
Screenshot		
Properties	Property	Value
	Type	REFERENCE
	Origin	FLAGCHIP
	Ref	/Icu/Icu/IcuGeneral/IcuFtuGtbGroup_0

3.2.4.12 IcuFtuPartitionRef

Variable	IcuFtuPartitionRef	
Description	Maps the Icu ftu instance to ECUC partitions	
Screenshot		
Properties	Property	Value
	Type	REFERENCE
	Origin	FLAGCHIP
	Ref	ASPathDataOfSchema:/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

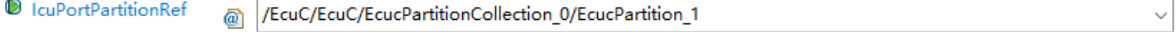
3.2.5 IcuPort

Container	IcuPort	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	Map
	MIN	0

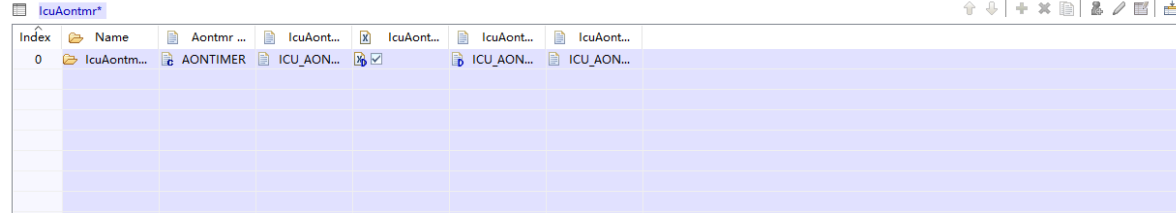
3.2.5.1 IcuPortModule

Variable	IcuPortModule	
Description	Select the physical Port Module.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	PORT_A, PORT_B, PORT_C, PORT_D, PORT_E, PORT_F, PORT_H, PORT_I

3.2.5.2 IcuPortPartitionRef

Variable	IcuPortPartitionRef	
Description	Maps the Icu port instance to ECUC partitions.	
Screenshot		
Properties	Property	Value
	Type	REFERENCE
	Origin	FLAGCHIP
	Ref	ASPathDataOfSchema:/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollectio n/EcucPartition

3.2.6 IcuAontmr

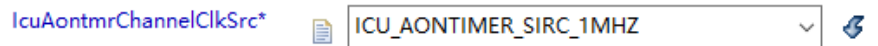
Container	IcuAontmr	
Description	N/A	
Screenshot		
Properties	Property	Value

	Type	Map
	MIN	0

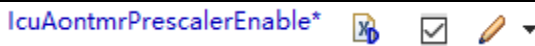
3.2.6.1 IcuAontmrModule

Variable	IcuAontmrModule	
Description	Select the physical Aontmr Module.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	AONTIMER

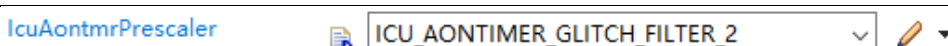
3.2.6.2 IcuAontmrChannelClkSrc

Variable	IcuAontmrChannelClkSrc	
Description	Select the clock source for the AONTIMER module for this platform.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	ICU_AONTIMER_SIRC_1MHZ, ICU_AONTIMER_RTC_CLK, ICU_AONTIMER_PCC_CLK

3.2.6.3 PrescalerEnable

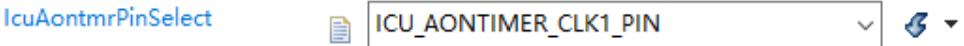
Variable	PrescalerEnable	
Description	When PBYP is set, the selected prescaler clock in Time Counter mode or selected input source in Pulse Counter mode directly clocks the CNR.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.6.4 IcuAontmrPrescaler

Variable	IcuAontmrPrescaler	
Description	The ICU module specific Glitch filter value.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP

	SymbolicNameValue	false
	Range	ICU_AONTIMER_GLITCH_FILTER_2, ICU_AONTIMER_GLITCH_FILTER_4, ICU_AONTIMER_GLITCH_FILTER_8, ICU_AONTIMER_GLITCH_FILTER_16, ICU_AONTIMER_GLITCH_FILTER_32, ICU_AONTIMER_GLITCH_FILTER_64, ICU_AONTIMER_GLITCH_FILTER_128, ICU_AONTIMER_GLITCH_FILTER_256, ICU_AONTIMER_GLITCH_FILTER_512, ICU_AONTIMER_GLITCH_FILTER_1024, ICU_AONTIMER_GLITCH_FILTER_2048, ICU_AONTIMER_GLITCH_FILTER_4096, ICU_AONTIMER_GLITCH_FILTER_8192, ICU_AONTIMER_GLITCH_FILTER_16384, ICU_AONTIMER_GLITCH_FILTER_32768

3.2.6.5 IcuAontmrPinSelect

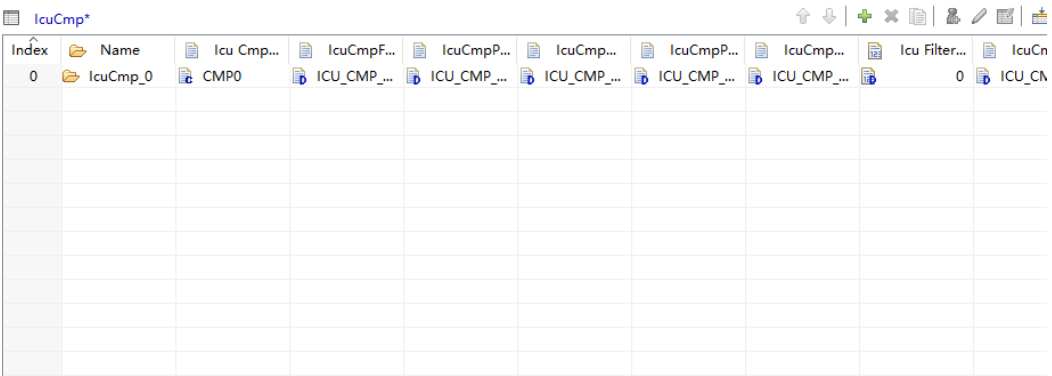
Variable	IcuAontmrPinSelect	
Description	Configures the input source to be used in Pulse Counter mode.	
Screenshot		
Properties	Property	Value
	Type	Variable : ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	ICU_AONTIMER_TRGSEL_OUTPUT, ICU_AONTIMER_CLK1_PIN, ICU_AONTIMER_CLK2_PIN, ICU_AONTIMER_CLK3_PIN

3.2.6.6 IcuAontmrPartitionRef

Variable	IcuAontmrPartitionRef	
Description	Maps the Icu Aontimer instance to ECUC partitions	
Screenshot		
Properties	Property	Value
	Type	REFERENCE
	Origin	FLAGCHIP
	Ref	ASPathDataOfSchema:/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

3.2.7 IcuCmp

Container	IcuCmp
Description	N/A

Screenshot		
Properties	Property	Value
	Type	Container

3.2.7.1 IcuCmp Hardware Module

Variable	PrescalerEnable	
Description	Select the physical Cmp Module.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	CMP0,CMP1,CMP2

3.2.7.2 IcuCmpFunctionMode

Variable	IcuCmpFunctionMode	
Description	Select the physical Cmp Module.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	ICU_CMP_MOD_DISABLE,ICU_CMP_MOD_CONTINUOUS,ICU_CMP_MOD_SAMPLE_NONFILTER_EXTCLK,ICU_CMP_MOD_SAMPLE_NONFILTER_INTCLK,ICU_CMP_MOD_SAMPLE_FILTER_EXTCLK,ICU_CMP_MOD_SAMPLE_FILTER_INTCLK,ICU_CMP_MOD_WINDOW,ICU_CMP_MOD_WINDOW_RE SAMPLE,ICU_CMP_MOD_WINDOW_FILTER

3.2.7.3 IcuCmpPositiveInputSrc

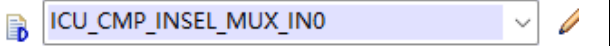
Variable	IcuCmpPositiveInputSrc	
Description	Select the input source of the positive input.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION

	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	ICU_CMP_INSRCSEL_DAC,ICU_CMP_INSRCSEL_MUX

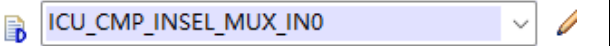
3.2.7.4 IcuCmpNegativeInputSrc

Variable	IcuCmpNegativeInputSrc	
Description	Select the input source of the negative input.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	ICU_CMP_INSRCSEL_DAC,ICU_CMP_INSRCSEL_MUX

3.2.7.5 IcuCmpPositiveInputMuxSelect

Variable	IcuCmpPositiveInputMuxSelect	
Description	Select the mux of the positive input. user should config the positive input src as ICU_CMP_INSRCSEL_MUX.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	ICU_CMP_INSEL_MUX_IN0,ICU_CMP_INSEL_MUX_IN1,ICU_CMP_INSEL_MUX_IN2,ICU_CMP_INSEL_MUX_IN3,ICU_CMP_INSEL_MUX_IN4,ICU_CMP_INSEL_MUX_IN5,ICU_CMP_INSEL_MUX_IN6,ICU_CMP_INSEL_MUX_IN7

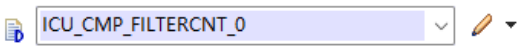
3.2.7.6 IcuCmpNegativeInputMuxSelect

Variable	IcuCmpNegativeInputMuxSelect	
Description	Select the mux of the negative input. user should config the positive input src as ICU_CMP_INSRCSEL_MUX.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	ICU_CMP_INSEL_MUX_IN0,ICU_CMP_INSEL_MUX_IN1,ICU_CMP_INSEL_MUX_IN2,ICU_CMP_INSEL_MUX_IN3,ICU_CMP_INSEL_MUX_IN4,ICU_CMP_INSEL_MUX_IN5,ICU_CMP_INSEL_MUX_IN6,ICU_CMP_INSEL_MUX_IN7

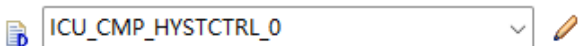
3.2.7.7 IcuCmpFilterPeriod

Variable	IcuCmpFilterPeriod	
Description	Specifies the sampling period (in bus clock cycles) of the comparator output filter	
Screenshot		
Properties	Property	Value
	Type	Variable: Integer
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	0-255

3.2.7.8 IcuCmpFilterCnt

Variable	IcuCmpFilterCnt	
Description	Select the CMP filter count numbers. Notice: The Filter counter may disable in some function mode.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	ICU_CMP_FILTERCNT_0,ICU_CMP_FILTERCNT_1,ICU_CMP_FILTERCNT_2,ICU_CMP_FILTERCNT_3

3.2.7.9 IcuCmpHystCtrlType

Variable	IcuCmpHystCtrlType	
Description	Select the Comparator hard block hysteresis control.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	ICU_CMP_HYSTCTRL_0,ICU_CMP_HYSTCTRL_1,ICU_CMP_HYSTCTRL_2,ICU_CMP_HYSTCTRL_3

3.2.7.10 IcuCmpStopModeEnable

Variable	IcuCmpStopModeEnable	
Description	user can enable the analog comparator or the DAC when in STOP mode	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

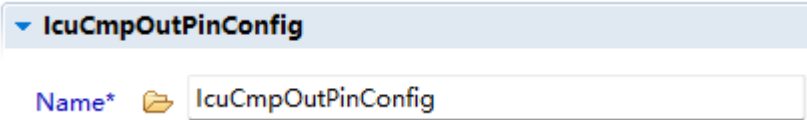
3.2.7.11 IcuInvCmpOut

Variable	IcuInvCmpOut	
Description	user can invert the analog comparator output (CMOOUT signal)	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

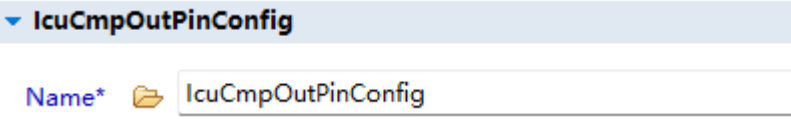
3.2.7.12 IcuCmpHighSpeedMode

Variable	IcuCmpHighSpeedMode	
Description	Highspeed mode will accelerate the conversion speed. But it also Increase power consumption.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.7.13 IcuCmpOutPinConfig

Variable	IcuCmpOutPinConfig	
Description	This container contains all configuration for the cmp out pin configuration	
Screenshot		
Properties	Property	Value
	Type	Variable: Container
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.7.14 IcuCmpOutPinConfig

Variable	IcuCmpOutPinConfig	
Description	This container contains all configuration for the cmp out pin configuration	
Screenshot		
Properties	Property	Value
	Type	Variable: Container
	Origin	FLAGCHIP
	SymbolicNameValue	false

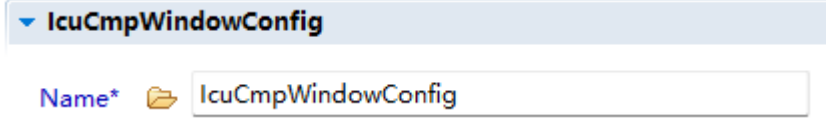
3.2.7.15 IcuCmpOutPinEnable

Variable	IcuCmpOutPinEnable	
Description	The CMPO output enable	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.7.16 IcuCmpDacReference

Variable	IcuCmpDacReference	
Description	The CMPO output source select.	
Screenshot		
Properties	Property	Value
	Type	Variable: String(Range)
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	ICU_CMP_FILTEROUT,ICU_CMP_WINOUT

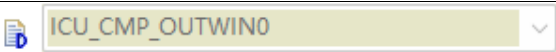
3.2.7.17 IcuCmpWindowConfig

Variable	IcuCmpWindowConfig	
Description	This container contains all configuration for the cmp window configuration	
Screenshot		
Properties	Property	Value
	Type	Variable: Container
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.7.18 IcuCmpWinOutUserDef

Variable	IcuCmpWinOutUserDef*	
Description	User can define CMPOUT_WIN level, when window is closed.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean(Range)
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.7.19 IcuCmpOutWinUserDefLevel

Variable	IcuCmpOutWinUserDefLevel	
Description	The CMPOUT_WIN defined by user.	
Screenshot		
Properties	Property	Value
	Type	Variable: String(Range)
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.7.20 IcuCmpoClseWindowEnable

Variable	IcuCmpoClseWindowEnable	
Description	During active window, user can define that if CMP changes its output (CMPO), the window is closed.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean(Range)
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.7.21 IcuCmpOCloseWinEvent

Variable	IcuCmpOCloseWinEvent	
Description	The CMP event caused window close.	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean(Range)
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	ICU_CMP_RISINGEDGE,ICU_CMP_FALLINGEDGE,ICU_CMP_BOTHEDGES

3.2.7.22 IcuCmpInvSampleEnable

Variable	IcuCmpInvSampleEnable	
Description	invert the WINDOW/SAMPLE signal,(only use for window mode)	
Screenshot		
Properties	Property	Value
	Type	Variable: Boolean(Range)
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.7.23 IcuCmpPartitionRef

Variable	IcuCmpPartitionRef	
Description	Maps the Icu Cmp instance to ECUC partitions	

Screenshot		
Properties	Property	Value
	Type	REFERENCE
	Origin	FLAGCHIP
	Range	ASPathDataOfSchema:/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

3.2.8 IcuGeneral

Container	IcuGeneral	
Description	Configuration of general ICU parameters.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.8.1 IcuDevErrorDetect

Variable	IcuDevErrorDetect	
Description	Switches the Development Error Detection and Notification on or off.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.8.2 IcuReportWakeupSource

Variable	IcuReportWakeupSource	
Description	Switch for enabling Wakeup source reporting.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.8.3 IcuOverflowTimeStatistics

Variable	IcuOverflowTimeStatistics	
Description	Switch for enabling overflow time statistics.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.8.4 IcuMultiCoreSupport

Variable	IcuMultiCoreSupport	
Description	Switch for enabling MutilCore Feature	
Screenshot		
Properties	Property	Value

	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

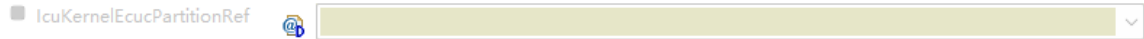
3.2.8.5 IcuFtuGtbSupport

Variable	IcuFtuGtbSupport	
Description	The FTU supports a global time base through the Global Time Base (GTB) feature. If enable, the ftu CNT won't start before calling Icu_StartGlobalTime() .	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.8.6 IcuTsDmaSupport

Variable	IcuTsDmaSupport	
Description	True:Icu Timestamp function supports dma.False :Icu Timestamp function don't support dma	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.8.7 IcuKernelEcucPartitionRef

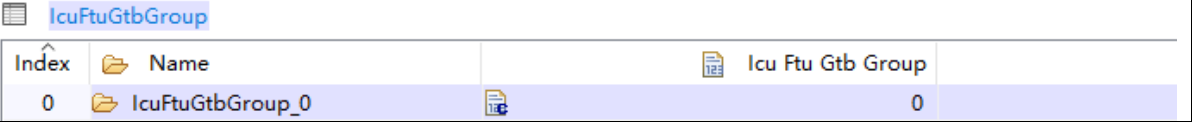
Variable	IcuKernelEcucPartitionRef	
Description	Maps the ICU kernel to zero or one ECUC partitions to assign the driver kernel to a certain core. The ECUC partition referenced is a subset of the ECUC partitions where the ICU driver is mapped to.	
Screenshot		
Properties	Property	Value
	Type	REFERENCE
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.8.8 IcuEcucPartitionRef

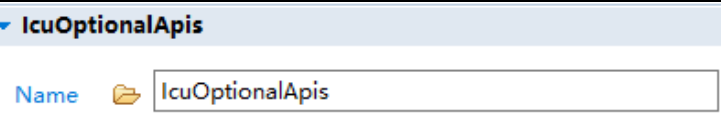
Variable	IcuEcucPartitionRef	
Description	Maps the Icu driver to zero or multiple ECUC partitions to make the driver API available in the according partition	
Screenshot		
Properties	Property	Value
	Type	REFERENCE
	Origin	FLAGCHIP

	SymbolicNameValue	false
	Range	ASPathDataOfSchema:/AUTOSAR/EcuDefs/EcuC/EcucPartitionCollection/EcucPartition

3.2.8.9 IcuFtuGtbGroup

Variable	IcuFtuGtbGroup	
Description	Configuration of a ftu gtb group	
Screenshot		
Properties	Property	Value
	Type	LIST
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range:	0~7

3.2.9 IcuOptionalApis

Container	IcuOptionalApis	
Description	This container contains all configuration switches for configuring optional API services of the ICU driver.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.9.1 IcuDelInitApi

Variable	IcuDelInitApi	
Description	Adds / removes the service Icu_DelInit() from the code.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.9.2 IcuDisableWakeupApi

Variable	IcuDisableWakeupApi	
Description	Adds / removes the service Icu_DisableWakeup() from the code.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.9.3 IcuEdgeCountApi

Variable	IcuEdgeCountApi	
Description	Adds / removes all services related to the edge counting functionality as listed below, from the code: Icu_ResetEdgeCount(), Icu_EnableEdgeCount(), Icu_DisableEdgeCount(), Icu_GetEdgeNumbers().	
Screenshot	IcuEdgeCountApi    	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.9.4 IcuEnableWakeupApi

Variable	IcuEnableWakeupApi	
Description	Adds / removes the service Icu_EnableWakeup() from the code.	
Screenshot	IcuEnableWakeupApi    	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.9.5 IcuGetDutyCycleValuesApi

Variable	IcuGetDutyCycleValuesApi	
Description	Adds / removes the service Icu_GetDutyCycleValues() from the code.	
Screenshot	IcuGetDutyCycleValuesApi    	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.9.6 IcuGetInputStateApi

Variable	IcuGetInputStateApi	
Description	Adds / removes the service Icu_GetInputState() from the code.	
Screenshot	IcuGetInputStateApi    	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.9.7 IcuGetTimeElapsedApi

Variable	IcuGetTimeElapsedApi	
Description	Adds / removes the service Icu_GetTimeElapsed() from the code.	

Screenshot	IcuGetTimeElapsedApi  ☒  	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.9.8 IcuGetVersionInfoApi

Variable	IcuGetVersionInfoApi	
Description	Adds / removes the service Icu_GetVersionInfo() from the code.	
Screenshot	IcuGetVersionInfoApi  ☒  	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.9.9 IcuSetModeApi

Variable	IcuSetModeApi	
Description	Adds / removes the service Icu_SetMode() from the code.	
Screenshot	IcuSetModeApi  ☒  	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.9.10 IcuSignalMeasurementApi

Variable	IcuSignalMeasurementApi	
Description	Adds / removes the services Icu_StartSignalMeasurement() and Icu_StopSignalMeasurement() from the code.	
Screenshot	IcuSignalMeasurementApi  ☒  	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.9.11 IcuTimestampApi

Variable	IcuTimestampApi	
Description	Adds / removes all services related to the timestamping functionality as listed below from the code: Icu_StartTimestamp(), Icu_StopTimestamp(), Icu_GetTimestampIndex().	
Screenshot	IcuTimestampApi  ☒  	
Properties	Property	Value
	Type	Variable: BOOLEAN

	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.9.12 IcuWakeupFunctionalityApi

Variable	IcuWakeupFunctionalityApi	
Description	Adds / removes the service Icu_CheckWakeup() from the code.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.9.13 IcuEdgeDetectApi

Variable	IcuEdgeDetectApi	
Description	Adds / removes the services Icu_EnableEdgeDetection() and Icu_DisableEdgeDetection() from the code.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.10 IcuNonAUTOSAR

Container	IcuNonAUTOSAR	
Description	Enabling the settings of this section will configure the driver in a mode not compliant with AUTOSAR requirements	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.10.1 IcuGetInputLevelApi

Variable	IcuGetInputLevelApi	
Description	Add / removes Icu_GetInputLevel API from the code	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.10.2 IcuOverflowNotificationApi

Variable	IcuOverflowNotificationApi	
Description	Add / removes Overflow Notification functionality.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.11 CommonPublishedInformation

Container	CommonPublishedInformation	
Description	Common container, aggregated by all modules. It contains published information about vendor and versions.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.11.1 ArReleaseMajorVersion

Variable	ArReleaseMajorVersion	
Description	Major version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	4

3.2.11.2 ArReleaseMinorVersion

Variable	ArReleaseMinorVersion	
Description	Minor version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	6

3.2.11.3 ArReleaseRevisionVersion

Variable	ArReleaseRevisionVersion	
-----------------	--------------------------	--

Description	Revision version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	0

3.2.11.4 SwMajorVersion

Variable	SwMajorVersion	
Description	Major version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	0

3.2.11.5 SwMinorVersion

Variable	SwMinorVersion	
Description	Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	3

3.2.11.6 SwPatchVersion

Variable	SwPatchVersion	
Description	Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	0

3.2.11.7 ModuleId

Variable	ModuleId	
Description	Module ID of this module from Module List.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	122

3.2.11.8 VendorId

Variable	VendorId	
Description	Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.	
Screenshot		
Properties	Property	Value
	Type	Variable : INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Range	174

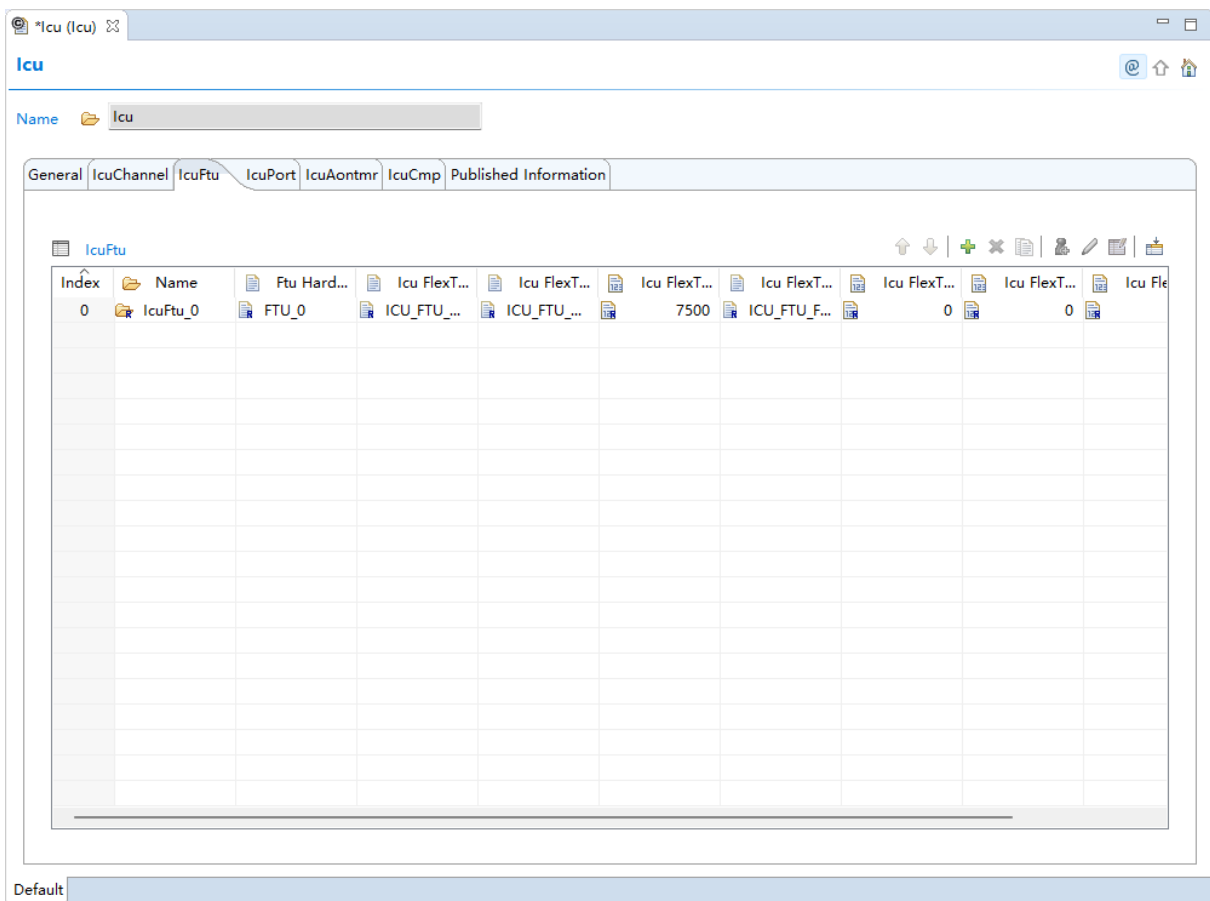
Chapter 4 Configuration Guides

4.1 Configuration Item Constraint

- 1) Each ICU channel must use a different physical channel.
- 2) Only FTU channels support ICU_MODE_SIGNAL_MEASUREMENT.
- 3) Only FTU channels support ICU_MODE_TIMESTAMP

4.2 ICU Usage Common Steps

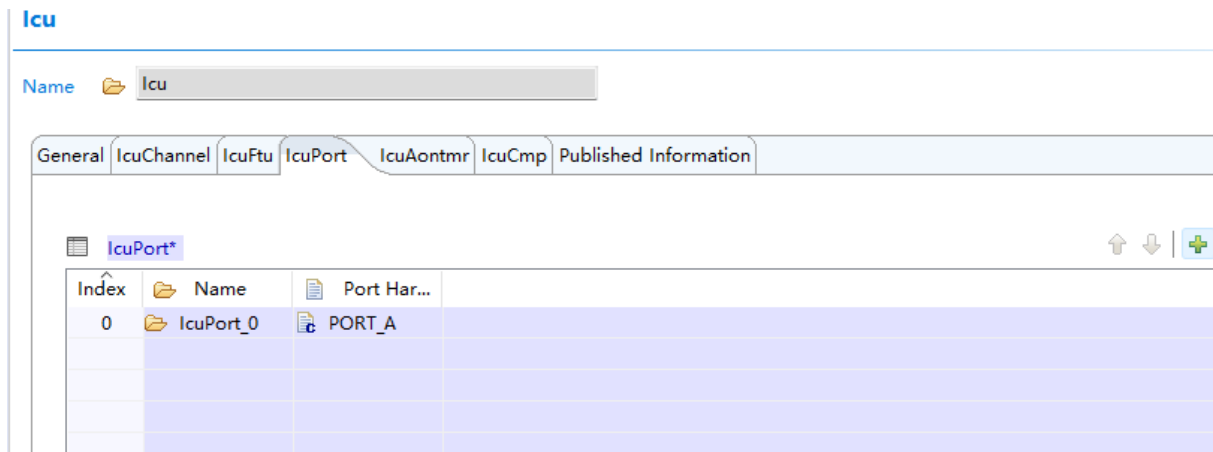
- 1) Set up the ICU hardware module.
 - If FTU is used.
 - a) Add an object in the IcuFtu tab.



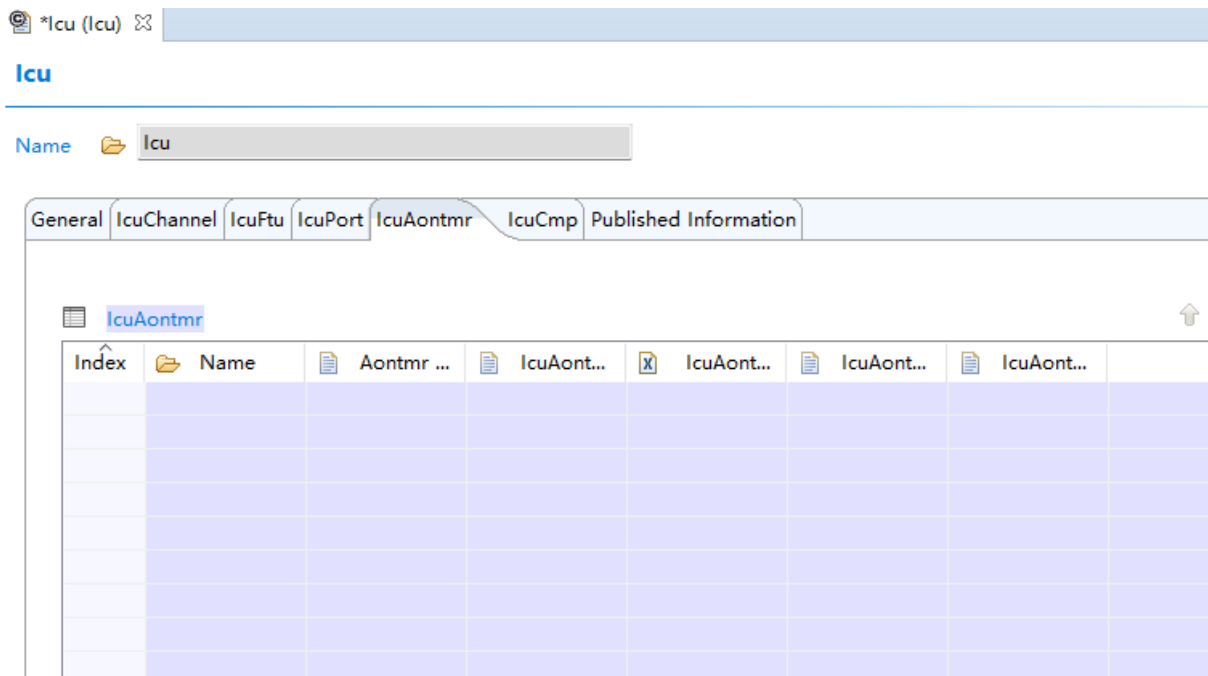
- b) Set up the FTU module. Includes hardware module, clock source, prescaler, period, channel filter prescaler, input filter value of channel 0~3.

- [illegible]

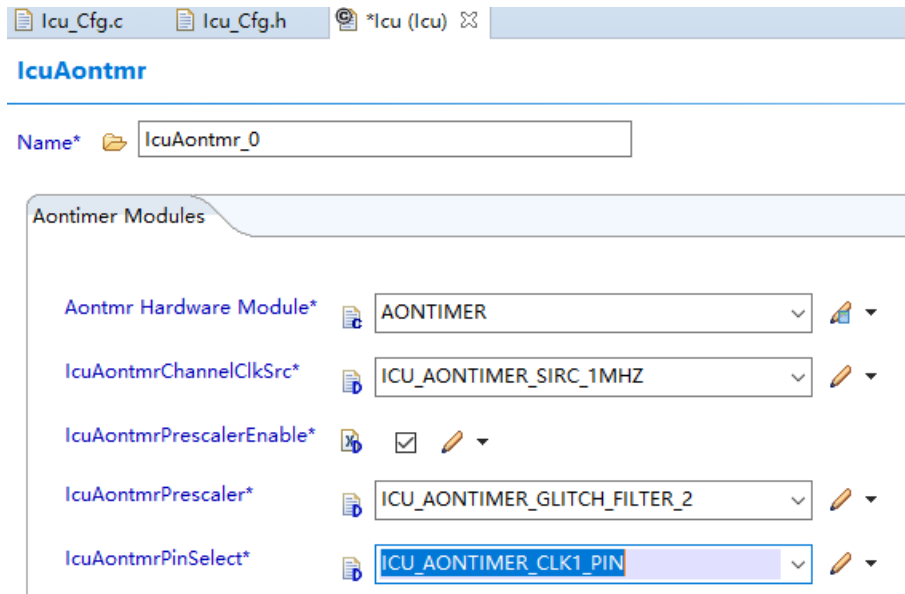
- Icu_Cfg.c Icu_Cfg.h *Icu (Icu) ✕
- ## IcuPort
- Name*  IcuPort_0
- Port Modules
- Port Hardware Module*  PORT_A ▼  ▼



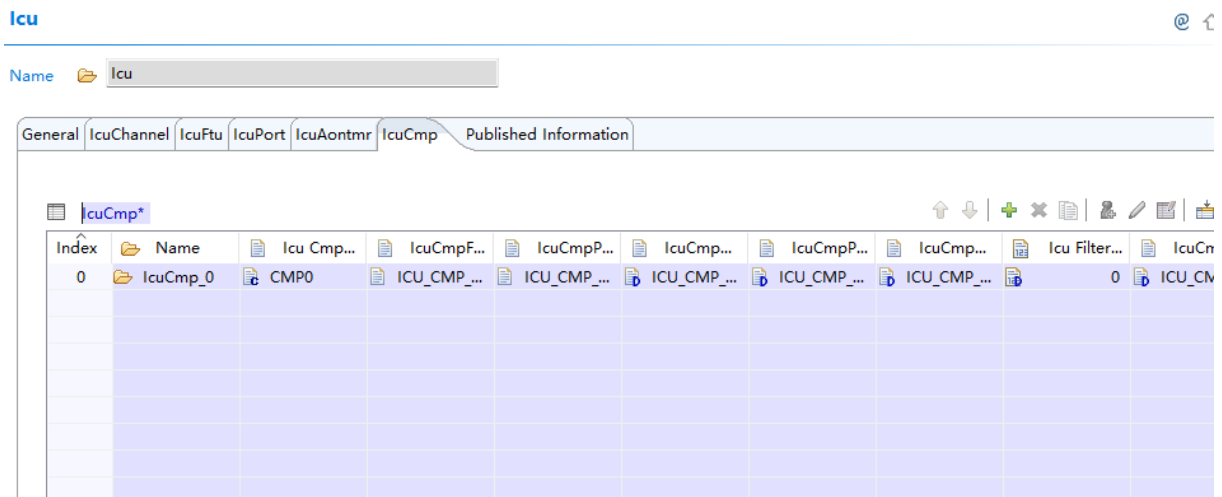
- If AONTIMER is used
 - a) Add an object in the IcuAontmr tab.



- b) Set up the AONTIMER module, including hardware module, clock source, prescaler enable, glitch filter value, and input source.



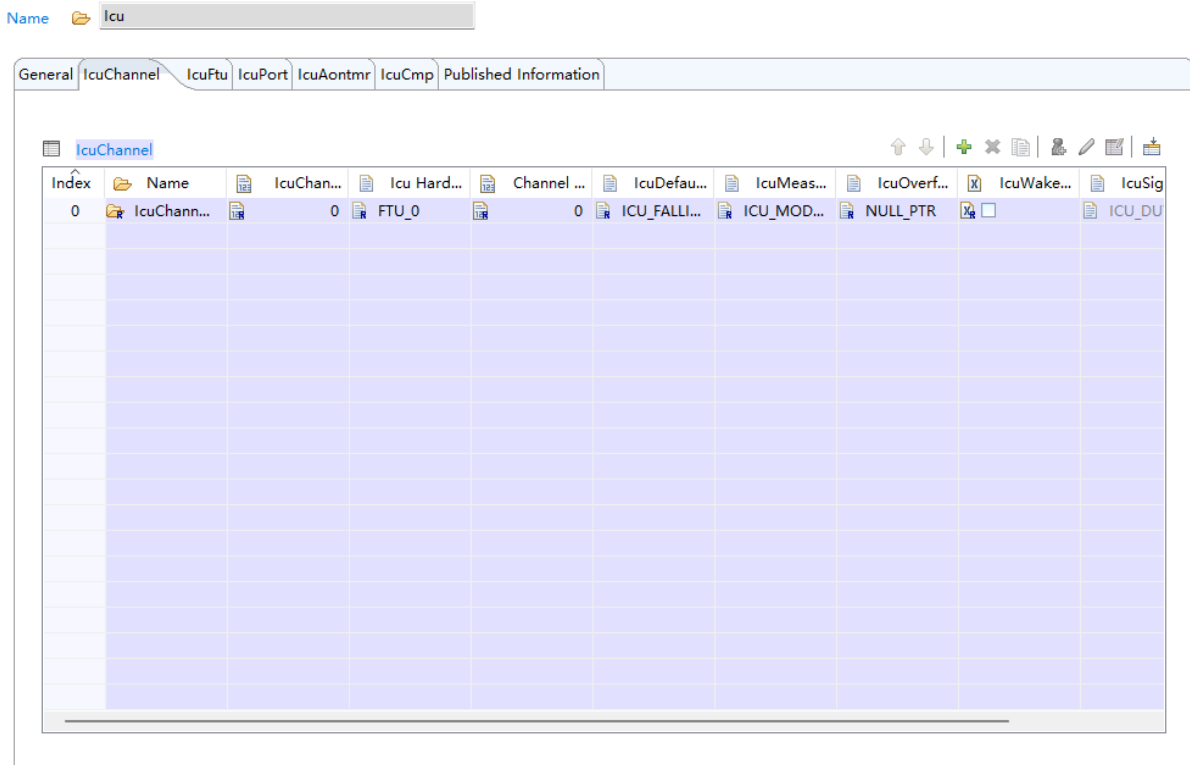
- If CMP is used.
 - Add an object in the Cmp tab.



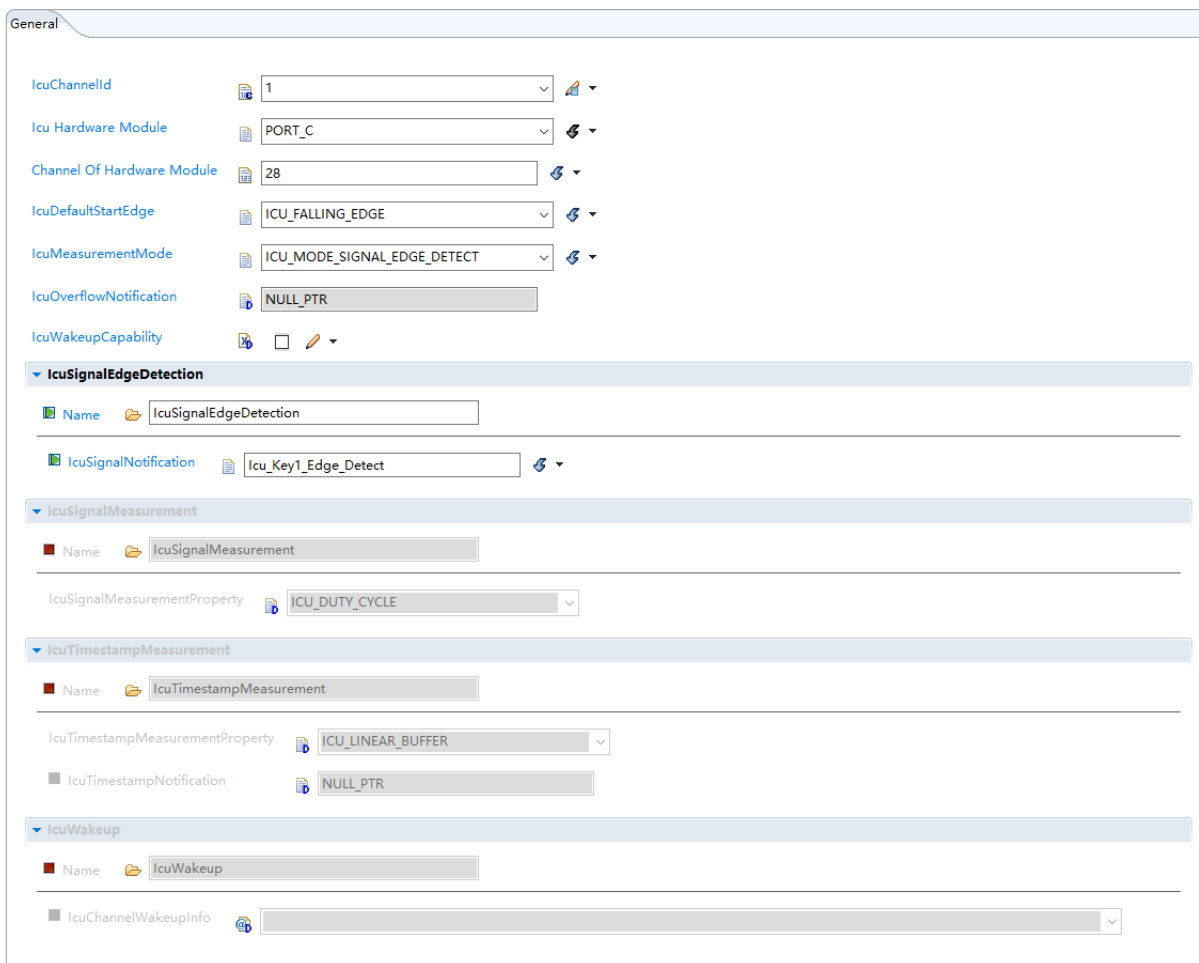
- Set up the Cmp module, including hardware module, function mode, inputsrc and MuxSelect.

2) Configure an ICU channel.

- Add an object in IcuChannel tab.



b) Configure each option of the ICU channel.



- c) Wakeup-capability:Enable IcuWakeupCapability and select a wakeup source which is configured in EcuM module.

IcuWakeupCapability*    ▼

IcuWakeup

 **Name*** 

 **IcuChannelWakeupInfo***   ▼

- d) If IcuMeasurementMode is set to ICU_MODE_SIGNAL_EDGE_DETECT,IcuSignalEdgeDetection must be enabled. Input the notification function name in IcuSignalNotification

IcuMeasurementMode*   ▼

IcuSignalEdgeDetection

 **Name*** 

 **IcuSignalNotification***   ▼

- e) If IcuMeasurementMode is set to ICU_MODE_SIGNAL_MEASUREMENT,IcuSignalMeasurement must be enabled.And select the IcuSignalMeasurementProperty.

IcuMeasurementMode*   ▼

IcuSignalMeasurement

 **Name*** 

IcuSignalMeasurementProperty*   ▼

- f) If IcuMeasurementMode is set to ICU_MODE_TIMESTAMP,IcuTimestampMeasurement must be enabled.Then select the IcuTimestampMeasurementProperty and input the notification function name in IcuTimestampNotification.

IcuMeasurementMode*   ▼

IcuTimestampMeasurement

 **Name*** 

IcuTimestampMeasurementProperty*   ▼

 **IcuTimestampNotification***   ▼

3) General Configuration:

- Select the General tab.
- Input the number of objects to be added to the IcuChannel tab.

▼

IcuConfigSet

Name

IcuConfigSet

IcuMaxChannel*

1

▼

- c) Enable or disable the development error detection, wakeup source reporting, overflow time statistics.

▼

IcuGeneral

Name

IcuGeneral

IcuDevErrorDetect

☒

IcuReportWakeupSource

☐

IcuOverflowTimeStatistics

☒

- d) Enabling or disable the settings of this section will configure the driver in a mode not compliant with AUTOSAR requirements.

▼

IcuNonAUTOSAR

Name

IcuNonAUTOSAR

IcuGetInputLevelApi

☒

IcuOverflowNotificationApi

☐

- e) Choose which API will be used.

▼

IcuOptionalApis

Name

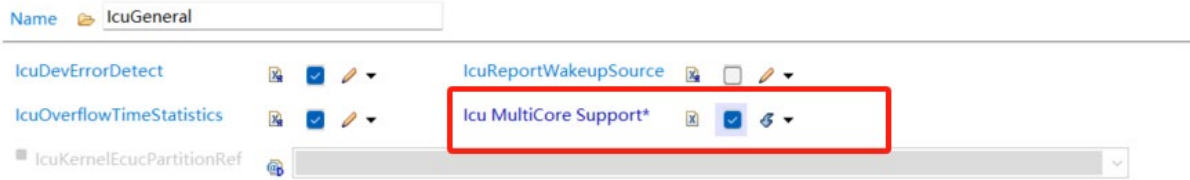
IcuOptionalApis

IcuDelInitApi	☒	IcuDisableWakeupApi	☒
IcuEdgeCountApi	☒	IcuEnableWakeupApi	☒
IcuGetDutyCycleValuesApi	☒	IcuGetInputStateApi	☒
IcuGetTimeElapsedApi	☒	IcuGetVersionInfoApi	☒
IcuSetModeApi	☒	IcuSignalMeasurementApi	☒
IcuTimestampApi	☒	IcuWakeupFunctionalityApi	☒
IcuEdgeDetectApi	☒		

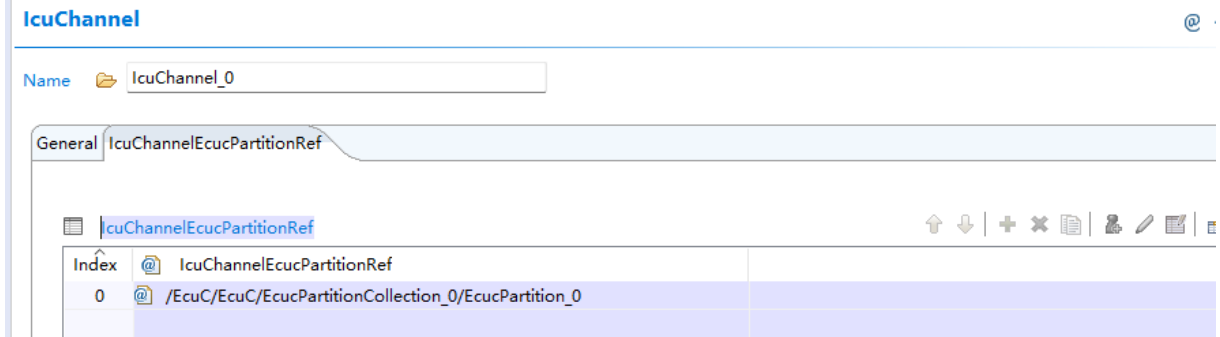
- 4) Generate configuration files.

- 5) MultiCoreFeature.

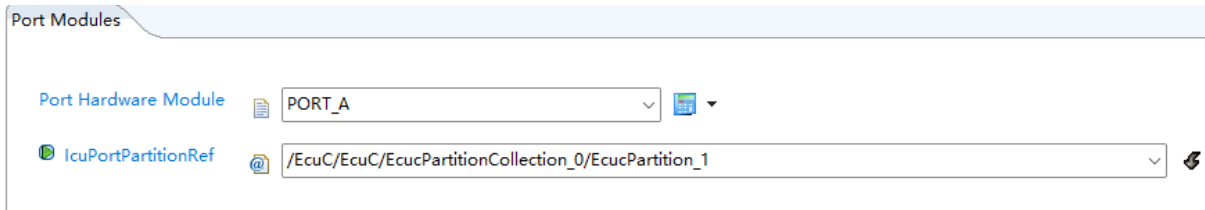
If Icu MultiCoreSupport is enable, each Channel should be distributed to a partition. Each HW unit should be distributed to a partition.



a) Channel.



b) HW unit.



c) Module.

